

From CLIP to Vision-Language Models

Following our CLIP lecture · ES 667 · Nipun Batra · IIT Gandhinagar

How Images Become Context for Language Generation

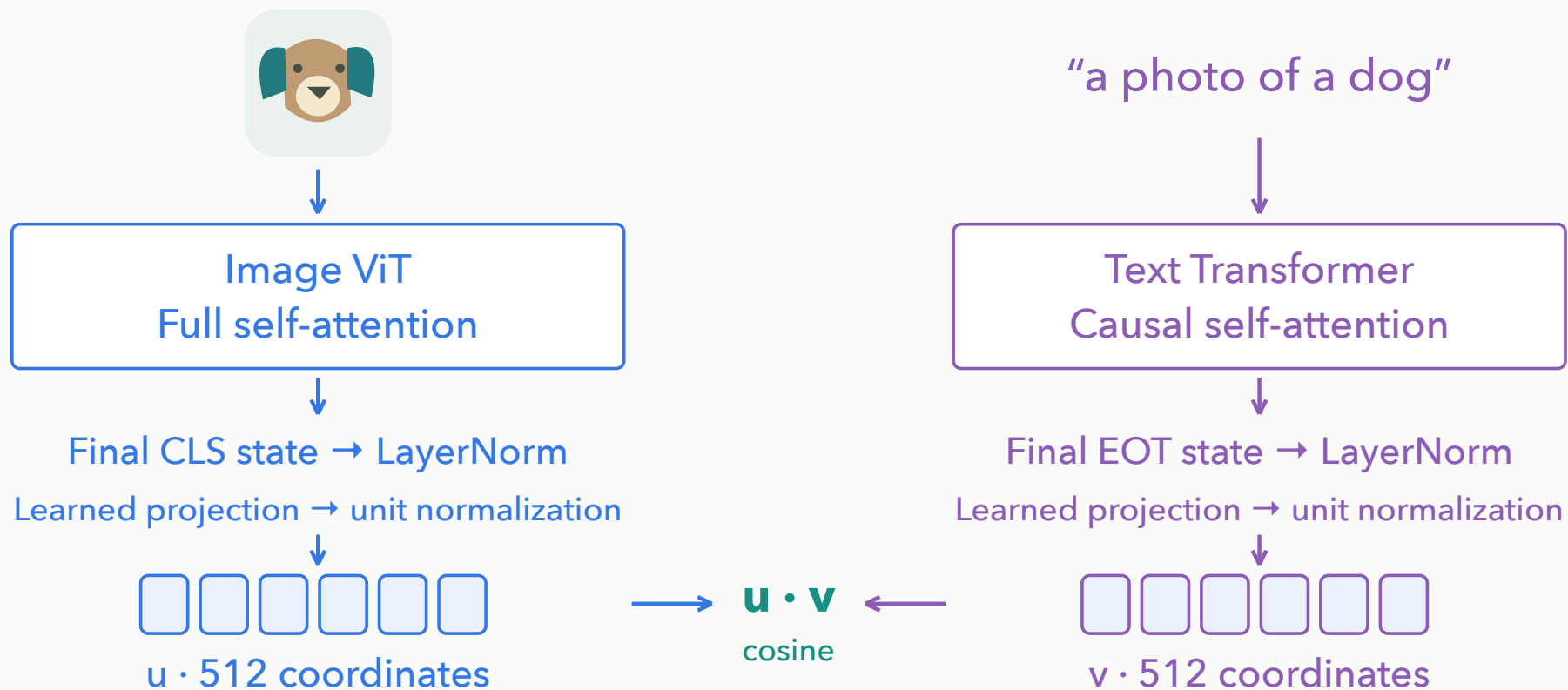
Last time: compare an image with descriptions.

Today: use the image to generate an answer.



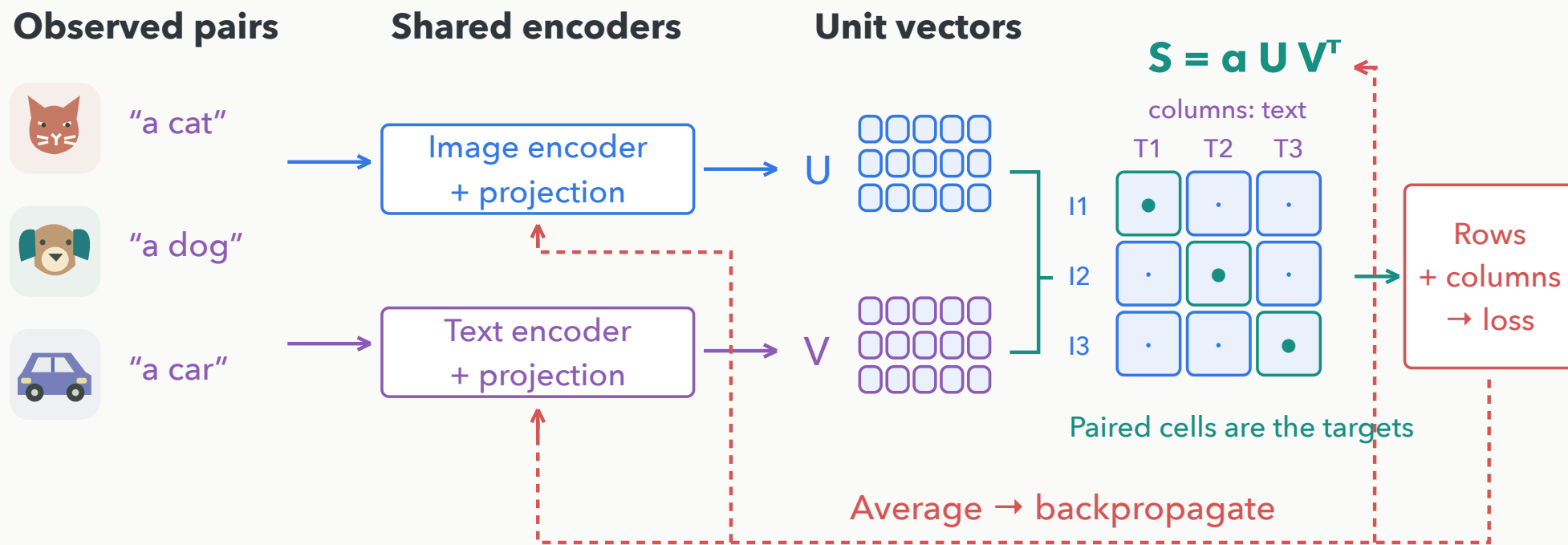
| How does visual information become context for next-token prediction?

Recall: two encoders, one comparison space



CLIP compares an image vector with a supplied text vector.

Recall: paired examples teach the match



Update both encoders, both projections and the score scale.

The matching image-caption pair is the target in both directions.

Recall: reuse the vectors for different tasks

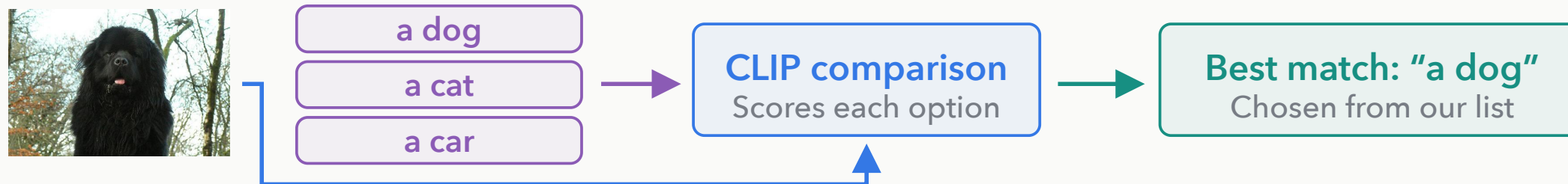


Reuse the frozen encoders. Choose which vectors to compare.

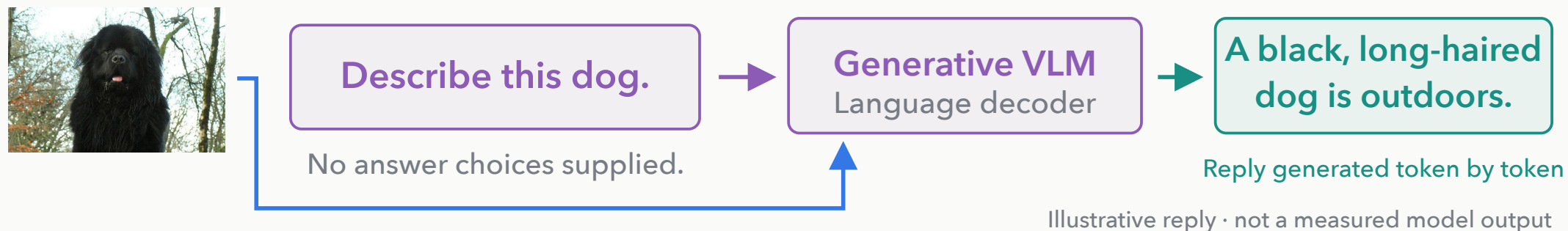
New descriptions give us new choices without retraining the encoders.

Next task: generate an answer from the image

CLIP · WE SUPPLY THE POSSIBLE DESCRIPTIONS



GENERATIVE VLM · WE SUPPLY AN IMAGE + A QUESTION



Illustrative reply · not a measured model output

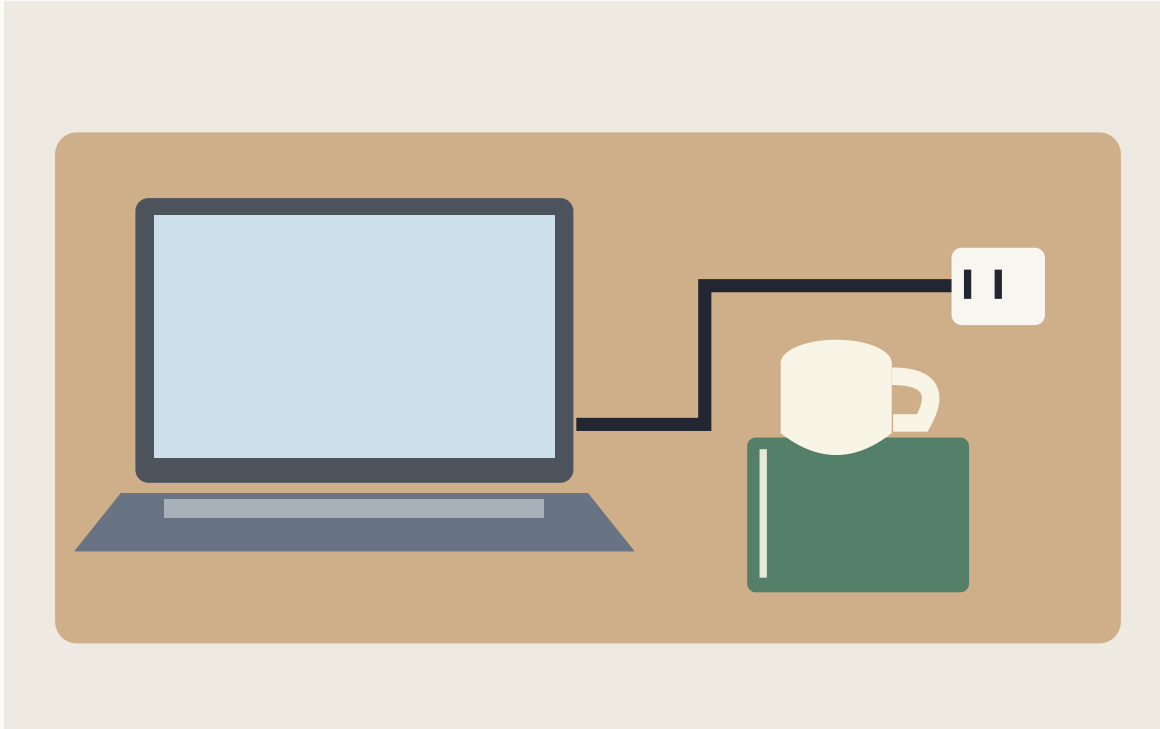
A language decoder writes the reply, using the image as context.

SECTION 01

What can a VLM do?

Start with the evidence an answer needs.

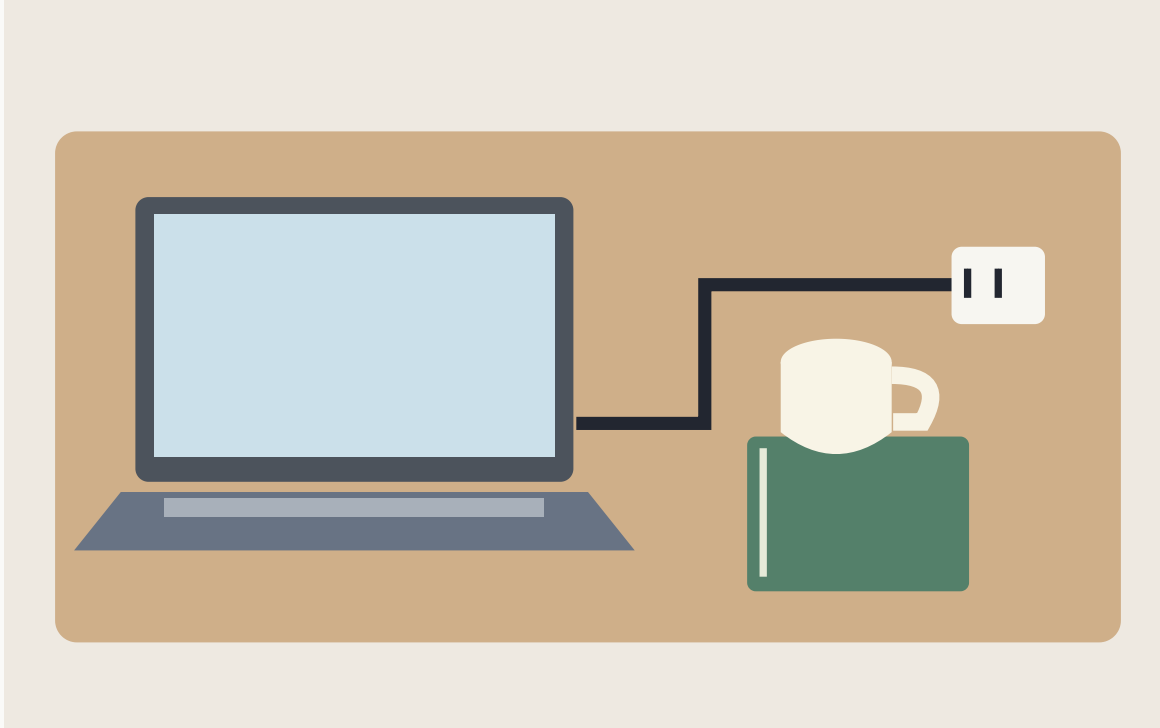
Can the answer use visible evidence?



QUESTION

Is the laptop plugged in?

Can the answer use visible evidence?

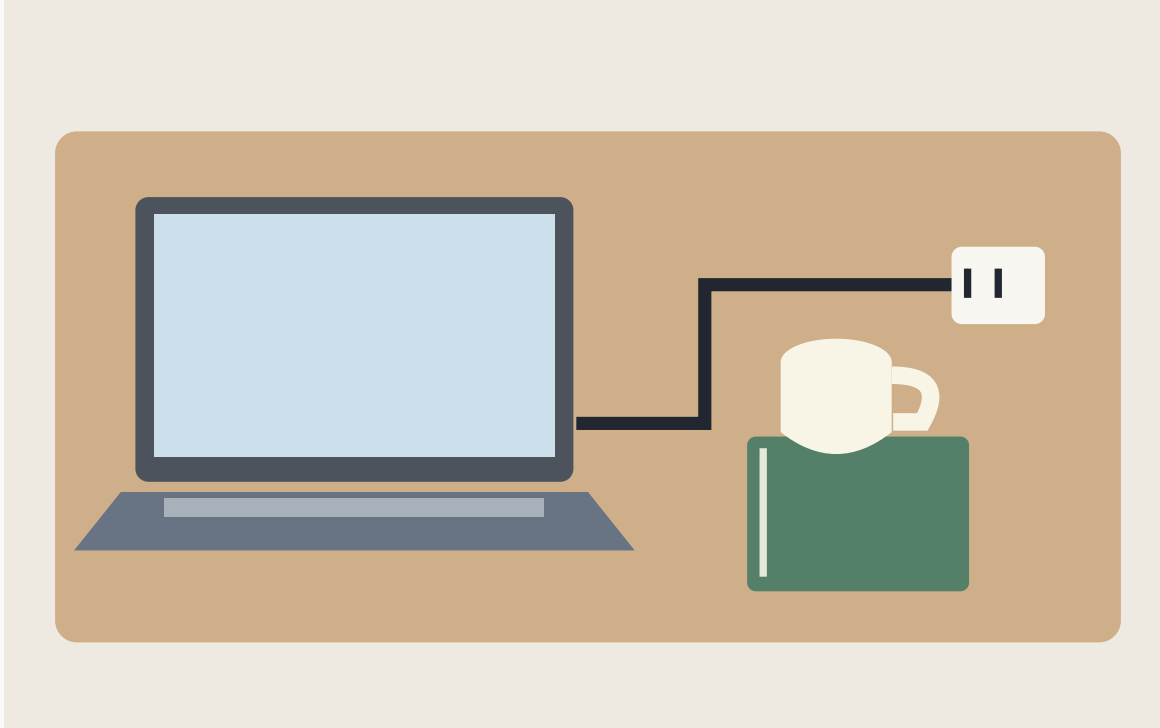


REFERENCE ANSWER

Is the laptop plugged in?

Yes. A cable connects the laptop to the socket.

Can the answer use visible evidence?



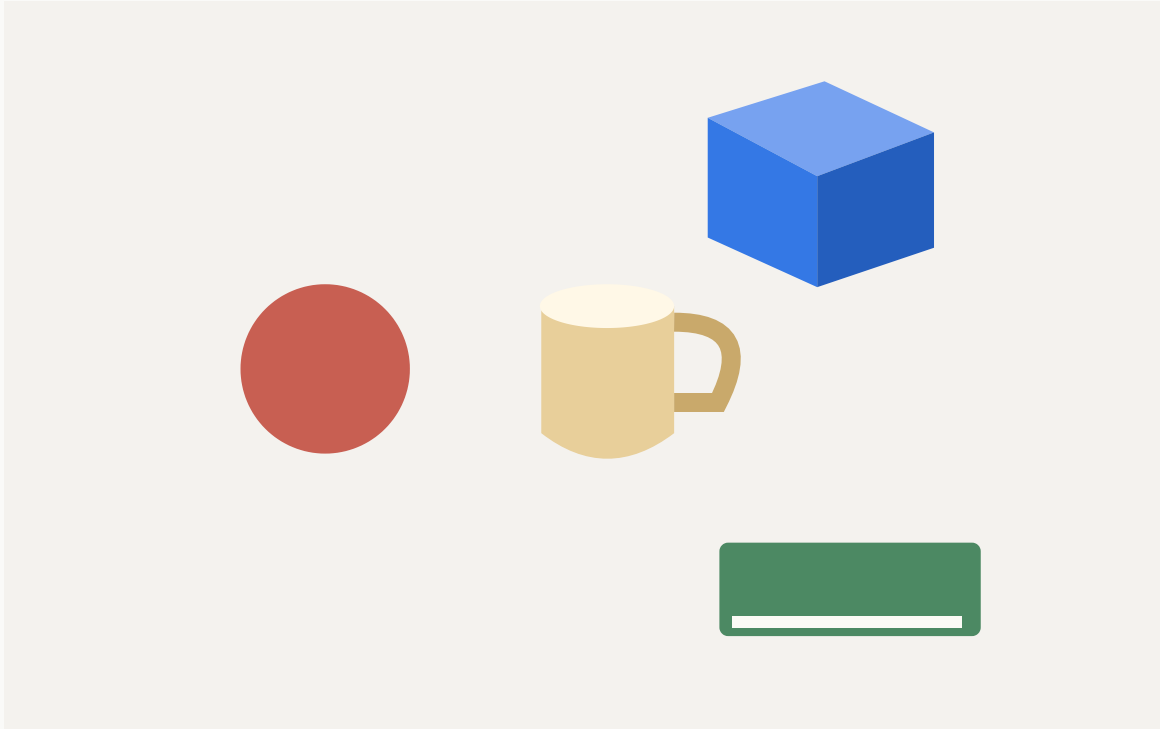
ARCHITECTURAL REQUIREMENT

Is the laptop plugged in?

Yes. A cable connects the laptop to the socket.

| Trace a connection between objects.

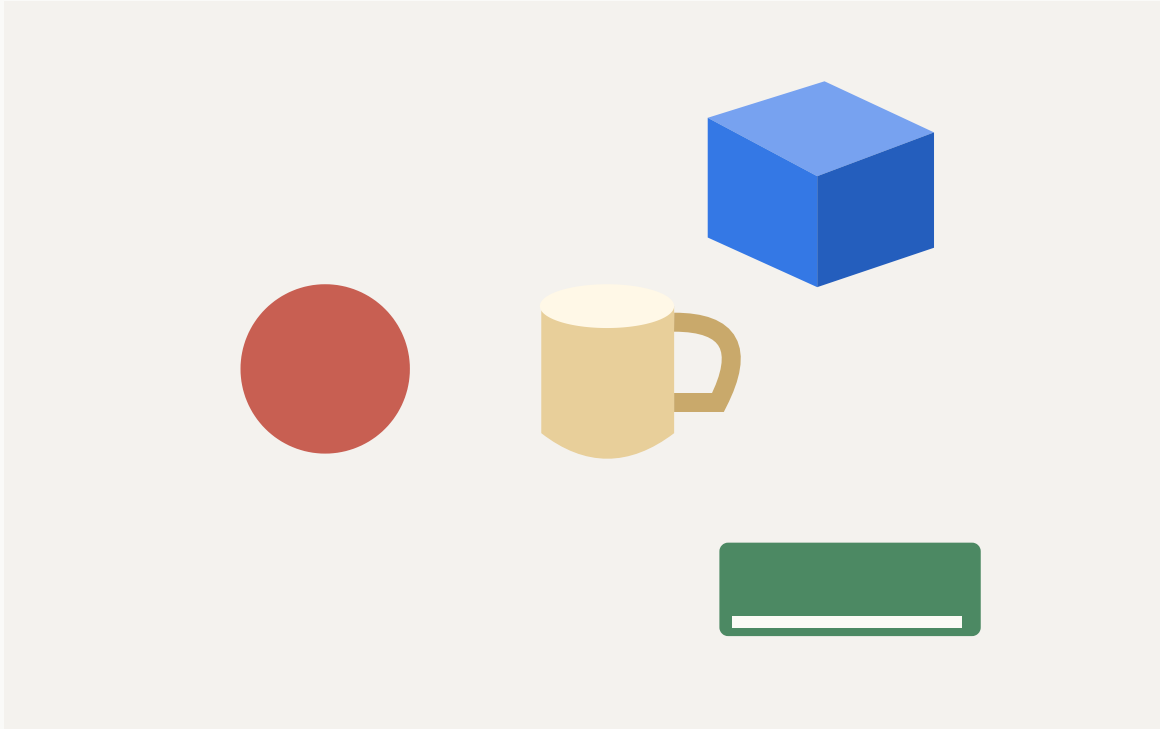
Identity is only part of the question



QUESTION

What is immediately left of the mug?

Identity is only part of the question

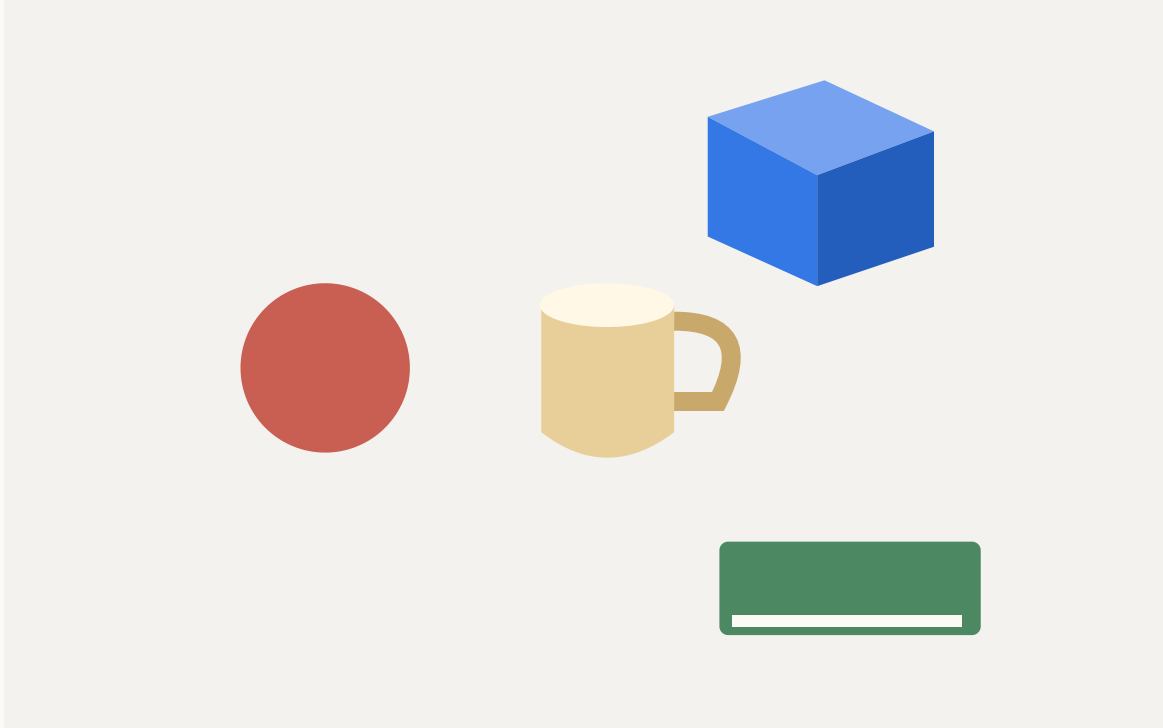


REFERENCE ANSWER

What is immediately left of the mug?

The red ball.

Identity is only part of the question



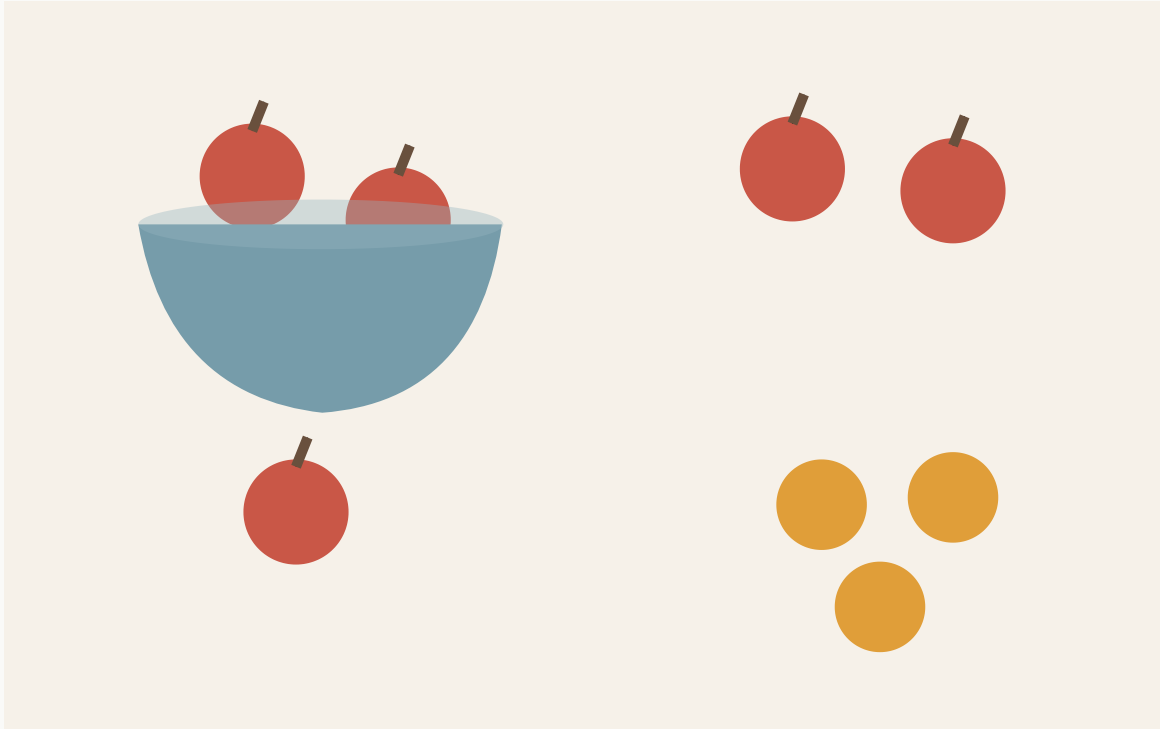
ARCHITECTURAL REQUIREMENT

What is immediately left of the mug?

The red ball.

| Keep object identity and spatial relations.

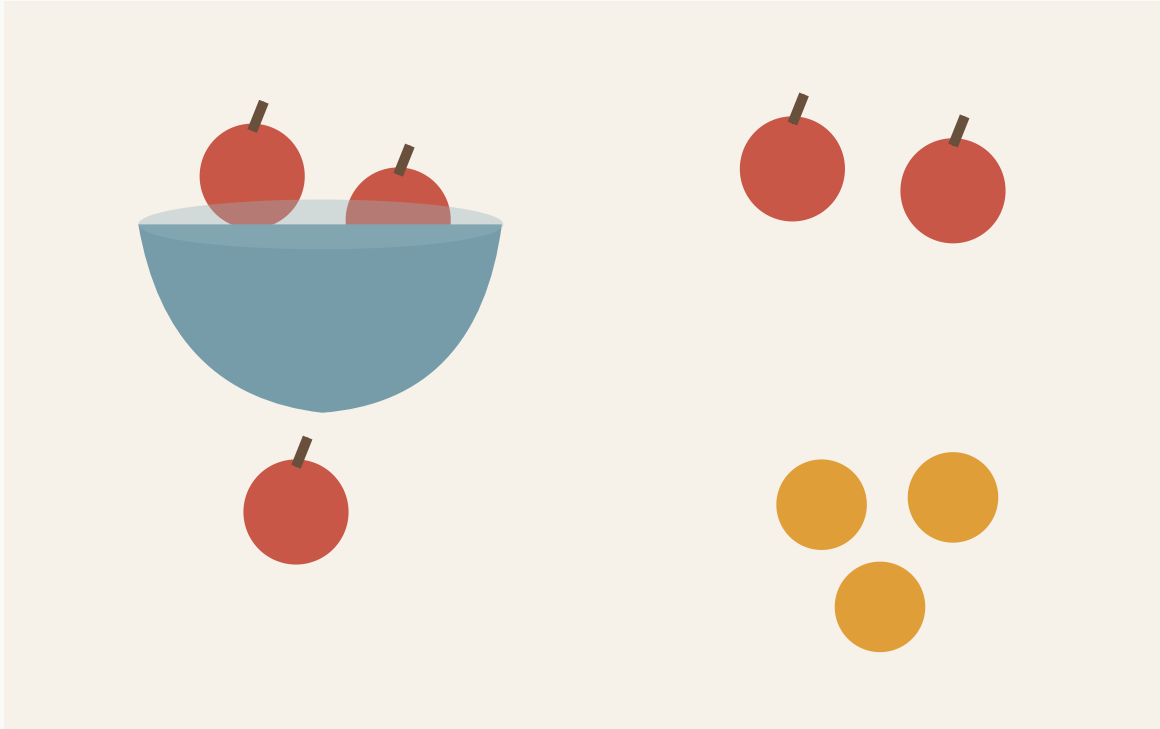
The question selects a subset



QUESTION

How many apples are outside the bowl?

The question selects a subset

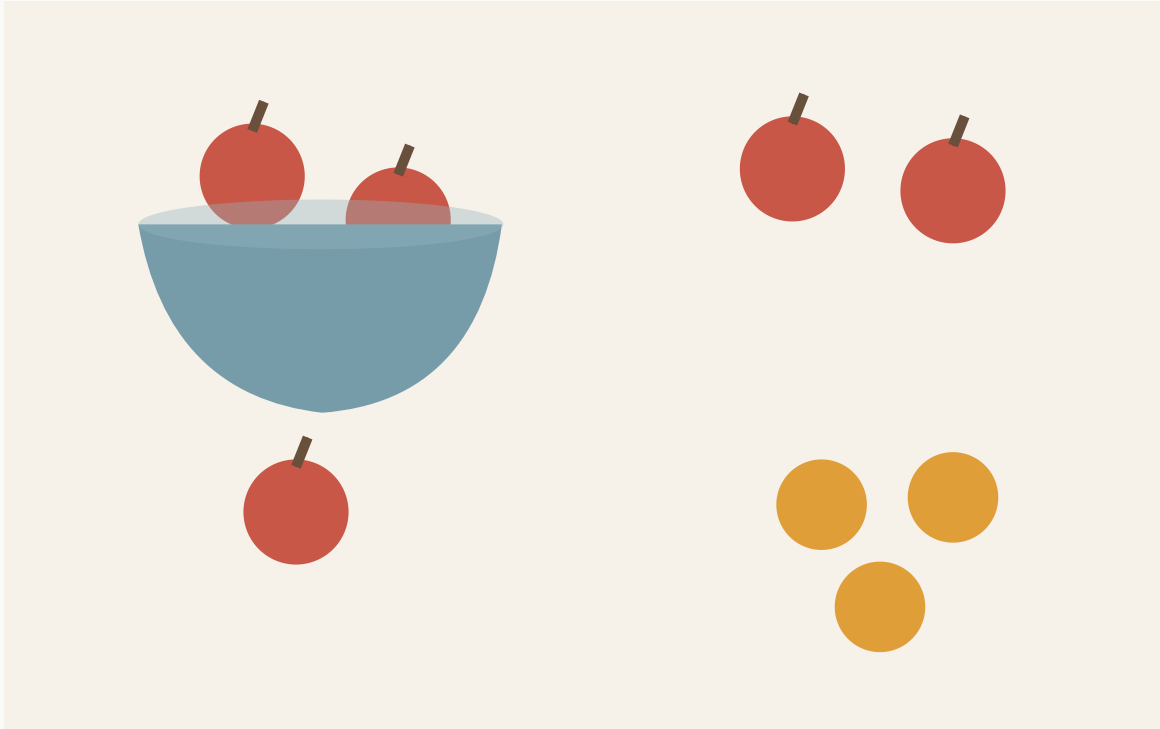


REFERENCE ANSWER

How many apples are outside the bowl?

Three.

The question selects a subset



ARCHITECTURAL REQUIREMENT

How many apples are outside the bowl?

Three.

Represent multiple objects and the region that selects them.

Read first, then calculate

CAMPUS STATIONERY

Invoice: ES667-042

Date: 01 October 2026

Notebooks (4 × ₹200) ₹800

Pens (10 × ₹20) ₹200

Subtotal ₹1,000

GST ₹180

TOTAL ₹1,180

QUESTION

What is the invoice ID?

What percentage of the subtotal is GST?

Read first, then calculate

CAMPUS STATIONERY

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Notebooks (4 × ₹200) ₹800

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Subtotal ₹1,000

GST ₹180

TOTAL ₹1,180

REFERENCE ANSWER

What is the invoice ID?

What percentage of the subtotal is GST?

ES667-042. GST is $180 / 1000 \times 100 = 18\%$.

Read first, then calculate

CAMPUS STATIONERY

Invoice: ES667-042

Date: 01 October 2026

Notebooks (4 × ₹200) ₹800

Pens (10 × ₹20) ₹200

Subtotal ₹1,000

GST ₹180

TOTAL ₹1,180

ARCHITECTURAL REQUIREMENT

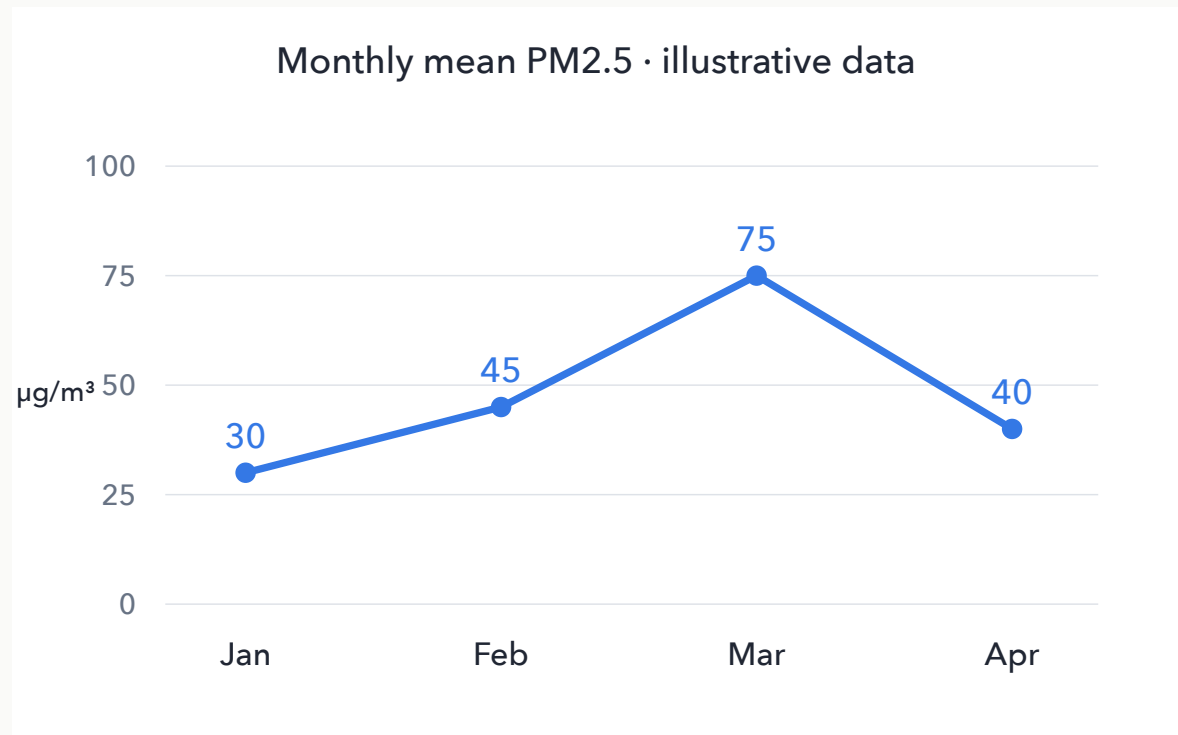
What is the invoice ID?

What percentage of the subtotal is GST?

ES667-042. GST is $180 / 1000 \times 100 = 18\%$.

| Preserve fine text, numbers and their roles.

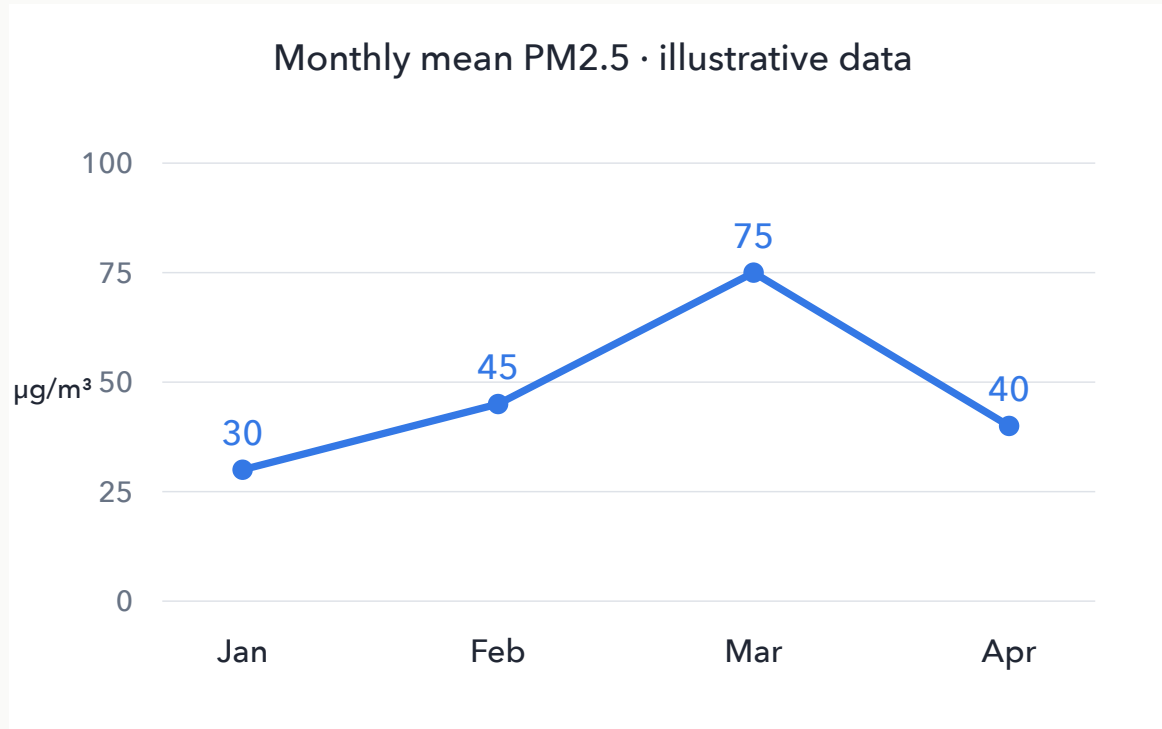
The labels change the answer



QUESTION

Which month is highest?
How much higher is March than January?

The labels change the answer

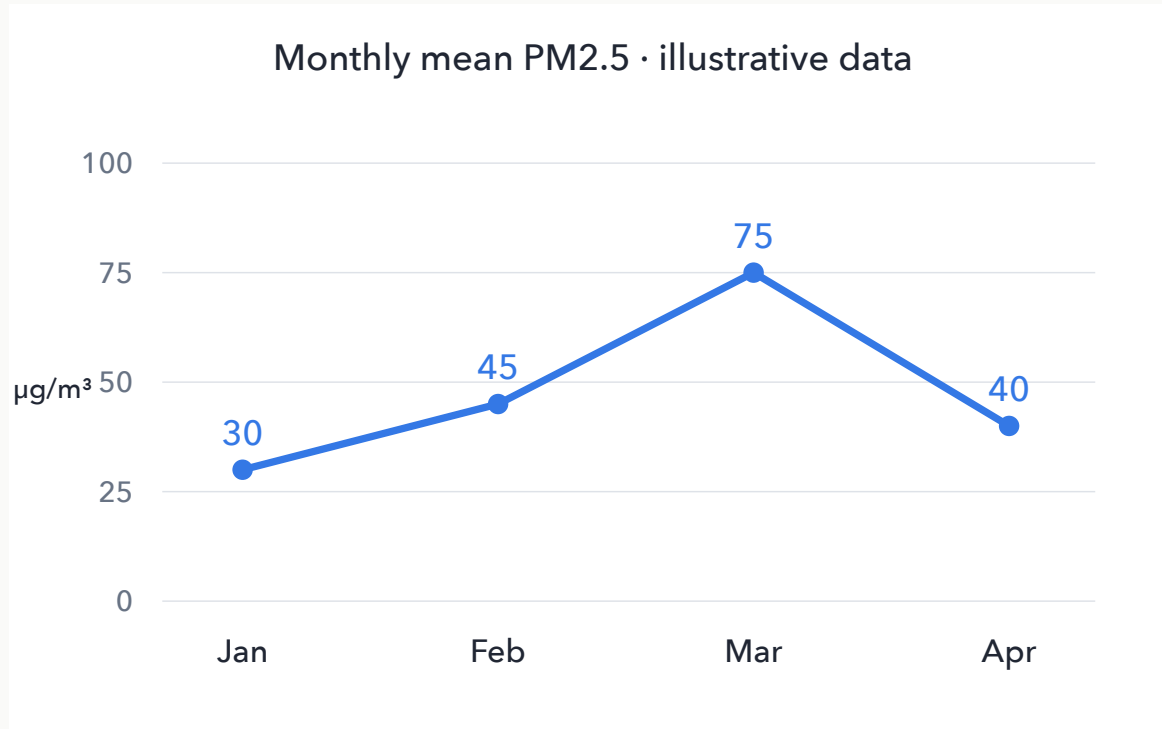


REFERENCE ANSWER

Which month is highest?
How much higher is March than January?

March. The difference is 45 $\mu\text{g}/\text{m}^3$.

The labels change the answer



ARCHITECTURAL REQUIREMENT

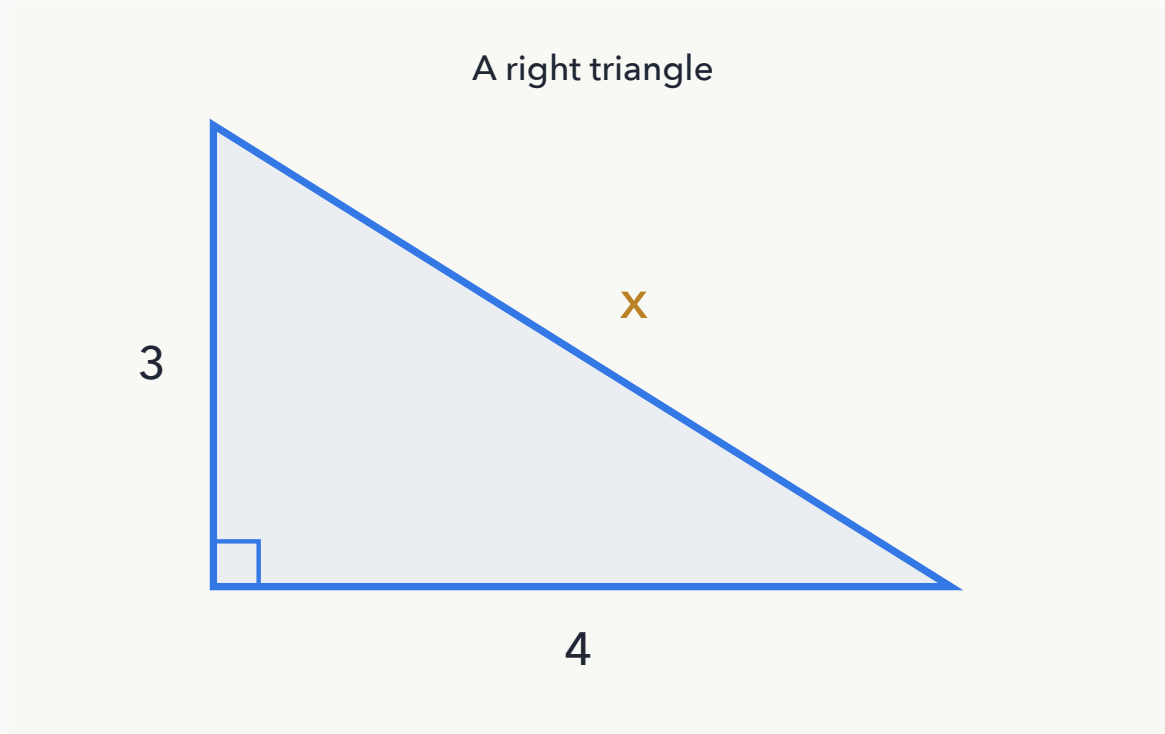
Which month is highest?

How much higher is March than January?

March. The difference is 45 μg/m³.

Bind chart labels to values and units.

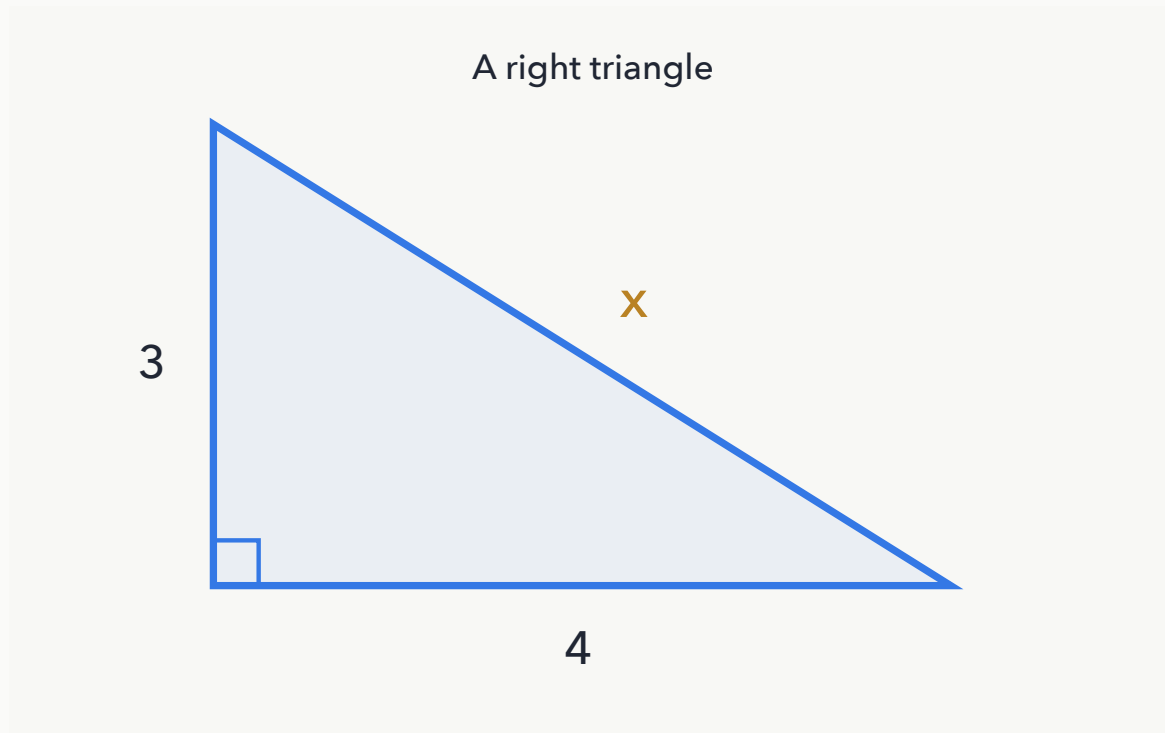
A diagram supplies conditions



QUESTION

What is x?

A diagram supplies conditions

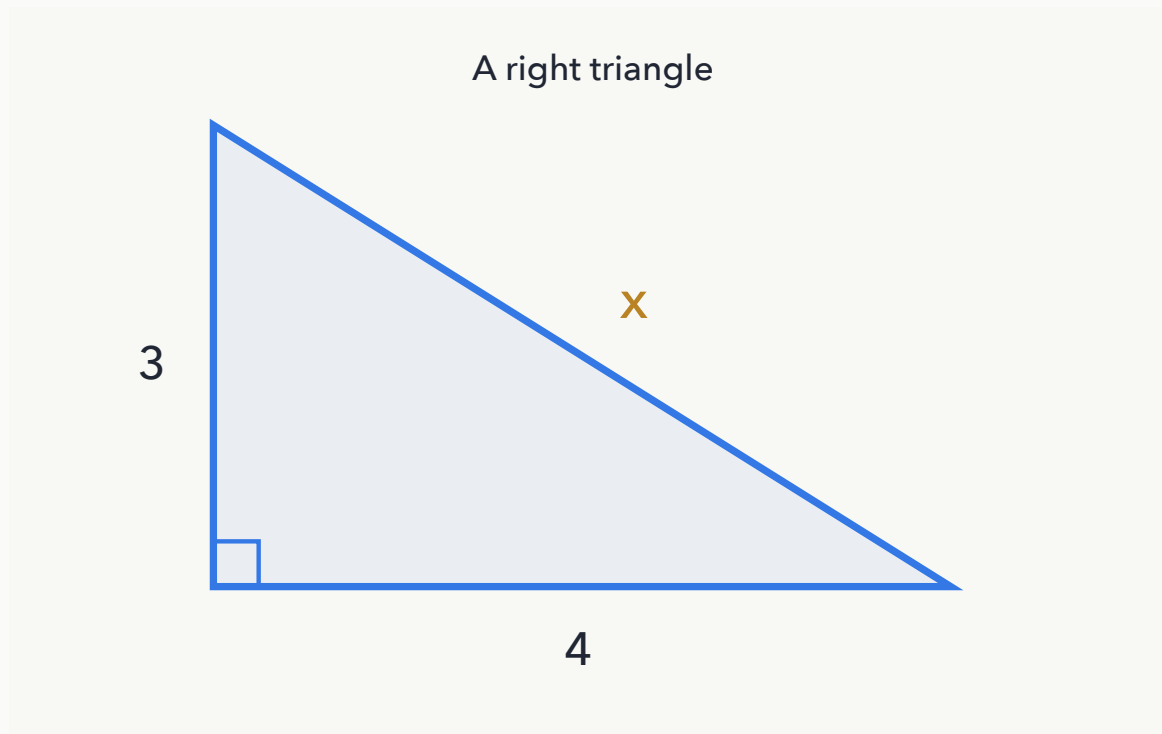


REFERENCE ANSWER

What is x?

$x = 5$, using the marked right angle and sides 3 and 4.

A diagram supplies conditions



ARCHITECTURAL REQUIREMENT

What is x?

x = 5, using the marked right angle and sides 3 and 4.

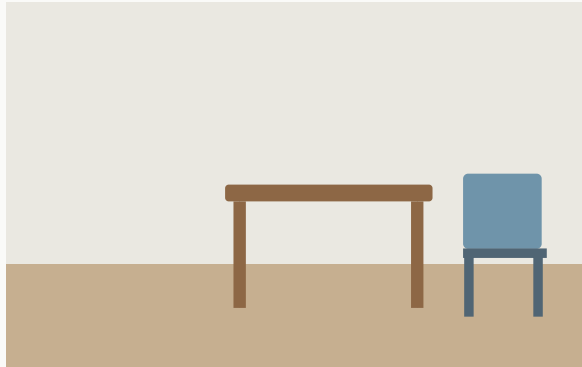
Combine diagram evidence with a mathematical rule.

Compare presence and position

A · BEFORE



B · AFTER



QUESTION

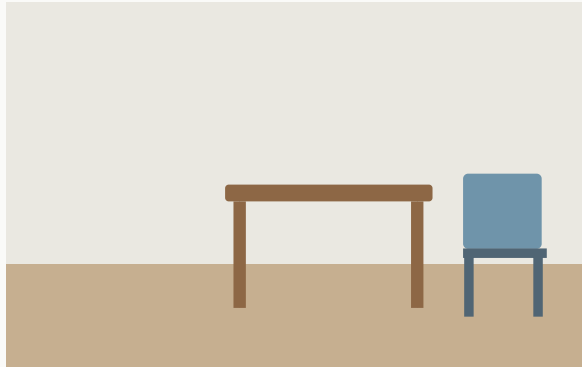
What changed from A to B?

Compare presence and position

A · BEFORE



B · AFTER



REFERENCE ANSWER

What changed from A to B?

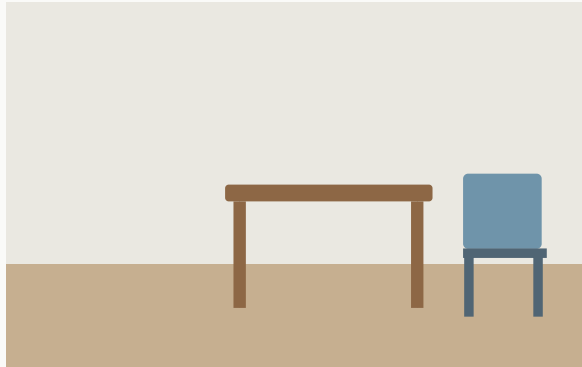
The chair moved to the right; the cup disappeared.

Compare presence and position

A · BEFORE



B · AFTER



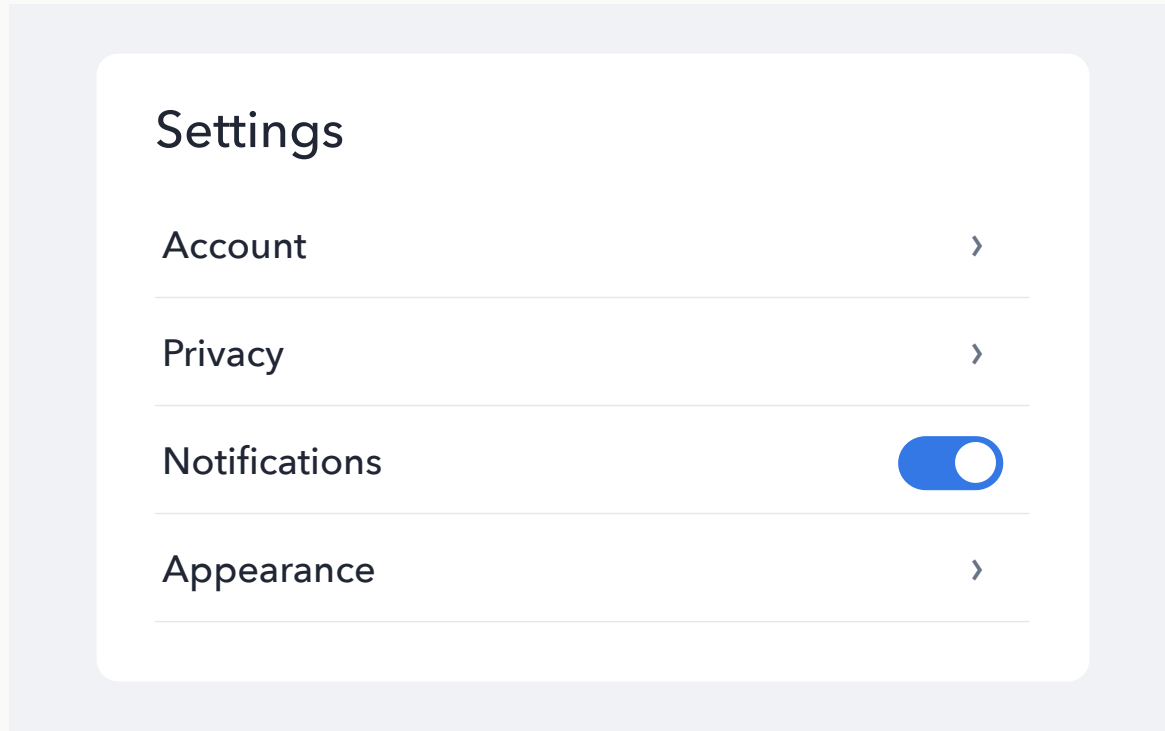
ARCHITECTURAL REQUIREMENT

What changed from A to B?

The chair moved to the right; the cup disappeared.

| Compare evidence across images.

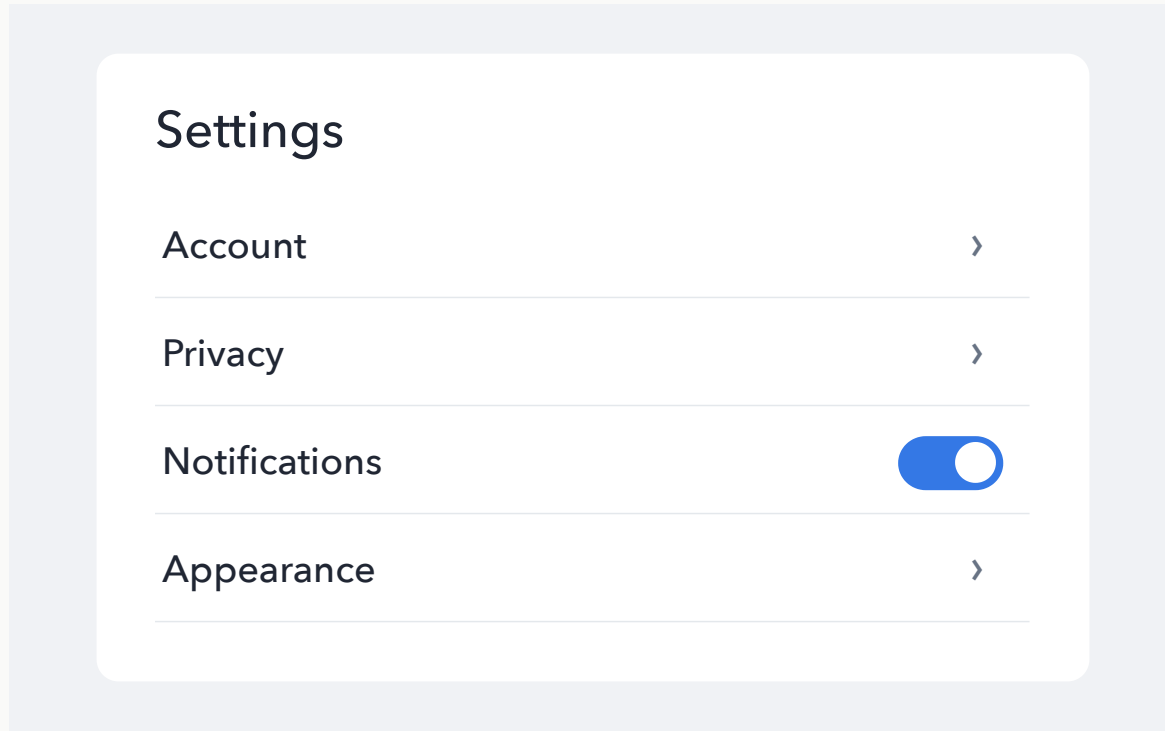
A screenshot is a spatial interface



QUESTION

Where should I click to disable notifications?

A screenshot is a spatial interface

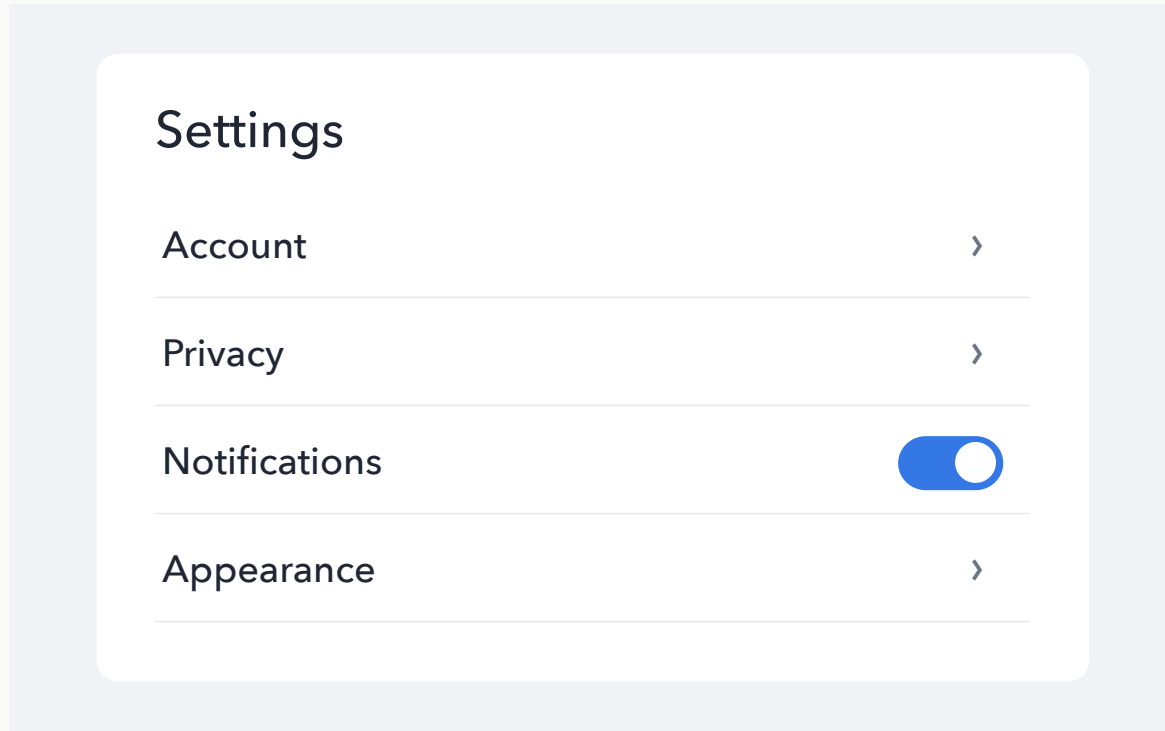


REFERENCE ANSWER

Where should I click to disable notifications?

The blue toggle on the Notifications row.

A screenshot is a spatial interface



ARCHITECTURAL REQUIREMENT

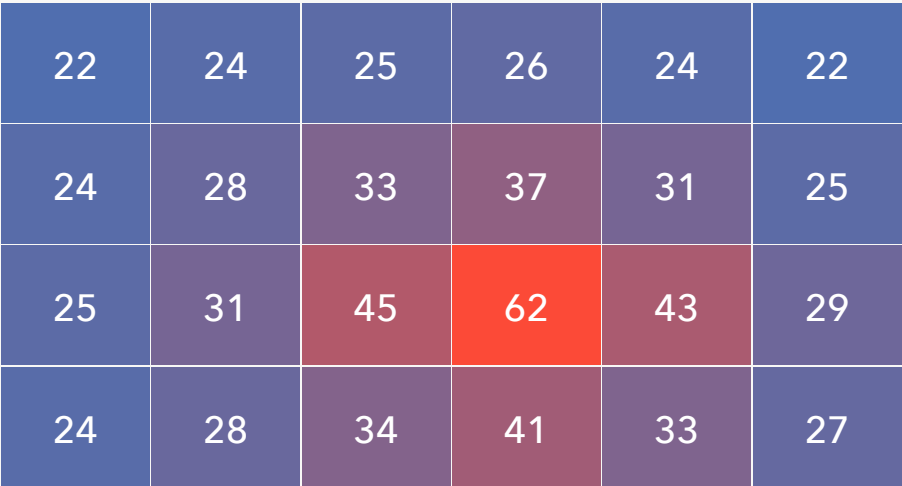
Where should I click to disable notifications?

The blue toggle on the Notifications row.

Locate a control relative to its label.

The image can be a scientific figure

Thermal map of a test plate (°C)



Synthetic measurements; no physical experiment claimed.

QUESTION

What is the hottest measured region?

The image can be a scientific figure

Thermal map of a test plate (°C)

22	24	25	26	24	22
24	28	33	37	31	25
25	31	45	62	43	29
24	28	34	41	33	27

Synthetic measurements; no physical experiment claimed.

REFERENCE ANSWER

What is the hottest measured region?

Row 3, column 4: 62 °C.

The image can be a scientific figure

Thermal map of a test plate (°C)

22	24	25	26	24	22
24	28	33	37	31	25
25	31	45	62	43	29
24	28	34	41	33	27

Synthetic measurements; no physical experiment claimed.

ARCHITECTURAL REQUIREMENT

What is the hottest measured region?

Row 3, column 4: 62 °C.

| Read a scale and local measurements.

The question can be wrong



QUESTION

What color is the bicycle?

The question can be wrong



REFERENCE ANSWER

What color is the bicycle?

No bicycle is visible.

The question can be wrong



ARCHITECTURAL REQUIREMENT

What color is the bicycle?

No bicycle is visible.

Let visual evidence override the wording of the question.

What must survive the visual representation?

QUESTION FAMILY	INFORMATION NEEDED
Counting / spatial	Multiple objects and their arrangement
OCR / charts	Fine detail, labels, values and units
UI / multiple images	Locations and correspondence across views
False premise	Evidence that can contradict the prompt

| Would one global CLIP vector always be enough?

Compare two small vision-language models



| A reference answer is not a model prediction.

Predict → compare → inspect both answers.

[Open the VLM lab ↗](#)

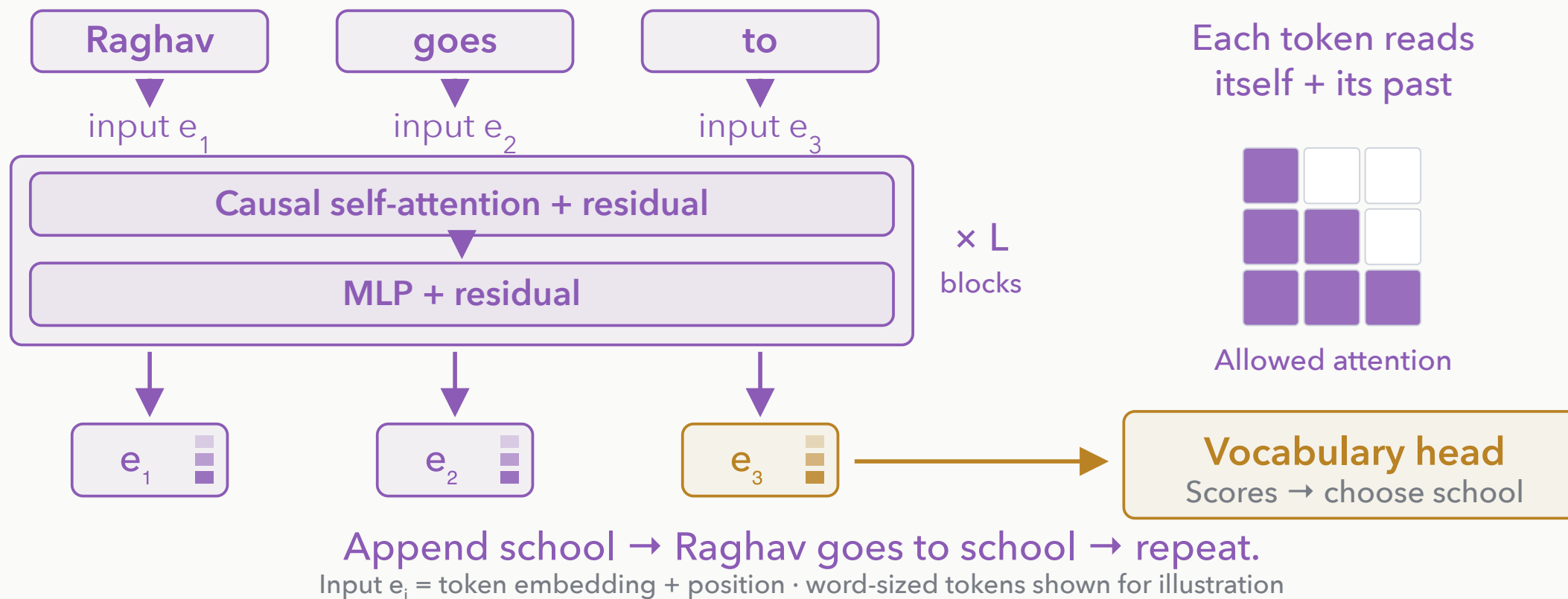
256M and 500M · same image and question.
Live or recorded answers; image-swap experiment.

SECTION 02

From matching to generation

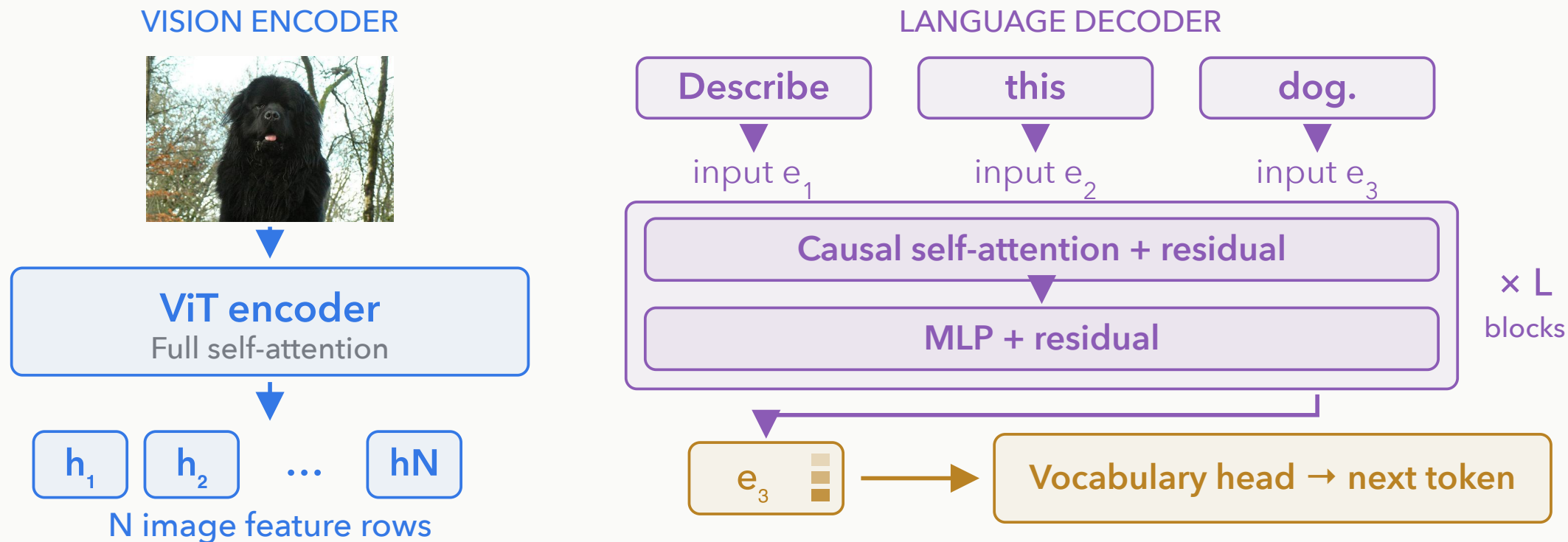
We have an image encoder. We also know a language decoder.

We already know this decoder from Beyond Attention



Read the last updated state $\rightarrow$ score the vocabulary $\rightarrow$ append a token $\rightarrow$ repeat.

Keep this decoder; give it access to the image



How can these image rows become context for this decoder?

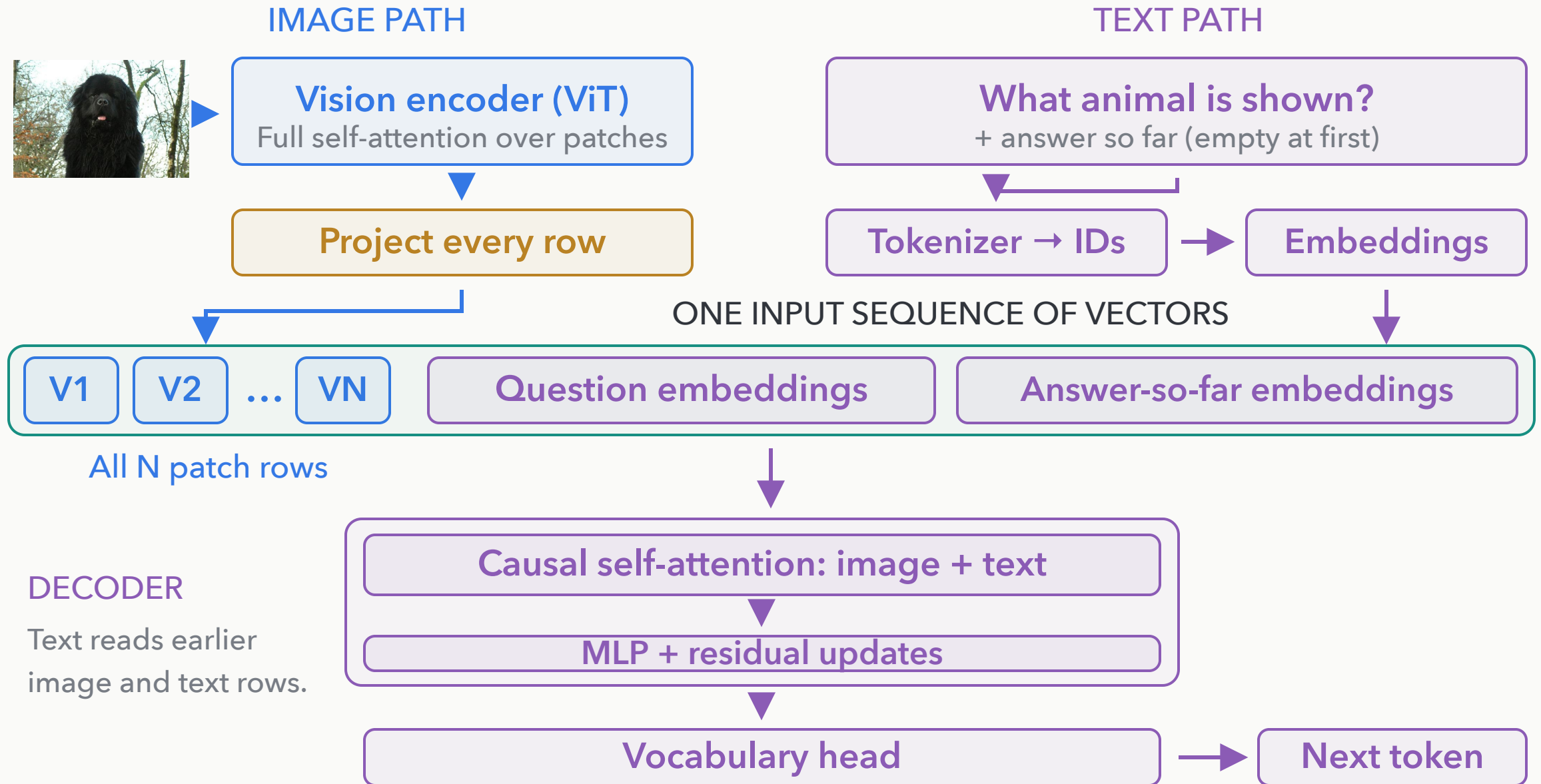
| We have both components. Next, choose how the decoder accesses the visual rows.

SECTION 03

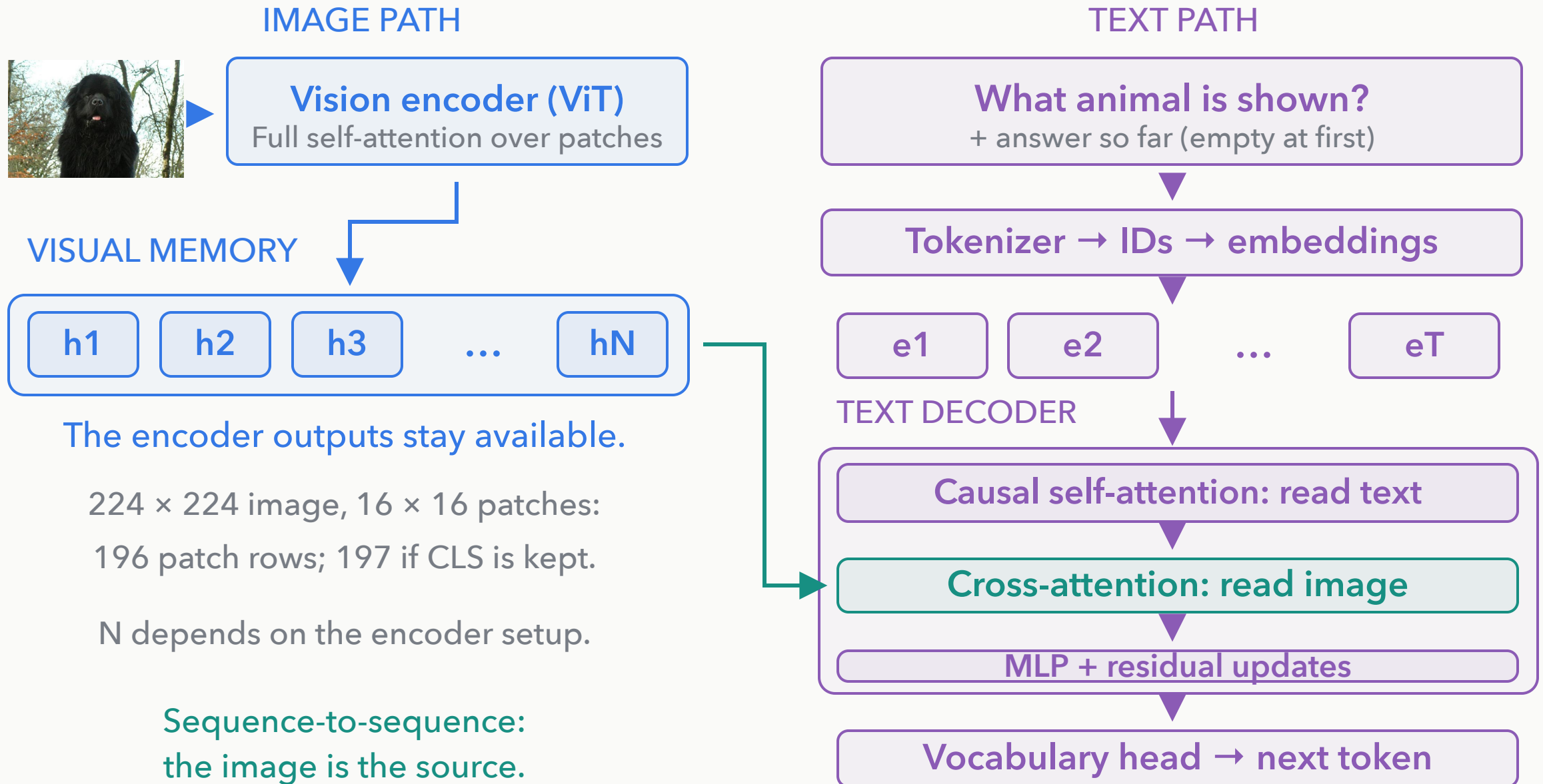
How can vision enter a language model?

First, see the three ideas. Then build each connection.

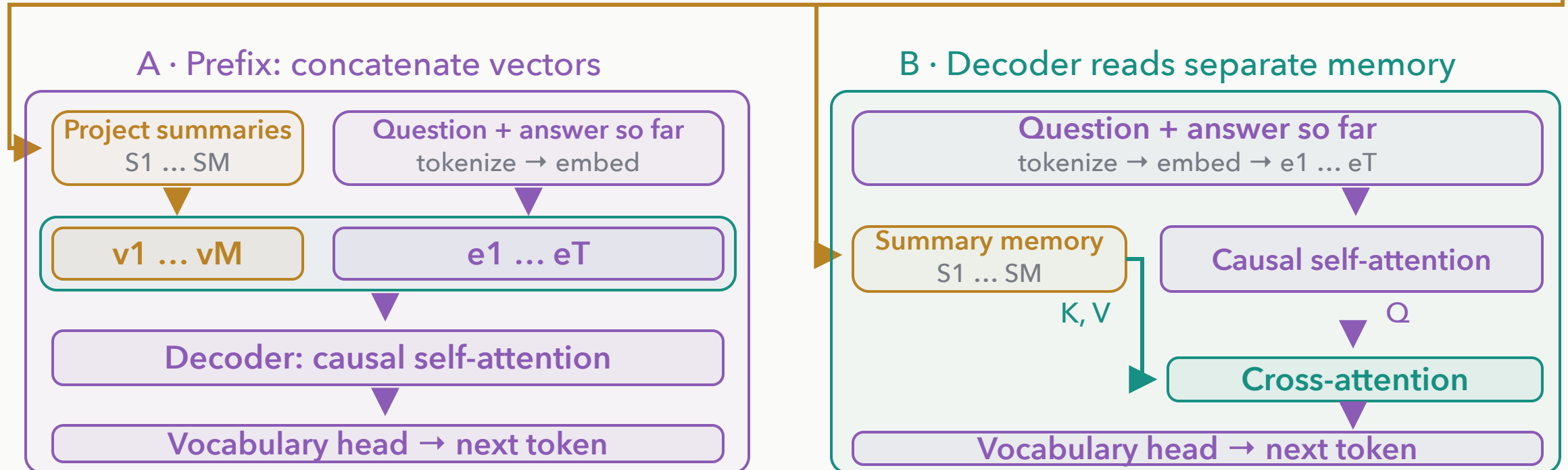
Idea 1 • prepend the image as visual tokens



Idea 2 • let the decoder look at a visual memory



Idea 3 · summarize the image, then use either connection

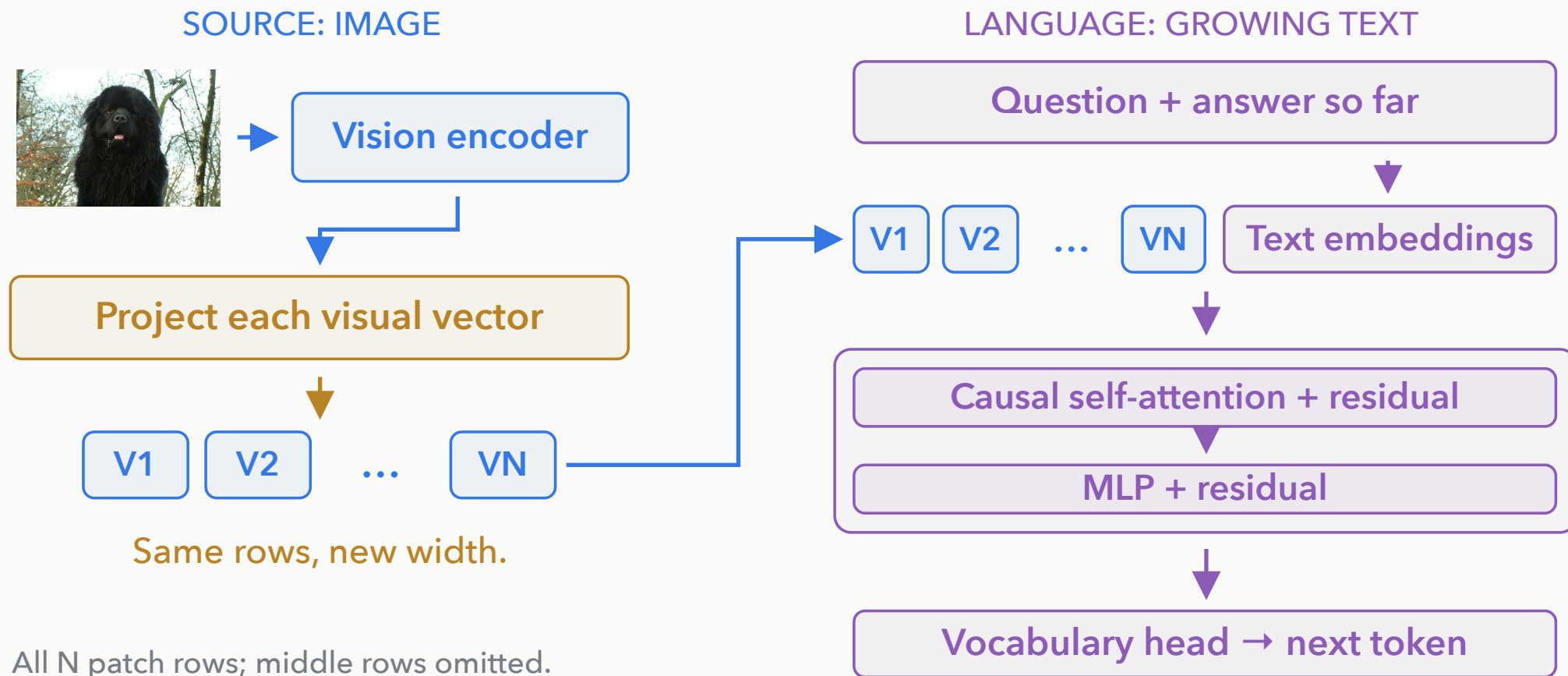


The map: two access routes, one compression choice

IDEA	HIGH-LEVEL DECISION	THEN WE WILL EXPLAIN...
1 · Visual prefix	Put visual vectors before text.	How to make them fit the decoder input.
2 · Separate memory	Let language read the image.	How a text state retrieves visual information.
3 · Shorter summary	Use fewer visual vectors.	How learned readers make the summaries.

| Ideas 1 and 2 choose the access route. Idea 3 can be used with either.

Idea 1 in detail · build the visual prefix



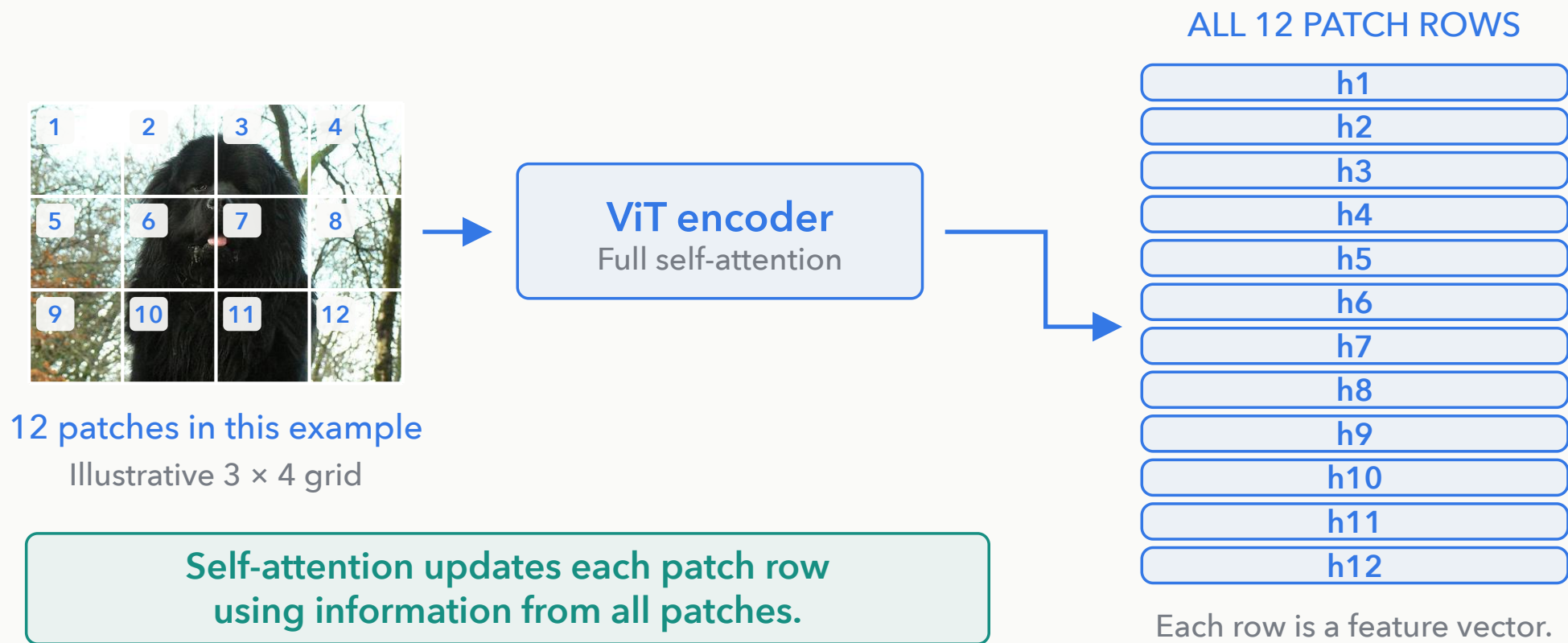
One sequence; the same causal self-attention can use both kinds of context.

What must we build to make this work?

JOB	WHY IT IS NEEDED
1. Keep image feature vectors	The decoder needs access to visual evidence.
2. Match the decoder input width	All input rows must have the same number of coordinates.
3. Place them before the question	Causal attention can read earlier context.

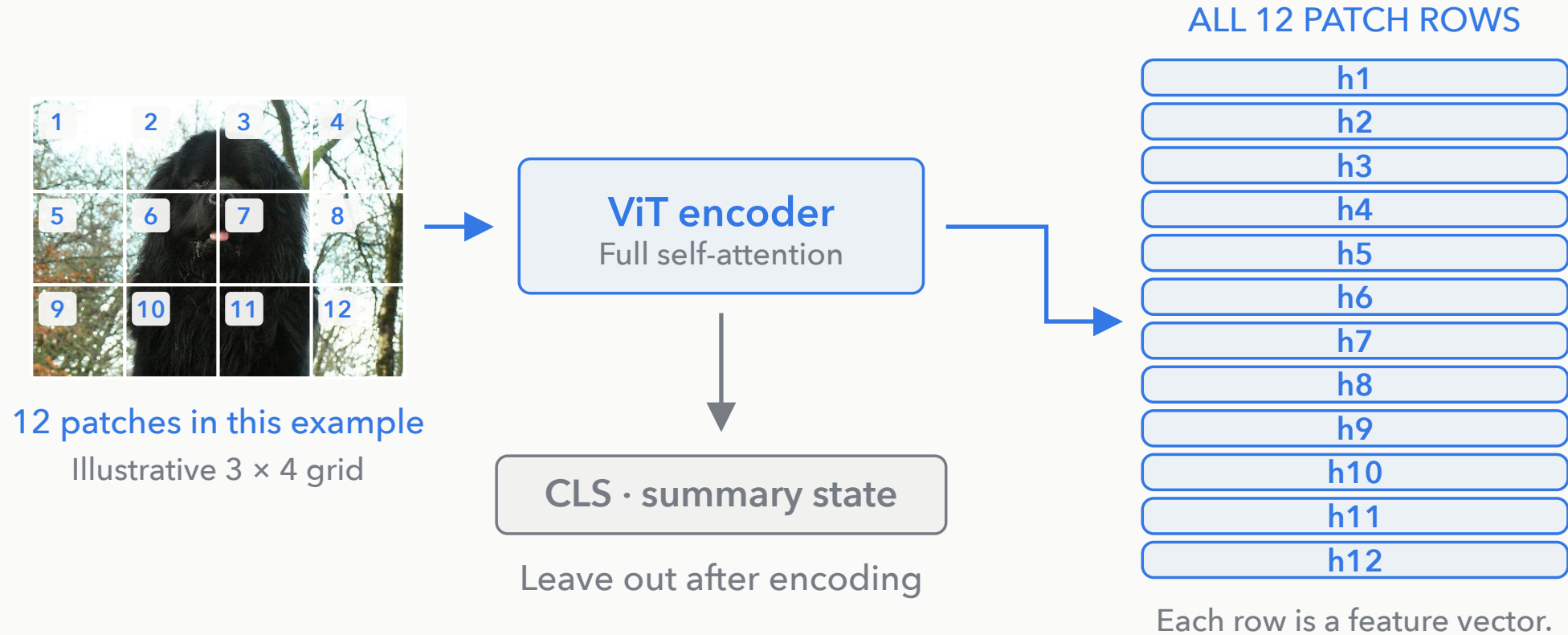
| We already have the decoder. Now build its visual input.

Recall ViT: every patch already has an output row



| Patch 1 has row h1, patch 2 has row h2, and so on through patch 12.

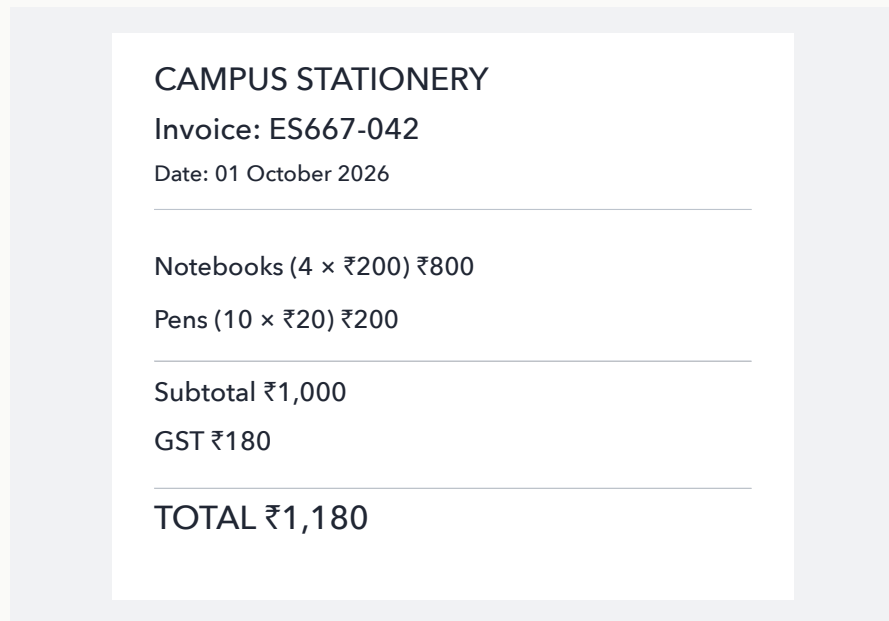
ViT outputs a CLS state and one row per patch



Forward all 12 patch rows to the projector; leave out the CLS output.

Pass the patch rows onward; leave out CLS

The question needs image details



Invoice ID? GST percentage?

SELECT FROM THE ViT OUTPUTS

CLS output

Leave out after encoding

12 patch rows

h1
h2
h3
h4
h5
h6
h7
h8
h9
h10
h11
h12

KEEP EVERY PATCH ROW

Project
each row

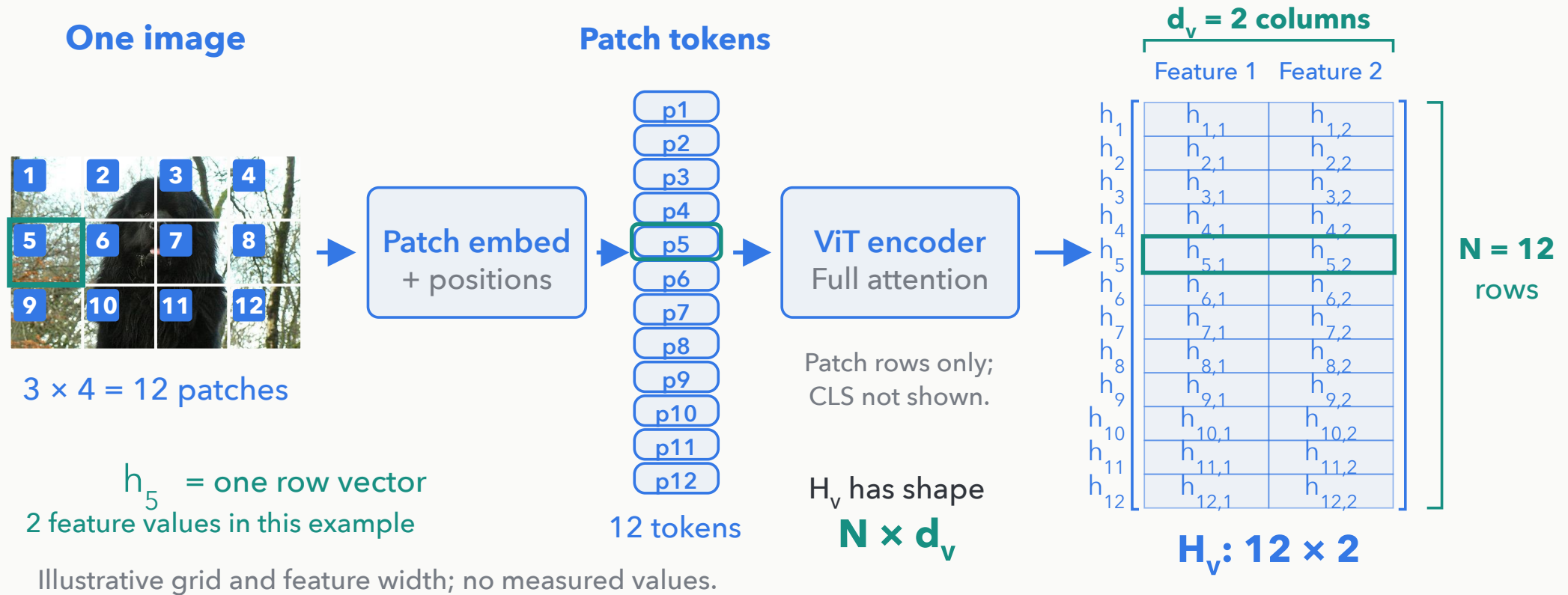
Visual prefix
12 vectors

Combine with text embeddings
for the language decoder.

Illustration: 12 patch rows; CLS left out.

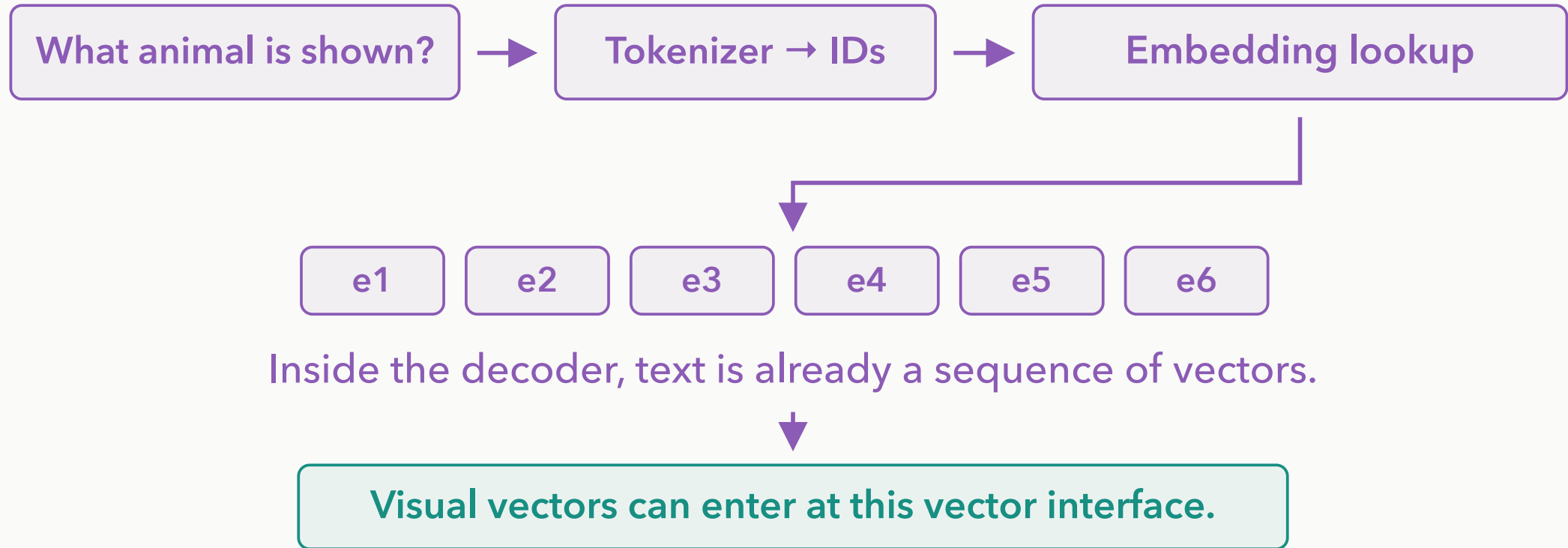
This design sends all N patch rows to the projector and omits the CLS output.

A visual state is one row of the feature matrix



One patch position $\rightarrow$ one output row. Each row has d_v feature values.

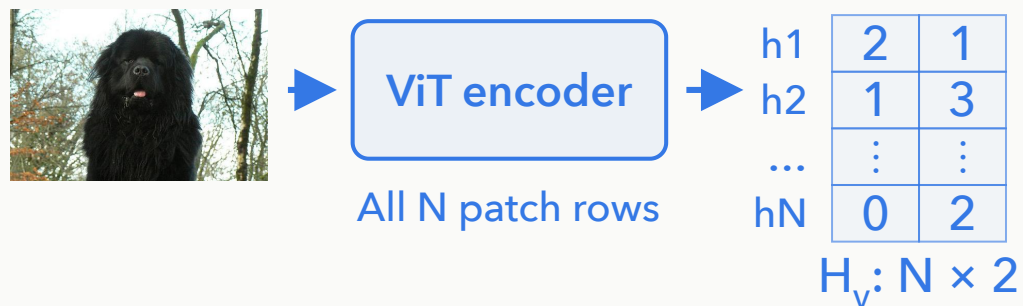
Recall: the decoder already consumes vectors



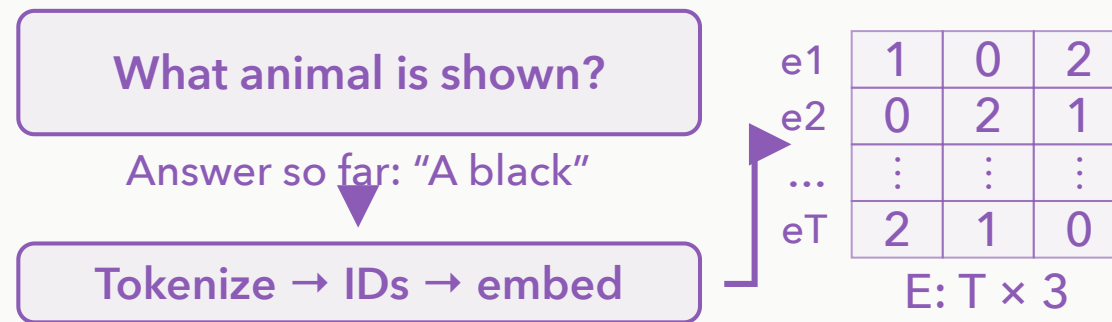
| Text IDs become embeddings before reaching the Transformer blocks.

The image rows have 2 coordinates; text rows have 3

IMAGE → FEATURE VECTORS



TEXT → EMBEDDING VECTORS



Toy widths and values; not measured from this image or text. Middle rows omitted.

Can these rows enter the same decoder?

h_1

2	1
---	---

≠ width

e_1

1	0	2
---	---	---

2 coordinates per image row

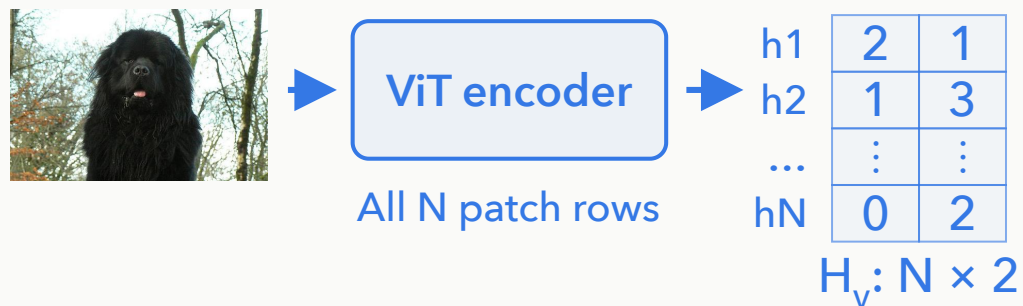
3 coordinates per text row

We need to turn each image row into a 3-coordinate vector.

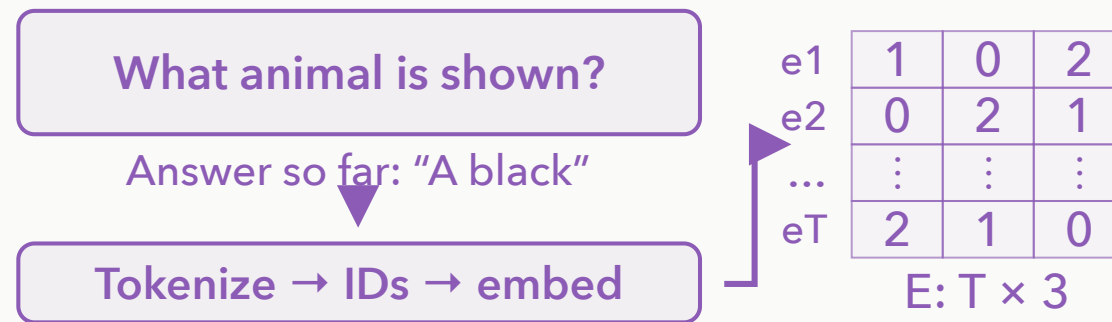
To form one input sequence, every vector must have the decoder's input width.

Add a projector to the image path

IMAGE → FEATURE VECTORS

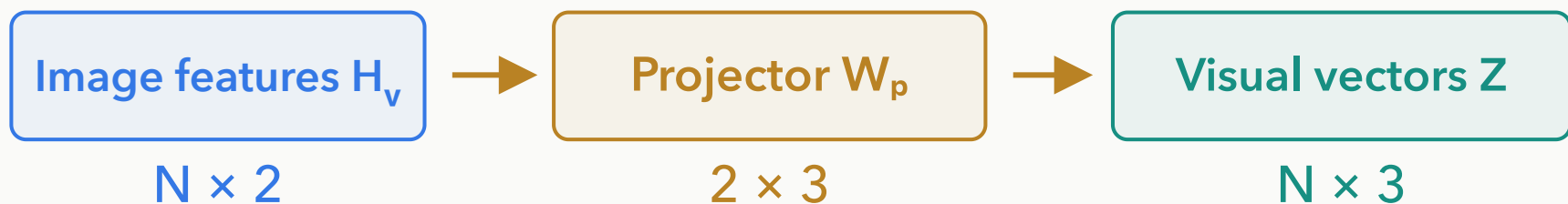


TEXT → EMBEDDING VECTORS



Toy widths and values; not measured from this image or text. Middle rows omitted.

Insert a shared projector on the image path

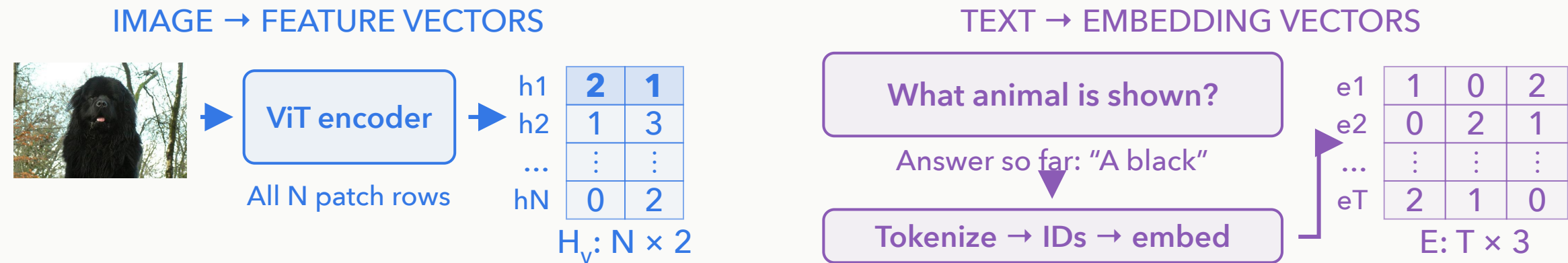


Text stays E
 $T \times 3$

$Z = H_V W_p$ · Keep N rows. Change 2 coordinates to 3.

| The projector changes $N \times 2$ image features into $N \times 3$ visual vectors.

Zoom into one row from the image



Toy widths and values; not measured from this image or text. Middle rows omitted.

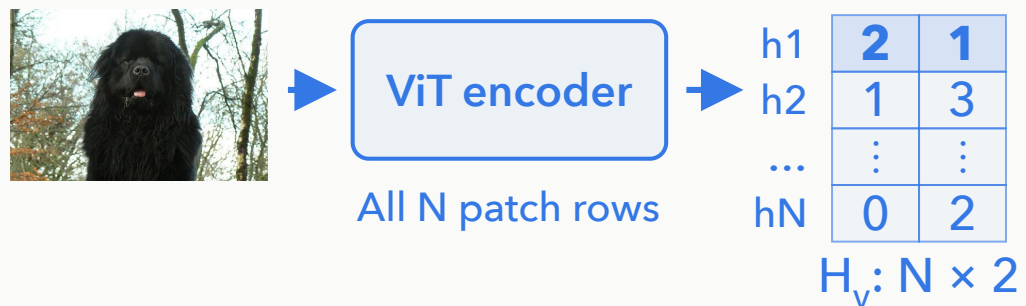


h1 is one of the image rows above. Its two coordinates are 2 and 1.

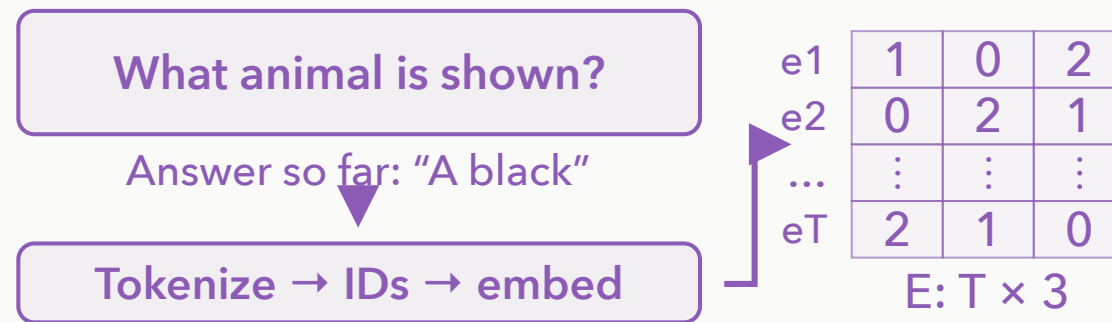
Take one 2-coordinate row from the image feature sequence.

Use a 2×3 matrix to change the width

IMAGE → FEATURE VECTORS



TEXT → EMBEDDING VECTORS



Toy widths and values; not measured from this image or text. Middle rows omitted.

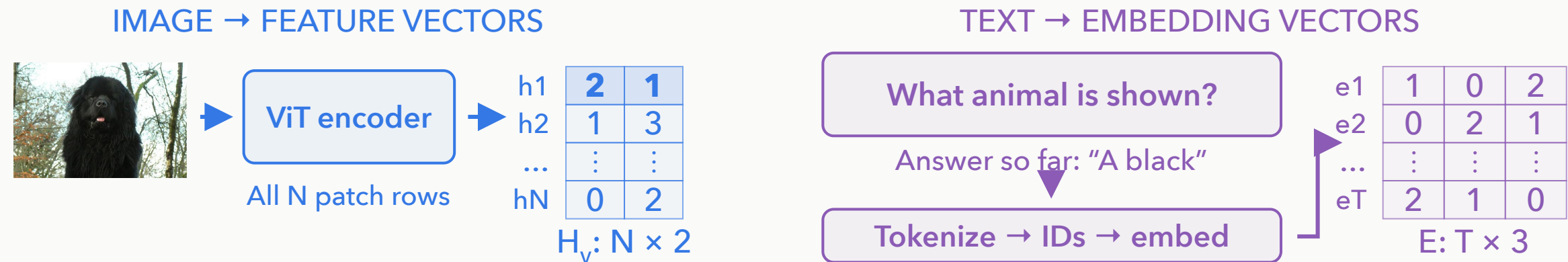
Zoom into h1

$$\begin{array}{c}
 \begin{array}{|c|c|} \hline 2 & 1 \\ \hline \end{array} \\
 1 \times 2
 \end{array}
 \times
 \begin{array}{c}
 W_p \\
 \begin{array}{|c|c|c|} \hline 1 & 0 & 1 \\ \hline 0 & 1 & 1 \\ \hline \end{array} \\
 2 \times 3
 \end{array}
 =
 \begin{array}{c}
 z1 \\
 \begin{array}{|c|c|c|} \hline ? & ? & ? \\ \hline \end{array} \\
 1 \times 3: \text{ same width as } e1
 \end{array}$$

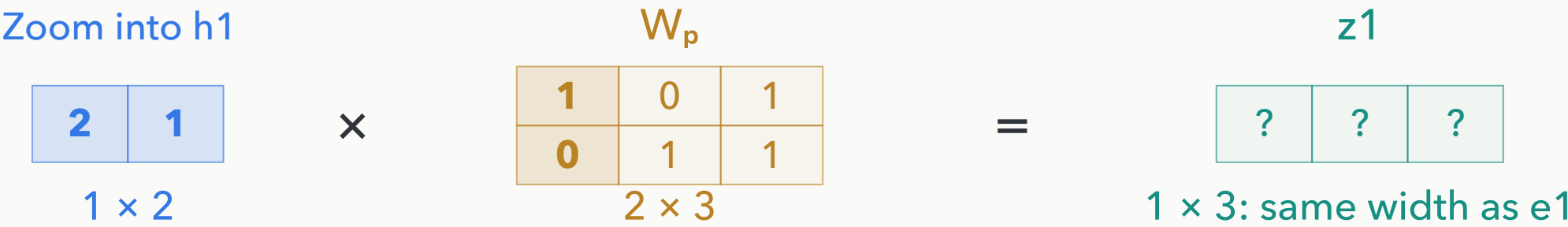
Each of the three columns of W_p will make one output coordinate.

Each column makes one of the 3 decoder-input coordinates.

The first column makes the first coordinate



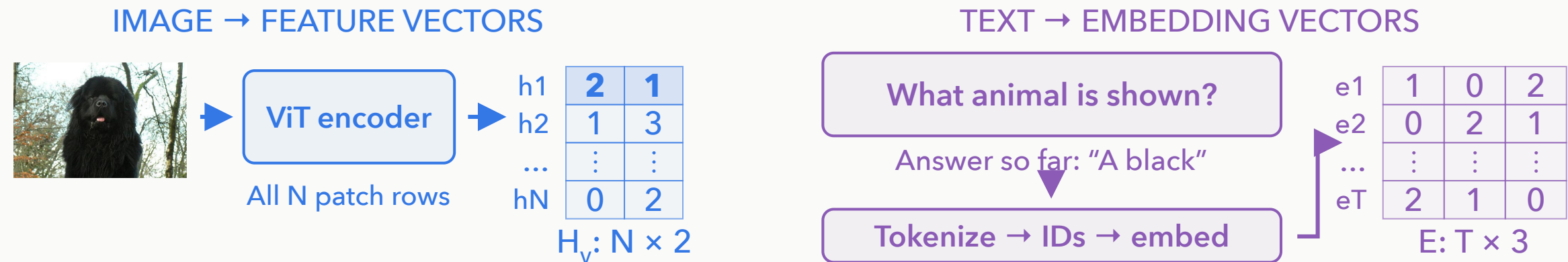
Toy widths and values; not measured from this image or text. Middle rows omitted.



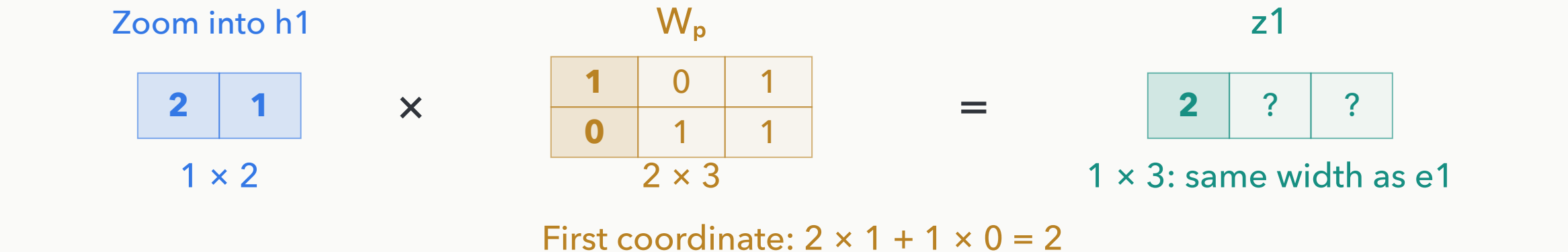
First column: how much of 2, and how much of 1, should we use?

| Which input coordinates does this column combine?

The first output coordinate is 2

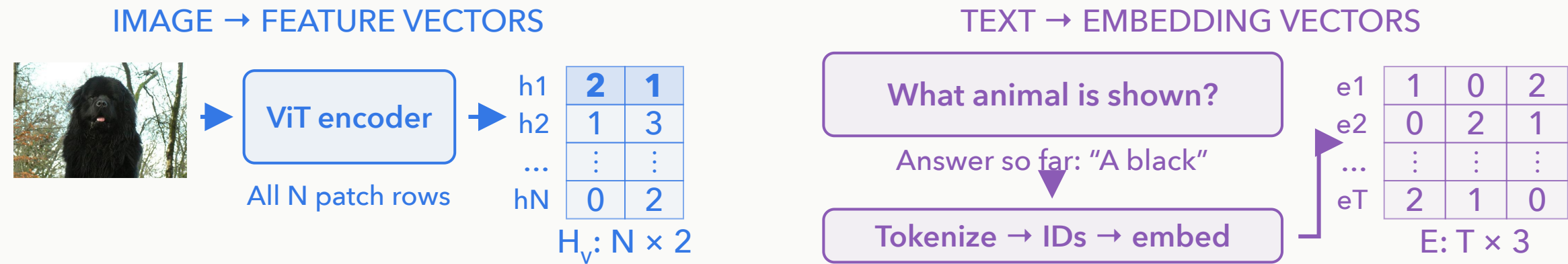


Toy widths and values; not measured from this image or text. Middle rows omitted.

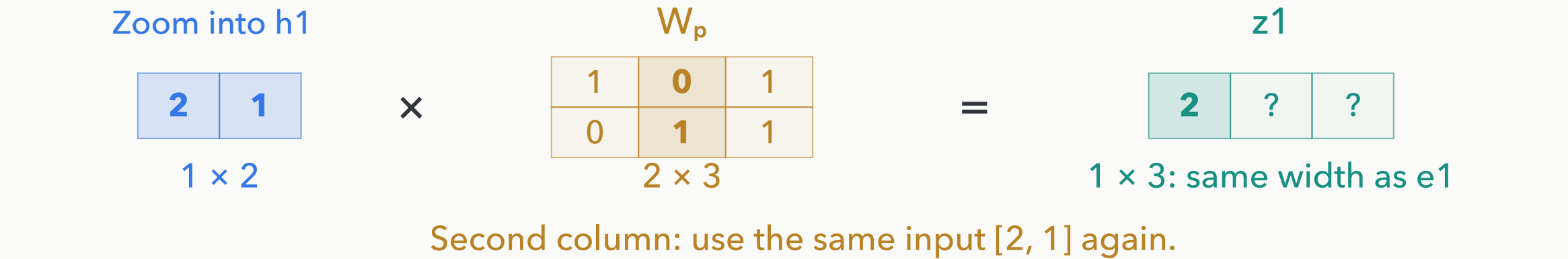


The first output coordinate copies the first input in this toy map.

The second column reuses the same image row



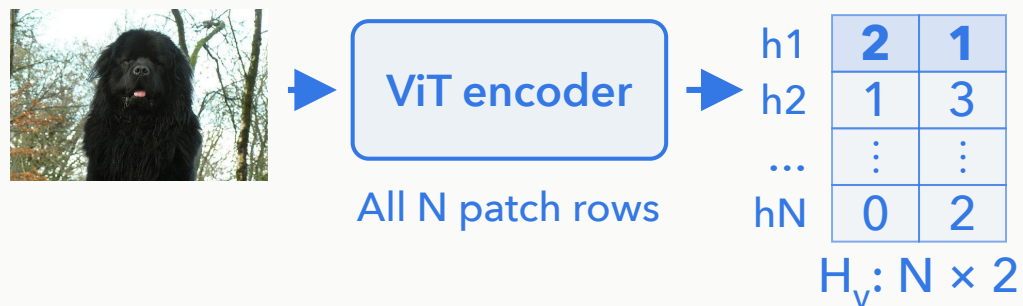
Toy widths and values; not measured from this image or text. Middle rows omitted.



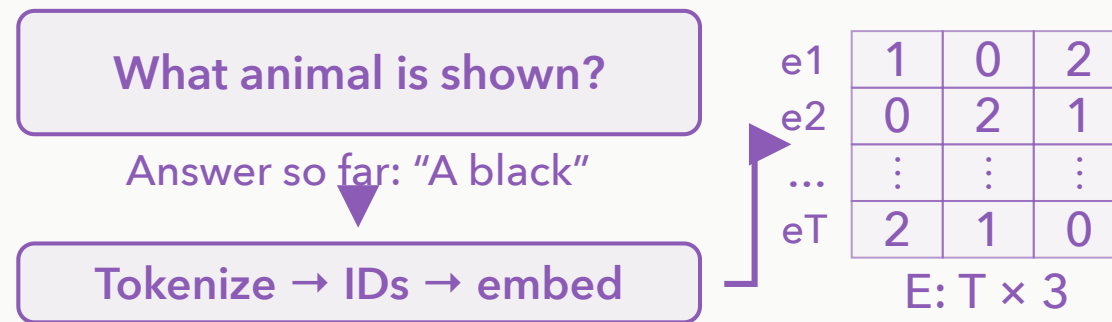
Now use the same input row with the second column.

The second output coordinate is 1

IMAGE → FEATURE VECTORS



TEXT → EMBEDDING VECTORS



Toy widths and values; not measured from this image or text. Middle rows omitted.

Zoom into h_1

$$\begin{array}{|c|c|} \hline 2 & 1 \\ \hline \end{array} \quad 1 \times 2 \quad \times \quad \begin{array}{|c|c|c|} \hline 1 & 0 & 1 \\ \hline 0 & 1 & 1 \\ \hline \end{array} \quad 2 \times 3 \quad = \quad \begin{array}{|c|c|c|} \hline 2 & 1 & ? \\ \hline \end{array} \quad 1 \times 3: \text{ same width as } e_1$$

W_p

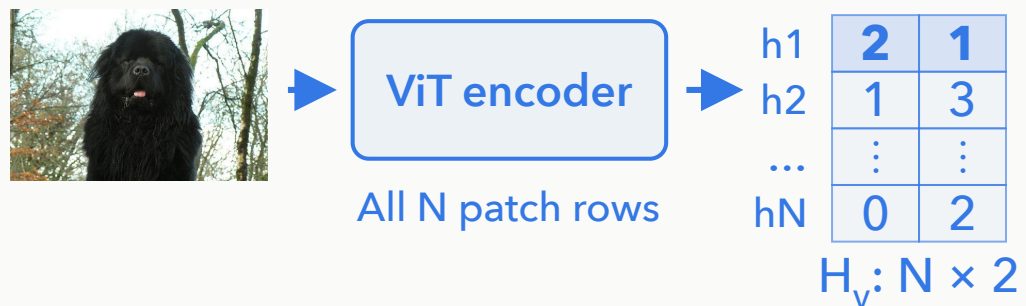
z_1

$$\text{Second coordinate: } 2 \times 0 + 1 \times 1 = 1$$

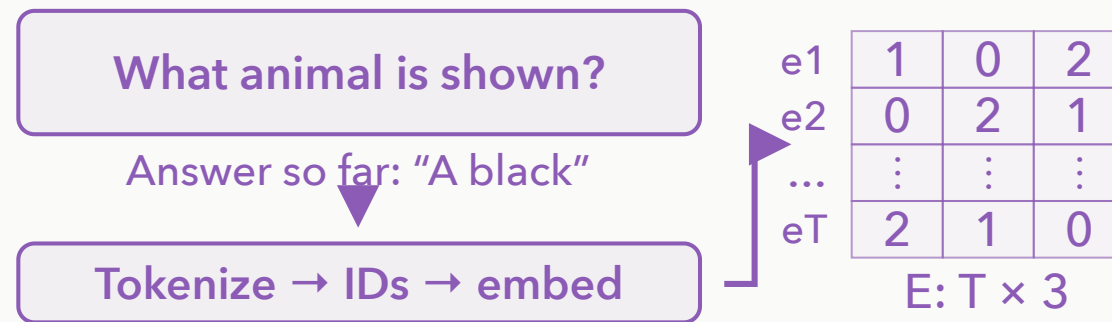
The second output coordinate copies the second input in this toy map.

The third column combines both input coordinates

IMAGE → FEATURE VECTORS



TEXT → EMBEDDING VECTORS



Toy widths and values; not measured from this image or text. Middle rows omitted.

Zoom into h_1

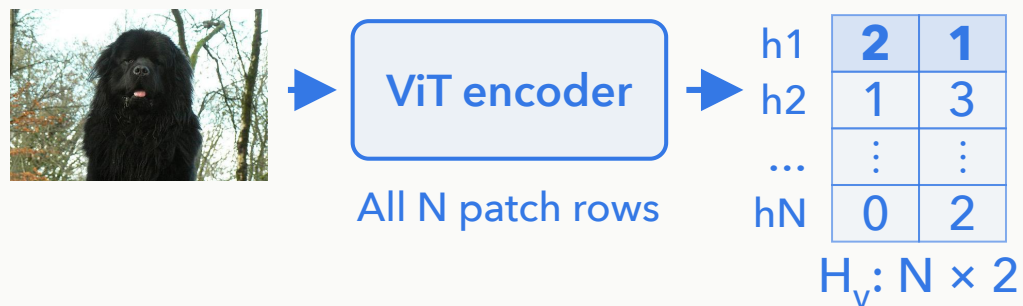
$$\begin{array}{c} \begin{array}{|c|c|} \hline 2 & 1 \\ \hline \end{array} \\ 1 \times 2 \end{array} \times \begin{array}{c} W_p \\ \begin{array}{|c|c|c|} \hline 1 & 0 & 1 \\ \hline 0 & 1 & 1 \\ \hline \end{array} \\ 2 \times 3 \end{array} = \begin{array}{c} z_1 \\ \begin{array}{|c|c|c|} \hline 2 & 1 & 3 \\ \hline \end{array} \\ 1 \times 3: \text{ same width as } e_1 \end{array}$$

$$\text{Third coordinate: } 2 \times 1 + 1 \times 1 = 3$$

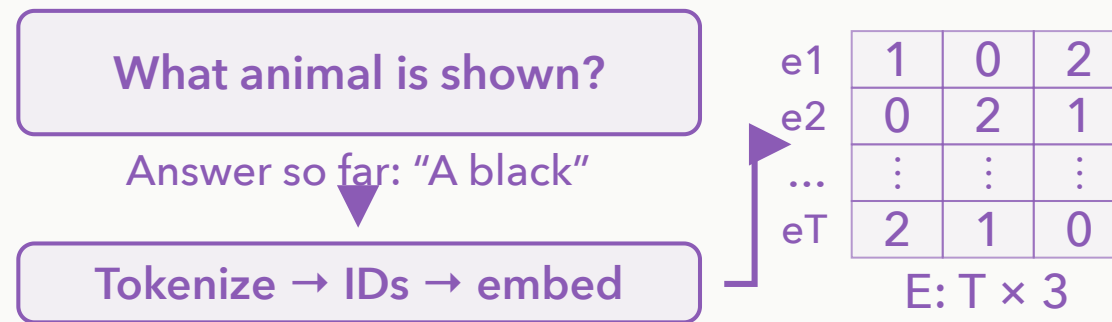
The third output coordinate adds the two inputs in this toy map.

Now this image vector has the required width

IMAGE → FEATURE VECTORS



TEXT → EMBEDDING VECTORS



Toy widths and values; not measured from this image or text. Middle rows omitted.

Zoom into h1

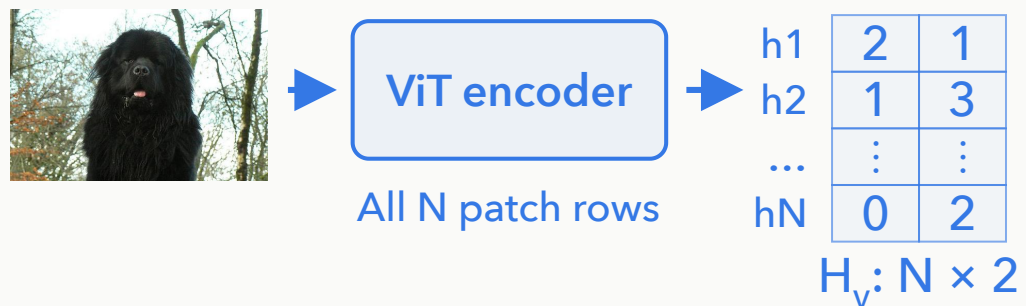
$$\begin{array}{c}
 \begin{array}{|c|c|} \hline 2 & 1 \\ \hline \end{array} \\
 1 \times 2
 \end{array}
 \times
 \begin{array}{c}
 W_p \\
 \begin{array}{|c|c|c|} \hline 1 & 0 & 1 \\ \hline 0 & 1 & 1 \\ \hline \end{array} \\
 2 \times 3
 \end{array}
 =
 \begin{array}{c}
 z1 \\
 \begin{array}{|c|c|c|} \hline 2 & 1 & 3 \\ \hline \end{array} \\
 1 \times 3: \text{same width as } e1
 \end{array}$$

The image row [2, 1] has become [2, 1, 3]. The text rows stay as they are.

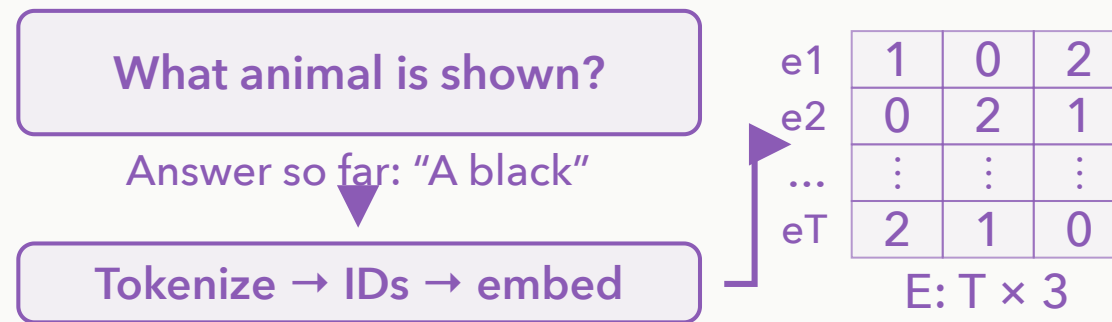
| The row [2, 1, 3] fits beside the 3-coordinate text embeddings.

Use the same matrix for all N image rows

IMAGE → FEATURE VECTORS



TEXT → EMBEDDING VECTORS



Toy widths and values; not measured from this image or text. Middle rows omitted.

Every image row

h_1	2	1
h_2	1	3
...	$\vdots$	$\vdots$
h_N	0	2

$N \times 2$

Same W_p

1	0	1
0	1	1

2×3

=

All N projected rows

z_1	2	1	3
z_2	1	3	4
...	$\vdots$	$\vdots$	$\vdots$
z_N	0	2	2

$N \times 3$

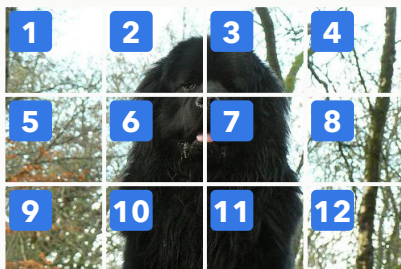
Width now matches E

N is unchanged.

N rows stay N rows; every projected row now has 3 coordinates.

Check: did the projector create more visual tokens?

Same image grid



$N = 3 \times 4 = 12$ patch rows

Image features H_v

12×2

From our toy ViT

Example row h_1

2	1
---	---

$d_v = 2$ features per row

Illustrative width, as before

From the text embeddings

Example row e_1

1	0	2
---	---	---

$d_l = 3$ features per row

Illustrative width, as before

Shared projector W_p

2×3

Projected features Z

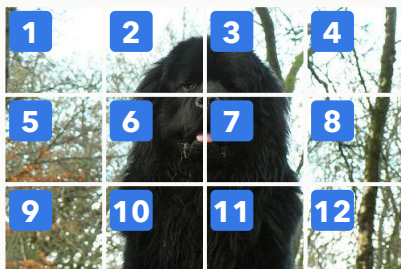
$? \times 3$

How many rows will Z have?

Predict the output shape before advancing.

Still 12 rows; each row now has 3 features

Same image grid



$N = 3 \times 4 = 12$ patch rows

Image features H_v

12×2

From our toy ViT

Example row h_1

2	1
---	---

$d_v = 2$ features per row

Illustrative width, as before

From the text embeddings

Example row e_1

1	0	2
---	---	---

$d_l = 3$ features per row

Illustrative width, as before

Shared projector W_p

2×3

Projected features Z

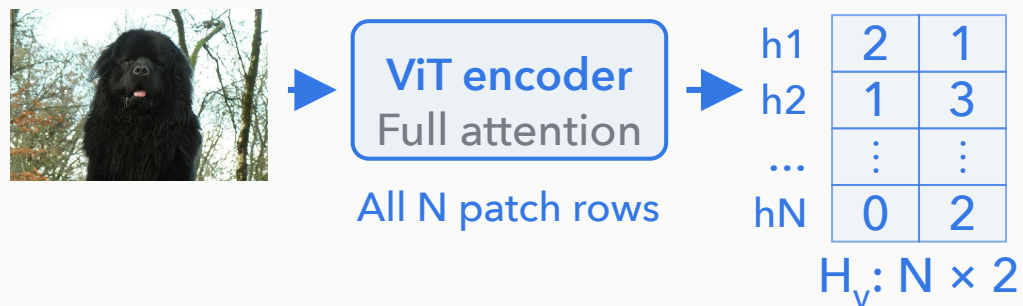
12×3

12 rows stay 12 rows. Each row changes from 2 features to 3.

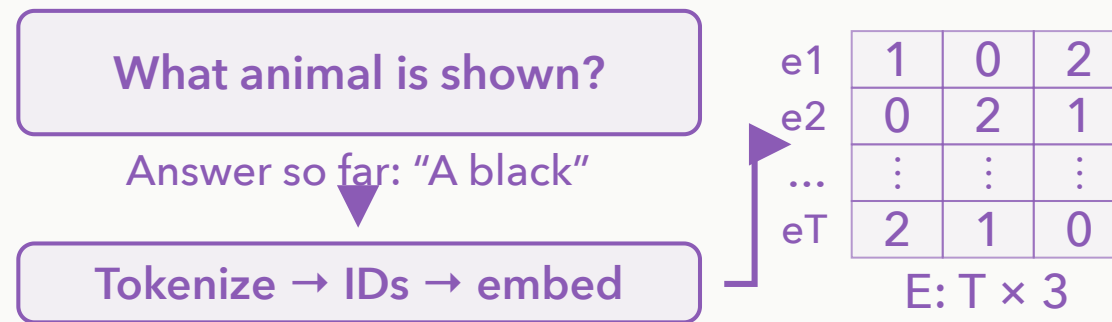
| The projector changes each row's features and keeps every patch position.

Now the two sequences fit one decoder input

IMAGE → FEATURE VECTORS



TEXT → EMBEDDING VECTORS



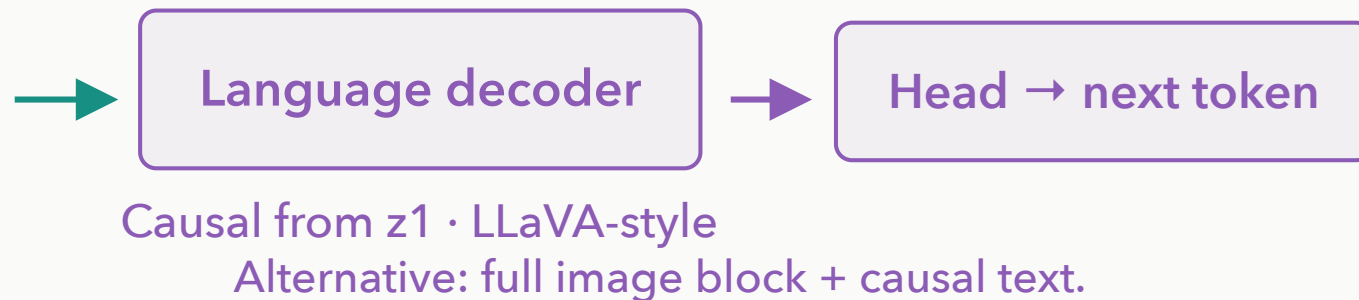
Toy widths and values; not measured from this image or text. Middle rows omitted.

Image → shared $W_p \rightarrow Z$
N projected rows

+ T text rows

$(N + T) \times 3$

z_1	2	1	3
...	⋮	⋮	⋮
z_N	0	2	2
e_1	1	0	2
...	⋮	⋮	⋮
e_T	2	1	0

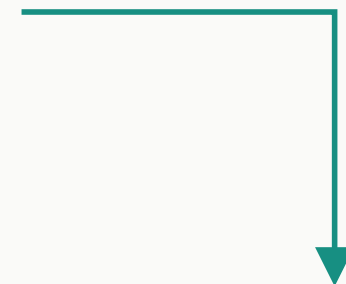
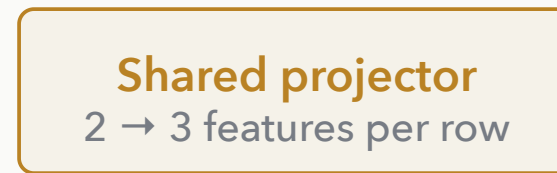
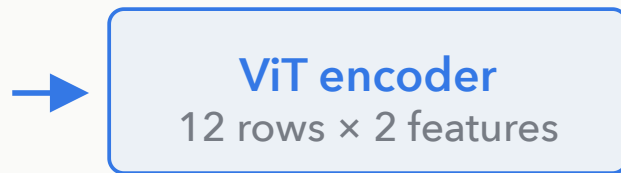


Our LLaVA-style example uses causal attention from the first image position.

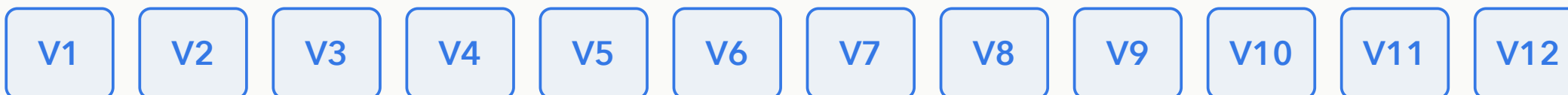
Start with all the projected image vectors



Same 12 patches



THE VISUAL PREFIX · ALL 12 ROWS

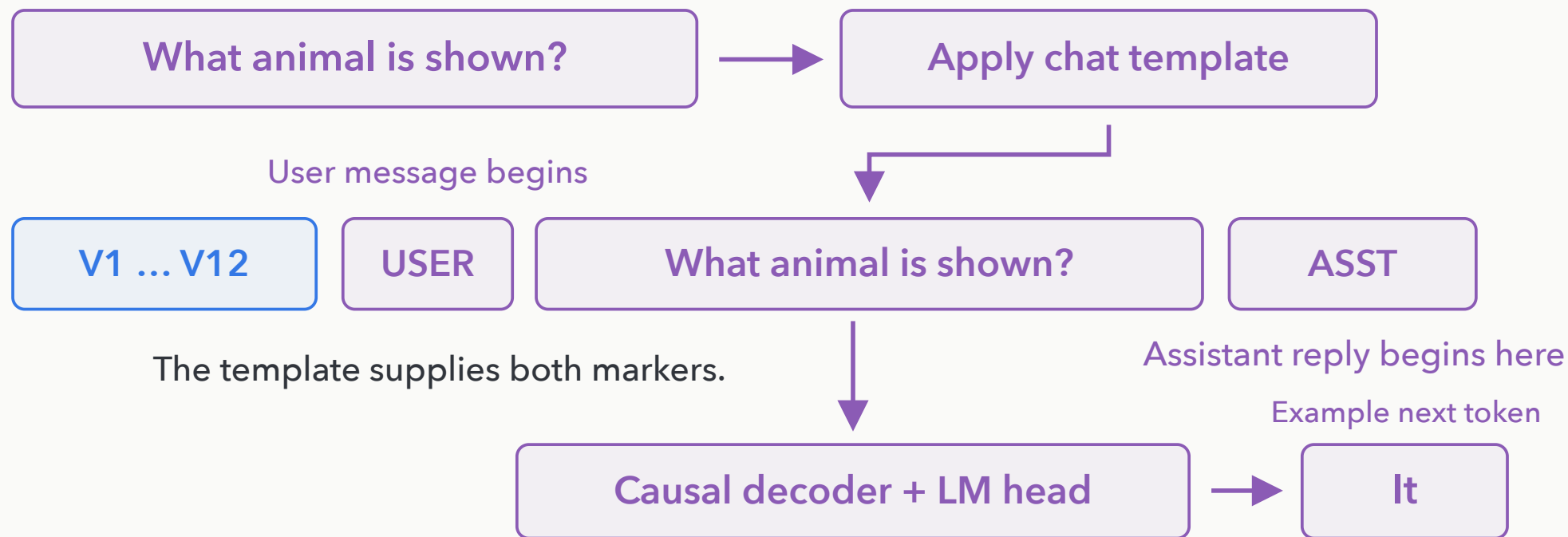


Each V_i is a vector with 3 features in our running example.

Next, put the formatted conversation after these vectors.

| Keep all 12 patch rows; each row has the decoder's input width.

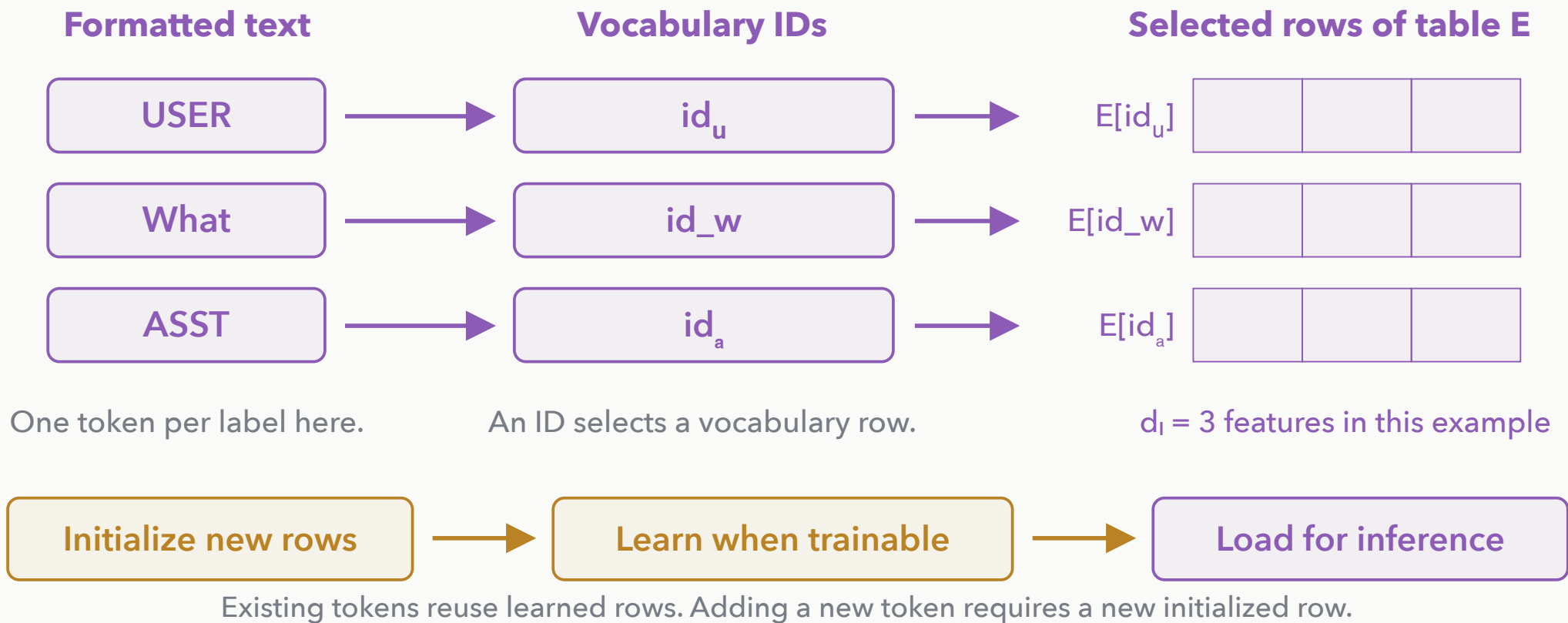
USER and ASST mark who is speaking



USER / ASST are teaching labels; a real role marker can span several tokens.

| The template supplies the role markers; generation continues after ASST.

Role tokens use the same embedding table



| A special role token is a vocabulary entry with its own learned embedding row.

Finish the question; start the assistant's answer

ONE SEQUENCE OF VECTORS

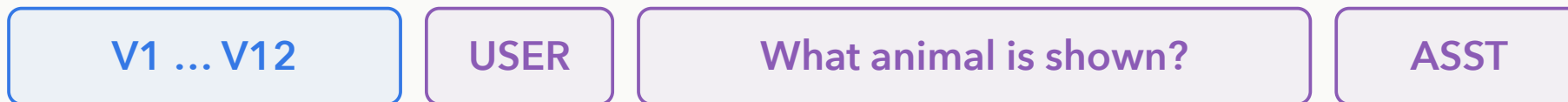


Illustration: 12 visual rows + 1 USER + 4 question tokens + 1 ASST.

0 ... 11

12

13 14 15 16

17

Position

USER is at 12; ASST is at 17. Positions do not restart at a role boundary.



Position-aware causal self-attention

LM head →

It

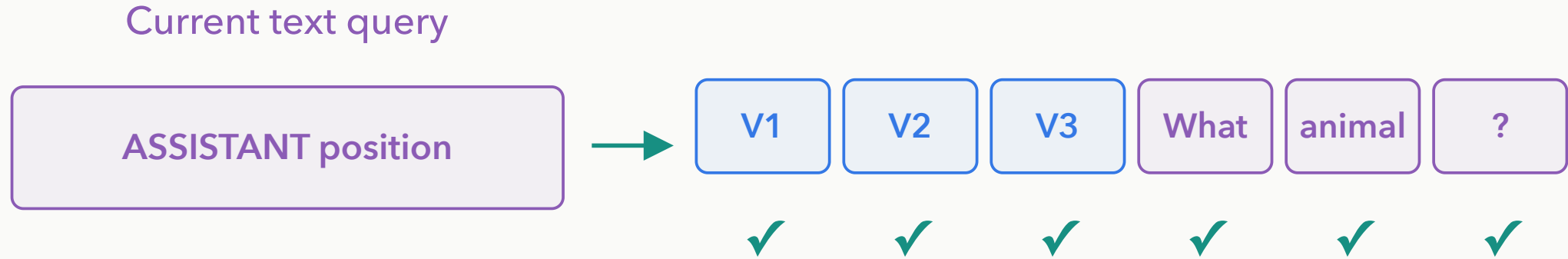
Additive scheme: $x_i = E[\text{token ID}_i] + P_i$, including role tokens.

LLaMA uses RoPE: position i rotates Q_i and K_i inside attention.

Next input
position: 18

Role tokens have sequence positions and participate in the same causal attention.

What may the current text position read?



All six are earlier context.

No future text position is available.

| The assistant's state can use both the picture and the question.

The decoder mask is a model choice

OUR DEFAULT · LLaVA-STYLE

Causal across image + text

Query ↓	Keys →						
	z1	z2	z3	T1	T2	T3	A
z1	Colored	Grey	Grey	Grey	Grey	Grey	Grey
z2	Colored	Colored	Grey	Grey	Grey	Grey	Grey
z3	Colored	Colored	Colored	Grey	Grey	Grey	Grey
T1	Grey	Grey	Grey	Colored	Grey	Grey	Grey
T2	Grey	Grey	Grey	Colored	Colored	Grey	Grey
T3	Grey	Grey	Grey	Colored	Colored	Colored	Grey
A	Grey	Grey	Grey	Colored	Colored	Colored	Colored

z1 reads itself; z2 reads z1 + z2.

Colored:
allowed

Grey:
blocked

Amber:
extra links

ALTERNATIVE · e.g. GEMMA 3

Full image block; causal text

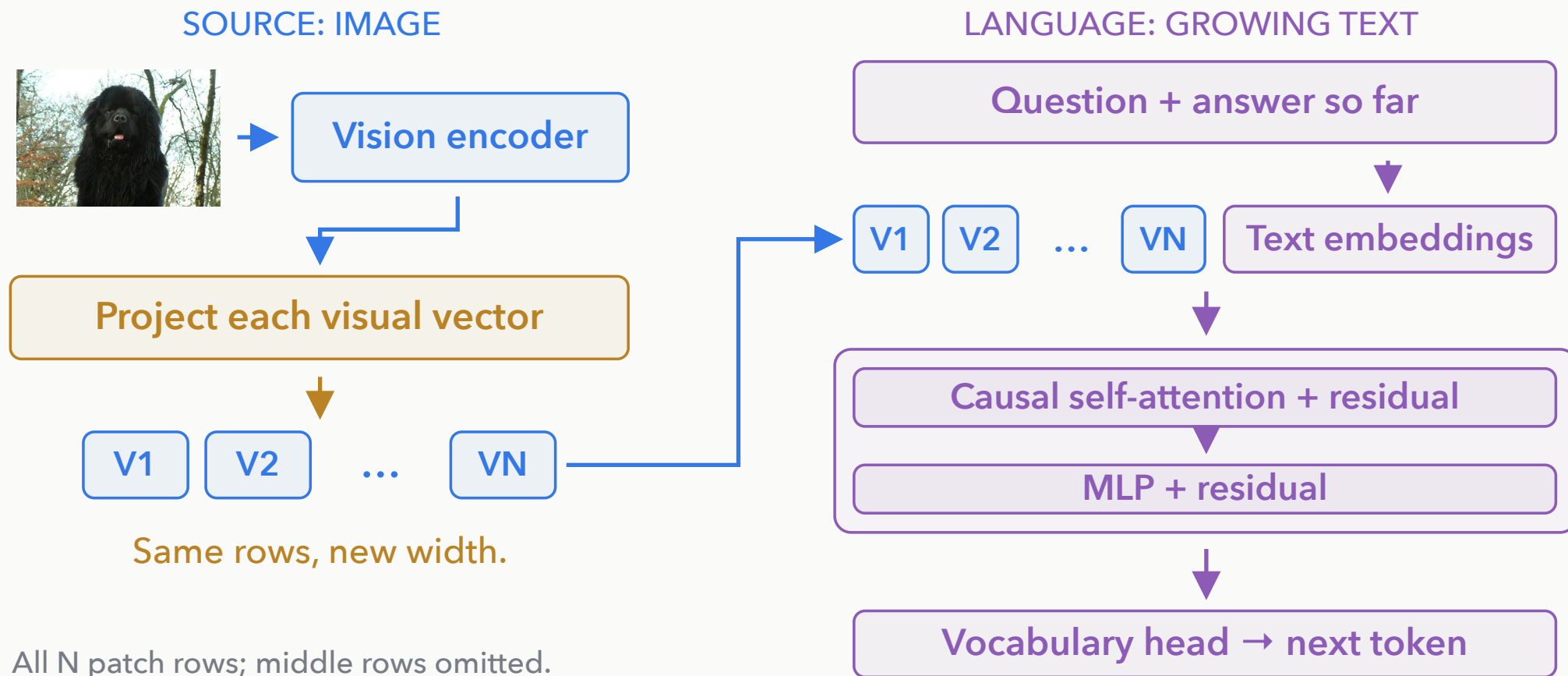
Query ↓	Keys →						
	z1	z2	z3	T1	T2	T3	A
z1	Colored	Amber	Amber	Grey	Grey	Grey	Grey
z2	Colored	Colored	Amber	Grey	Grey	Grey	Grey
z3	Colored	Colored	Colored	Grey	Grey	Grey	Grey
T1	Grey	Grey	Grey	Colored	Grey	Grey	Grey
T2	Grey	Grey	Grey	Colored	Colored	Grey	Grey
T3	Grey	Grey	Grey	Colored	Colored	Colored	Grey
A	Grey	Grey	Grey	Colored	Colored	Colored	Colored

z1 can read z1, z2 and z3.

z: image · T: prompt · A: assistant start. ViT mixed the image features earlier.

Both keep text causal. The difference is which image rows can read each other.

This is the basic LLaVA-style connection

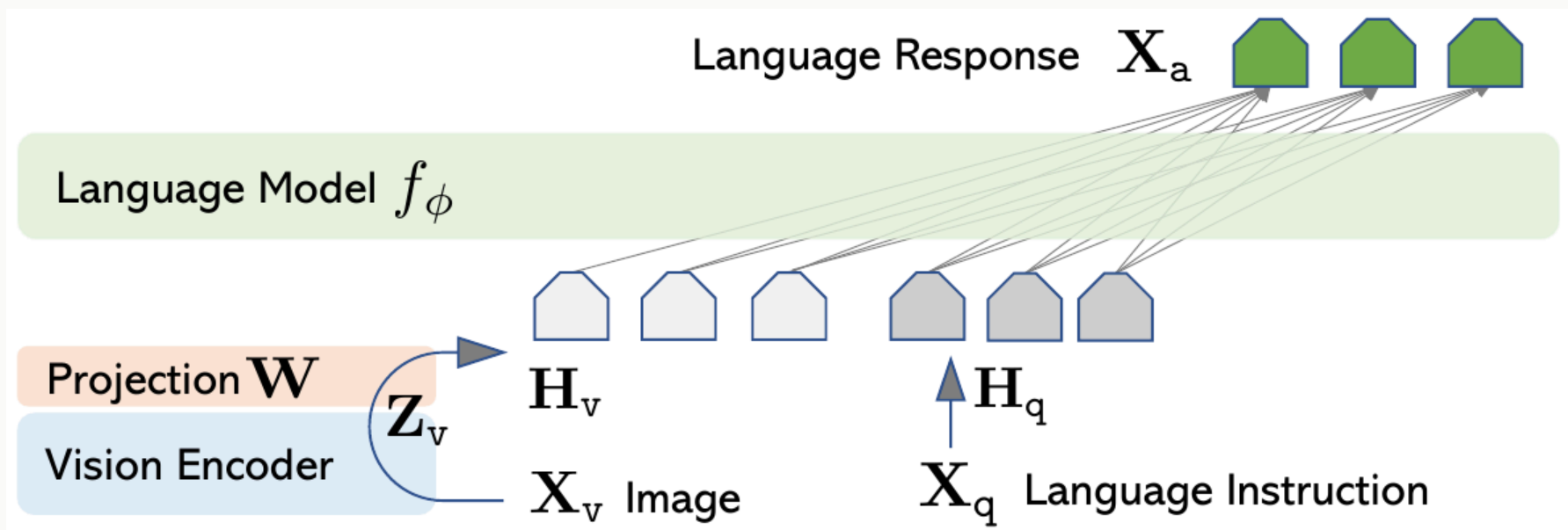


Projected patch vectors occupy earlier positions; they do not become word IDs.

Visual Instruction Tuning

Haotian Liu • Chunyuan Li • Qingyang Wu • Yong Jae Lee

[Read the paper ↗](#)



Original Figure 1 • Image features and instruction embeddings enter the same language model.

1 • Vision encoder

CLIP ViT-L/14 supplies image features Z_v .

2 • Projection W

The original paper uses a linear map to match the LM embedding width.

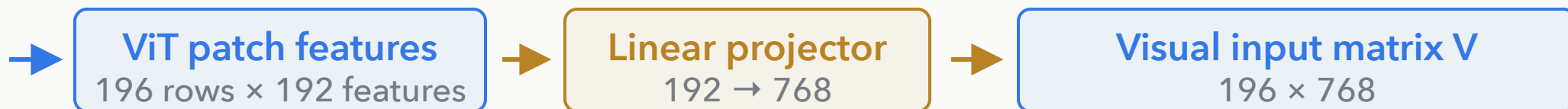
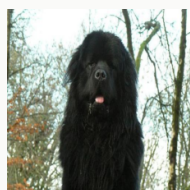
3 • Language model

Vicuna reads the visual vectors and text, then predicts the response.

Build the exact image-and-question input

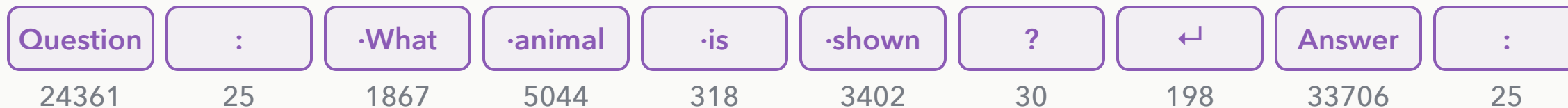
IDEA 1 · ALL PATCH ROWS → VISUAL PREFIX

Real shapes / token IDs · illustrative answer

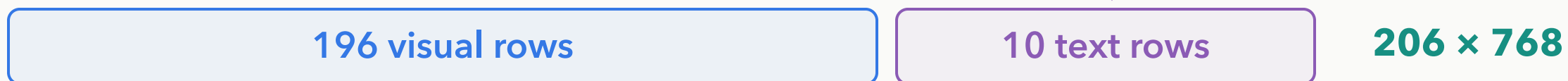


Notebook: $(224 \div 16)^2 = 196$ patches · DeiT-Tiny width 192 · DistilGPT2 width 768

Question: What animal is shown? ↵ Answer:



10 real token IDs; · marks a leading space, ↵ marks the newline.



196 visual rows + 10 prompt embeddings = 206 decoder positions.

Read the final ":" state, then score the vocabulary

IDEA 1 · ALL PATCH ROWS → VISUAL PREFIX

Hand-set toy head · not model scores



h_{205} is the final prompt ":" state after decoder blocks and final normalization.

Tiny worked head: choose $h = [1, 2]$, four vocabulary entries, and zero bias.



For ·A: $1 \times 1 + 2 \times 1 = 3$

Each output column scores one vocabulary entry using the same final state.

Turn the scores into one next-token choice

IDEA 1 · ALL PATCH ROWS → VISUAL PREFIX

Hand-set toy head · not model scores

softmax: exponentiate each score, then divide by their sum

Token	Logit	exp(logit)	Probability	
·A	3	20.086		0.8310
·The	1	2.718		0.1125
·dog	0	1.000		0.0414
EOS	-1	0.368		0.0152

Sum = 24.172

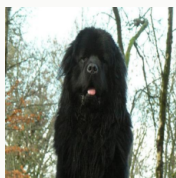
Sum = 1; displayed values rounded.

Greedy choice: " A" → token ID 317 → embedding lookup next

The four-entry calculation is a toy; the real vocabulary has 50,257 entries.

Feed the selected token back as an embedding

IDEA 1 · ALL PATCH ROWS → VISUAL PREFIX



What animal is shown?

Answer so far: empty

Selection returns an ID.

The decoder receives its embedding.

Real shapes / token IDs · illustrative answer

Selected "A" · ID 317



E[317]: one 768-wide row



E[317]

new position 206

196 image vectors

10 prompt embeddings



207 positions → same causal decoder → next token

Append E[317] for "A"; the image and question stay fixed.

Idea 1, end to end in one function

**image**

One image

question = "What animal is shown?"**answer_so_far** = "It is a" (empty at the start)

One next-token step · high-level pseudocode

Shapes · batch omitted

<code>def next_token(image, question, answer_so_far):</code>	
<code>patches = vit(image).patch_states</code>	$N \times d_V$
<code>visual = projector(patches)</code>	$N \times d_{LM}$
<code>ids = tokenize(question, answer_so_far)</code>	T token IDs
<code>text = embed(ids)</code>	$T \times d_{LM}$
<code>inputs = concat([visual, text], axis="tokens")</code>	$(N + T) \times d_{LM}$
<code>hidden = causal_decoder(inputs)</code>	$(N + T) \times d_{LM}$
<code>scores = lm_head(hidden[-1])</code>	Vocabulary scores
<code>return argmax(scores)</code>	Next text token ID

causal_decoder stacks causal self-attention + MLP blocks over *all* input positions.

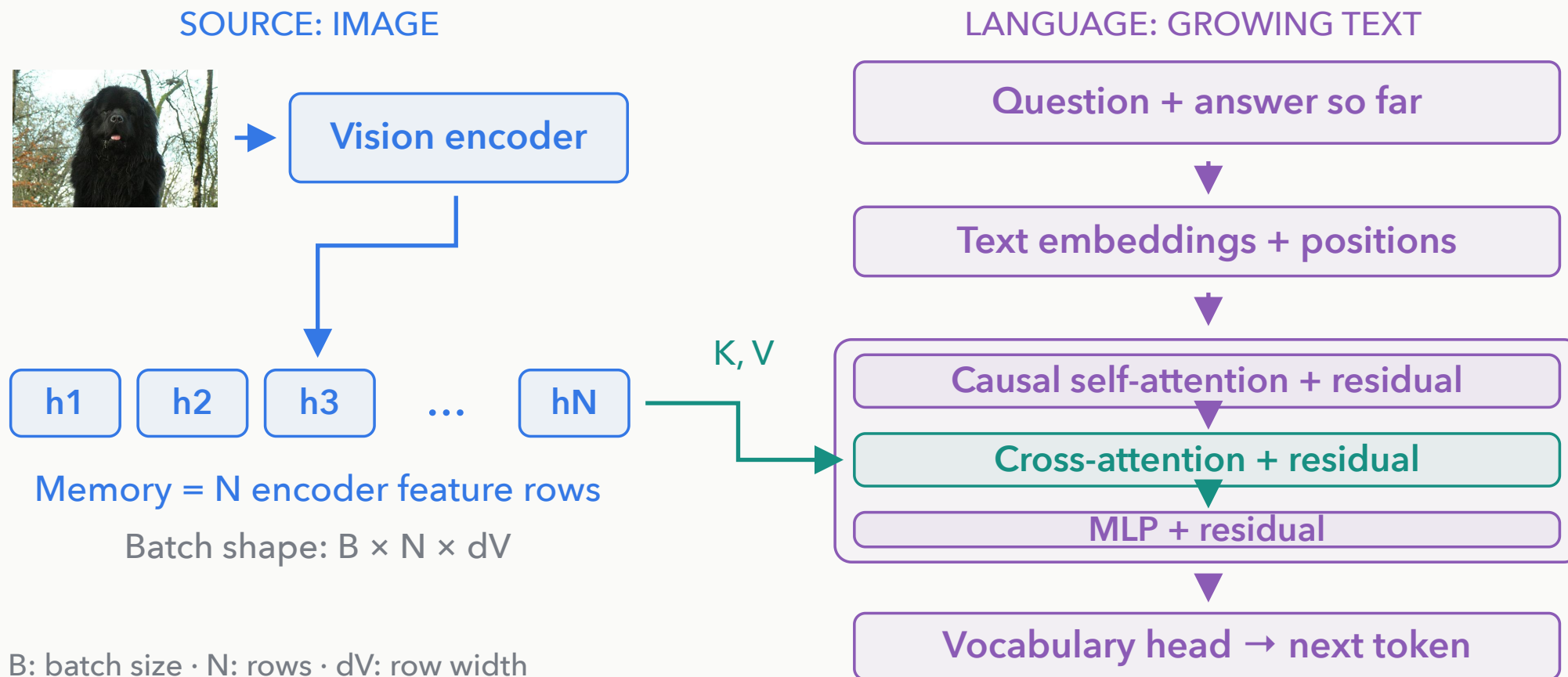
Display the returned token with the tokenizer; append its ID to the answer and predict again.

SECTION 04

Could vision remain separate?

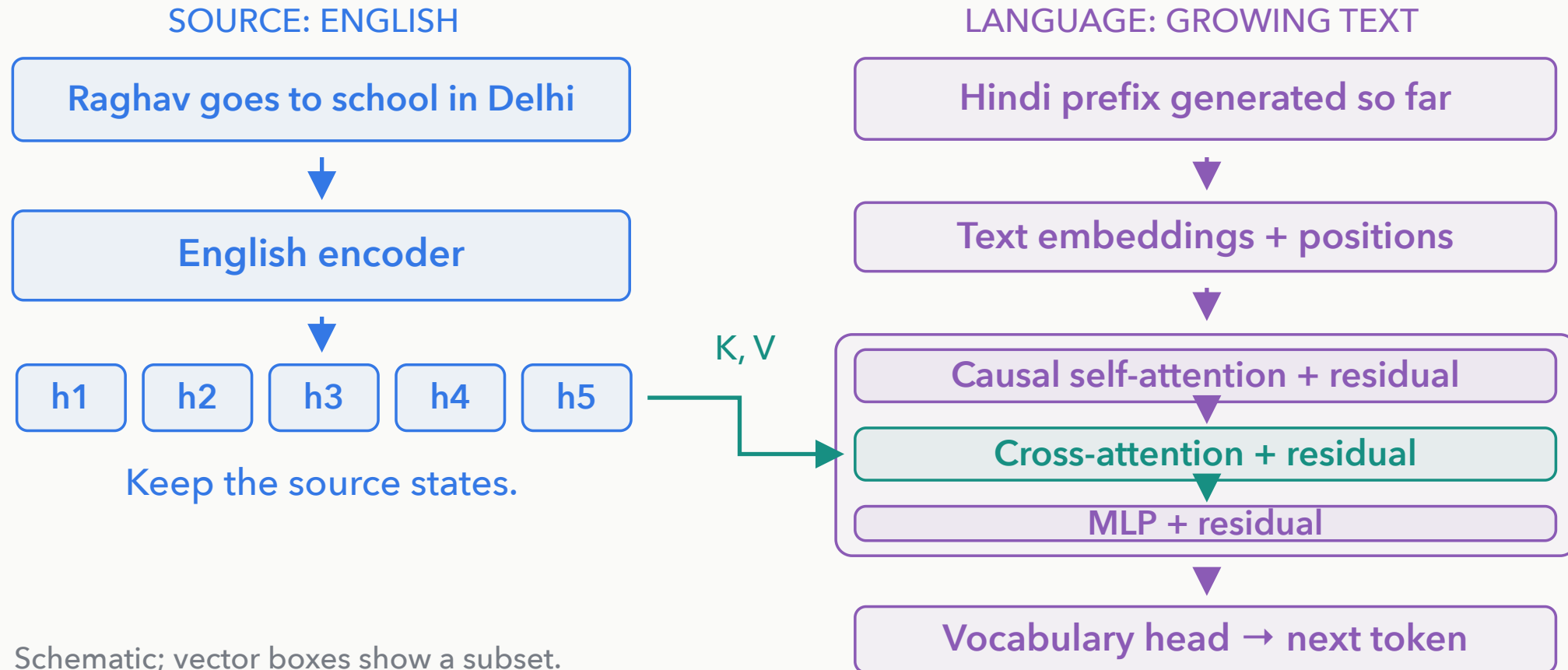
Return to Idea 2: how can language read separate visual memory?

Idea 2 in detail · read a separate visual memory



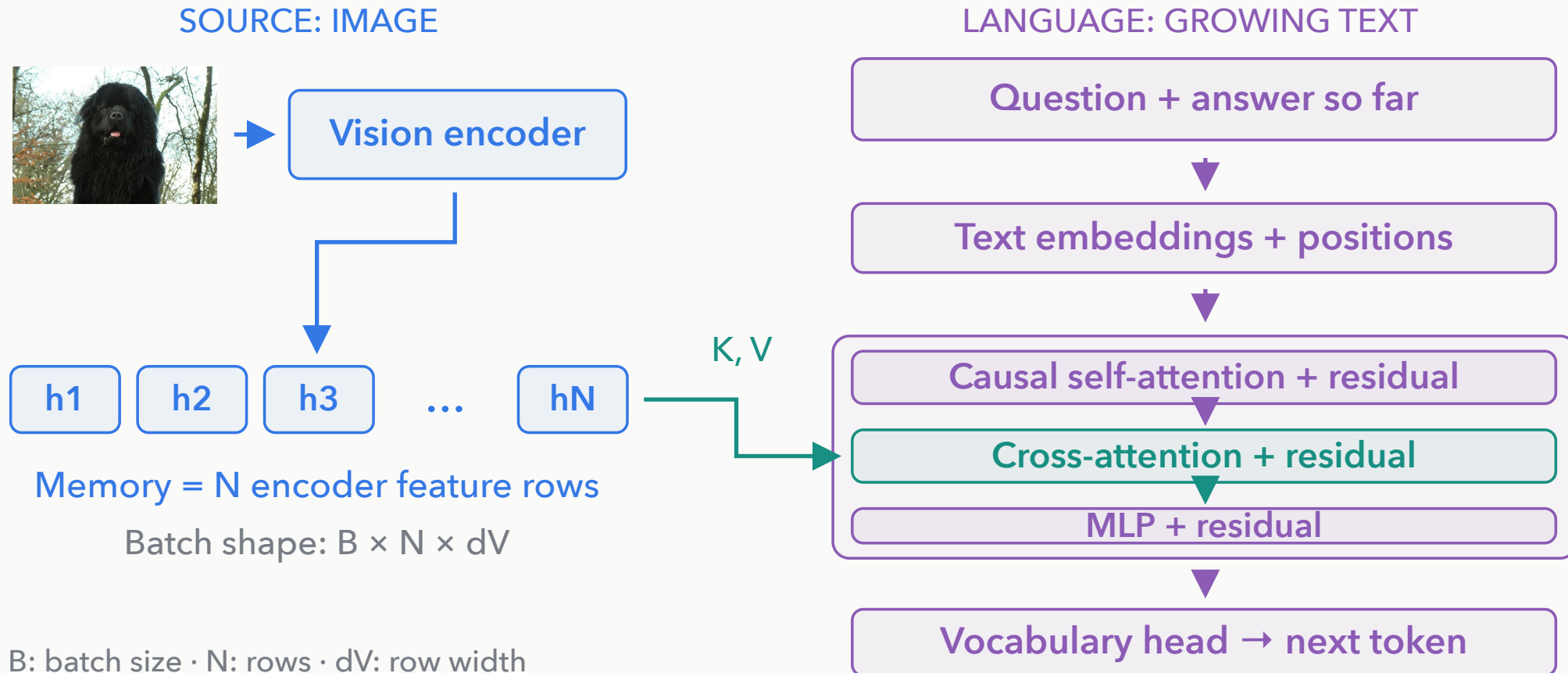
Causal self-attention reads text; cross-attention reads image memory.

We already did this with English-to-Hindi translation



English source states stayed available while the Hindi prefix grew.

Keep the decoder idea; change English to an image



| The source changes. The decoder still reads a separate memory.

Same image, two questions: what should we read?

ONE IMAGE



A dog, with trees behind it.

Authored examples;
not a recorded model run.

QUESTION A

What animal is shown?

Answer so far: "It is a"

Reference continuation: dog

QUESTION B

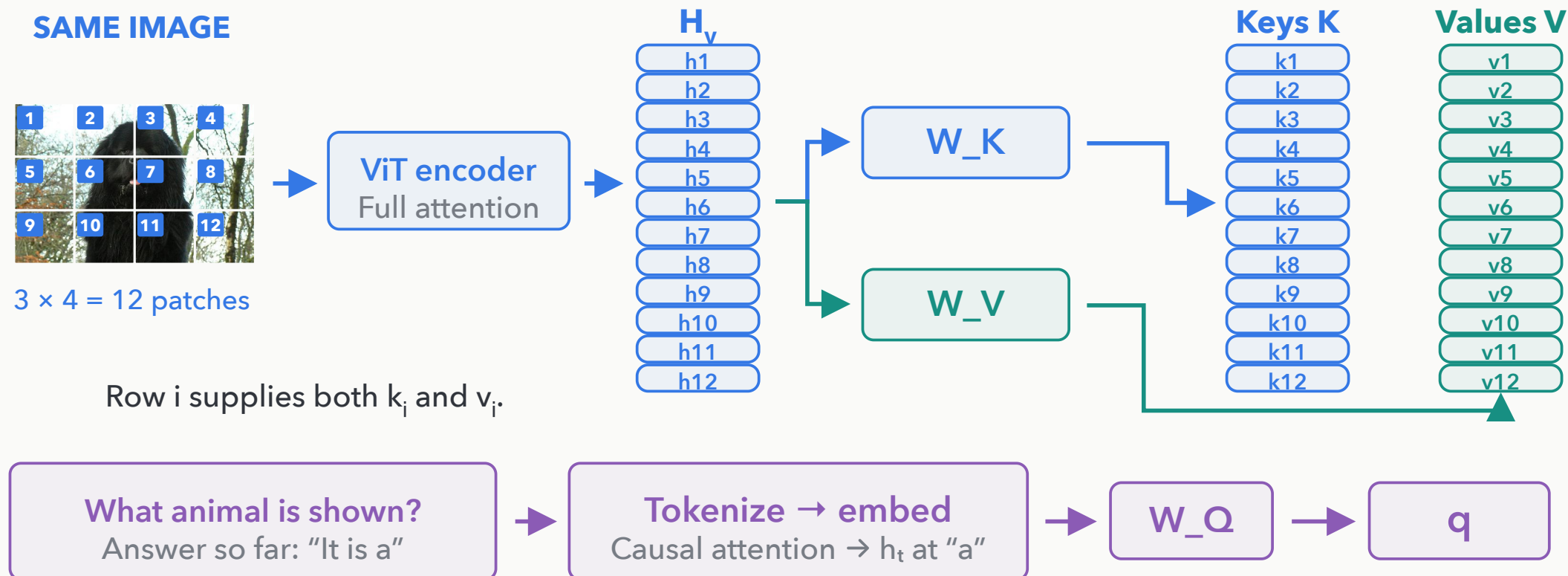
What is behind the dog?

Answer so far: "Behind it are"

Reference continuation: trees

Different questions need different visual evidence from the same image.

The text supplies Q; the image supplies K and V



Each image row supplies a key and value; the contextual state at "a" supplies q .

Use the keys to weight the values, then continue the answer



What animal is shown?

Answer so far:

It

is

a

State at "a" $\rightarrow W_Q \rightarrow q$
After causal self-attention

Includes the question + "It is".

1 · Compare q with every key

k1

k2

k3

k4

k5

k6

k7

k8

k9

k10

k11

k12

2 · Softmax gives weights $\alpha_1 \dots \alpha_{12}$, which sum to 1

α_1

α_2

α_3

α_4

α_5

α_6

α_7

α_8

α_9

α_{10}

α_{11}

α_{12}

x

x

x

x

x

x

x

x

x

x

x

x

v1

v2

v3

v4

v5

v6

v7

v8

v9

v10

v11

v12

$$c = \sum_i \alpha_i v_i$$

Update "a" state $\rightarrow$ LM head

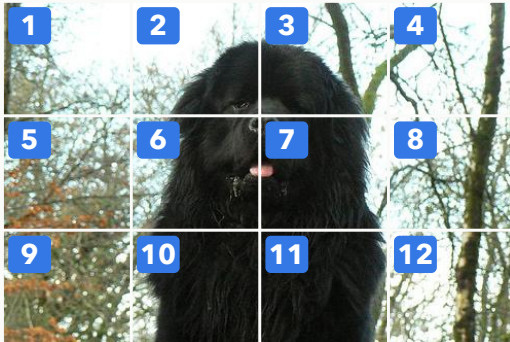
dog

Reference continuation

Update the state at "a"; the language head predicts the next token after it.

The state at "a" supplies the query

SAME IMAGE · ALL 12 PATCHES



What animal is shown?

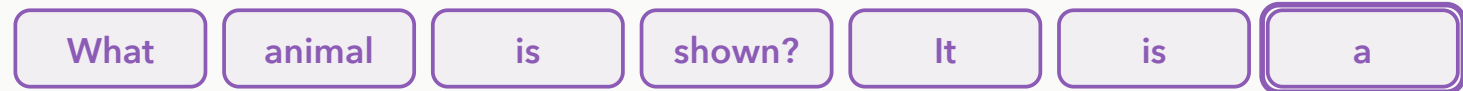
ANSWER SO FAR



Read the state at "a".

Toy numbers · one head
Not measured from this photo

QUESTION + ANSWER: READABLE TOKEN LABELS



Tokenizer → IDs → embeddings + position handling



Causal self-attention → current text state h_t

$$\begin{array}{c} h_t (1 \times 2) \\ \begin{array}{|c|c|} \hline 1 & 2 \\ \hline \end{array} \end{array} \times \begin{array}{c} W_Q (2 \times 2) \\ \begin{array}{|c|c|} \hline 0 & 1 \\ \hline 0.5 & -0.5 \\ \hline \end{array} \end{array} = \begin{array}{c} q (1 \times 2) \\ \begin{array}{|c|c|} \hline 1 & 0 \\ \hline \end{array} \end{array}$$

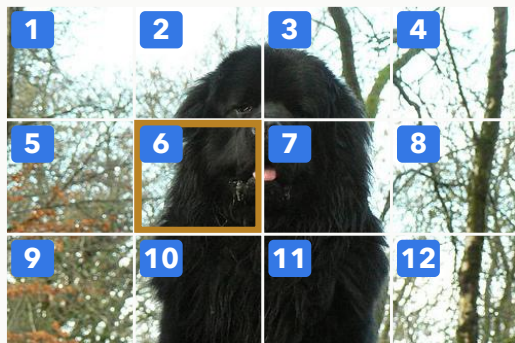
$$q = [1, 2] W_Q = [1, 0]$$

Toy h_t is assumed after the text sublayer; W_Q stays fixed.

Use the final supplied text state, after causal self-attention, to make q .

The same twelve image rows become twelve keys

SAME IMAGE · ALL 12 PATCHES



What animal is shown?

ANSWER SO FAR

It is a

Read the state at "a".

Toy numbers · one head
Not measured from this photo

ViT output H_v

12 rows × 2 features

1	3	0
2	2	1
3	2	1
4	3	0
5	1	1
6	0	3
7	1	3
8	2	1
9	1	0
10	0	2
11	0	2
12	2	0

 W_K

0	1
1	0

2 × 2

$$H_v W_K = K$$

Keys K

12 rows × 2 features

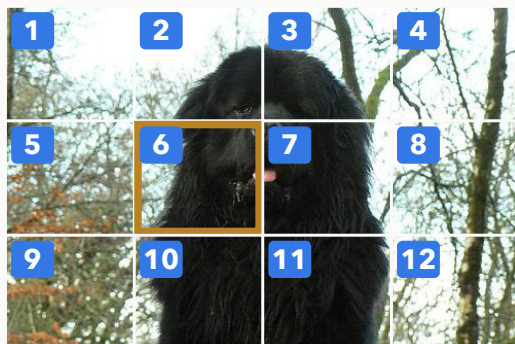
1	0	3
2	1	2
3	1	2
4	0	3
5	1	1
6	3	0
7	3	1
8	1	2
9	0	1
10	2	0
11	2	0
12	0	2

Row 6: $[0, 3] W_K = [3, 0]$

Apply W_K to every encoder row; inspect patch 6 without dropping the other rows.

Each image row also supplies a paired value

SAME IMAGE · ALL 12 PATCHES



What animal is shown?

ANSWER SO FAR

It is a

Read the state at "a".

Toy numbers · one head
Not measured from this photo

ViT output H_v

12 rows × 2 features

1	3	0
2	2	1
3	2	1
4	3	0
5	1	1
6	0	3
7	1	3
8	2	1
9	1	0
10	0	2
11	0	2
12	2	0

 W_v

1	0
1	2

2 × 2

$$H_v W_v = V$$

Values V

12 rows × 2 features

1	3	0
2	3	2
3	3	2
4	3	0
5	2	2
6	3	6
7	4	6
8	3	2
9	1	0
10	2	4
11	2	4
12	2	0

Row 6: $[0, 3] W_v = [3, 6]$

The key determines the match; its paired value supplies the features to mix.

Score every image row for this text position

SAME IMAGE · ALL 12 PATCHES



What animal is shown?

ANSWER SO FAR

It is a

Read the state at "a".

Toy numbers · one head

Not measured from this photo

$q = [1, 0]$ · one head has $d_k = 2$

Patch i	Key k_i	$q \cdot k_i$	Score $s_i = (q \cdot k_i) / \sqrt{2}$
1	[0, 3]	0	0.000
2	[1, 2]	1	0.707
3	[1, 2]	1	0.707
4	[0, 3]	0	0.000
5	[1, 1]	1	0.707
6	[3, 0]	3	2.121
7	[3, 1]	3	2.121
8	[1, 2]	1	0.707
9	[0, 1]	0	0.000
10	[2, 0]	2	1.414
11	[2, 0]	2	1.414
12	[0, 2]	0	0.000

Score scale · before softmax

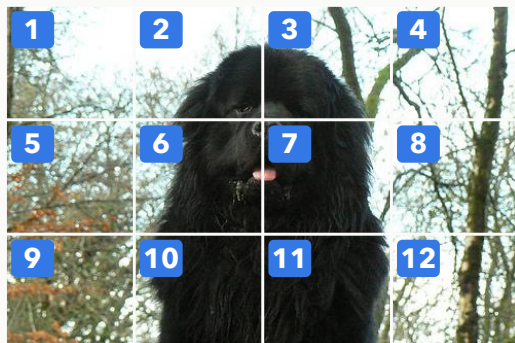
0.000 0.707 1.414 2.121

Patch 6: $(1 \times 3 + 0 \times 0) / \sqrt{2} = 2.121$

One query × twelve keys gives twelve scores; every patch stays in the computation.

Normalize across all twelve image positions

SAME IMAGE · ALL 12 PATCHES



What animal is shown?

ANSWER SO FAR

It is a

Read the state at "a".

Toy numbers · one head
Not measured from this photo

$$a_i = \exp(s_i) / \sum_j \exp(s_j)$$

Patch i	Score s_i	$\exp(s_i)$	Weight a_i
1	0.000	1.000	0.027
2	0.707	2.028	0.055
3	0.707	2.028	0.055
4	0.000	1.000	0.027
5	0.707	2.028	0.055
6	2.121	8.342	0.225
7	2.121	8.342	0.225
8	0.707	2.028	0.055
9	0.000	1.000	0.027
10	1.414	4.113	0.111
11	1.414	4.113	0.111
12	0.000	1.000	0.027

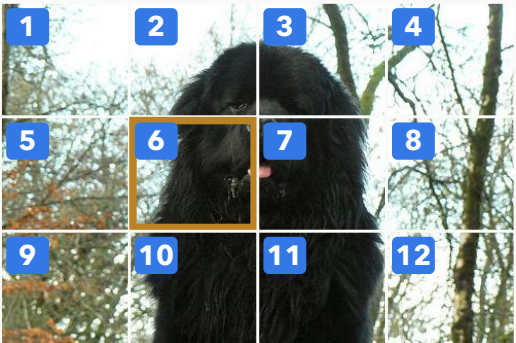
Patch 6: 8.342 / 37.023 = 0.225

$\sum_i a_i = 1$ exactly before rounding; these are not word probabilities.

Divide each exponential by the sum of all twelve exponentials.

Weight the paired values and add the twelve rows

SAME IMAGE · ALL 12 PATCHES



What animal is shown?

ANSWER SO FAR

It is a

Read the state at "a".

Toy numbers · one head

Not measured from this photo

$$c = aV = \sum_i a_i v_i$$

Patch i	Weight a_i	Value v_i	Contribution $a_i v_i$
1	0.027	[3.000, 0.000]	[0.081, 0.000]
2	0.055	[3.000, 2.000]	[0.164, 0.110]
3	0.055	[3.000, 2.000]	[0.164, 0.110]
4	0.027	[3.000, 0.000]	[0.081, 0.000]
5	0.055	[2.000, 2.000]	[0.110, 0.110]
6	0.225	[3.000, 6.000]	[0.676, 1.352]
7	0.225	[4.000, 6.000]	[0.901, 1.352]
8	0.055	[3.000, 2.000]	[0.164, 0.110]
9	0.027	[1.000, 0.000]	[0.027, 0.000]
10	0.111	[2.000, 4.000]	[0.222, 0.444]
11	0.111	[2.000, 4.000]	[0.222, 0.444]
12	0.027	[2.000, 0.000]	[0.054, 0.000]

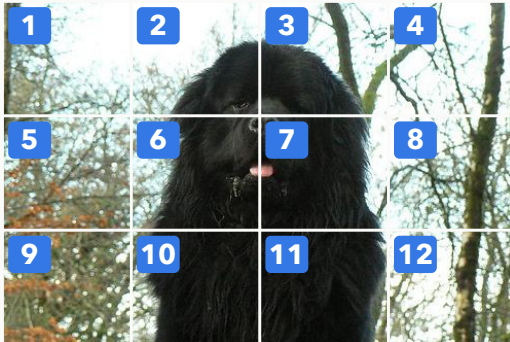
Add all 12 rows → $c = [2.867, 4.031]$

Shown to 3 decimal places; products and sums use full precision.

All twelve contributions add up to one visual message for the current text position.

Change the question; read the state at "are"

SAME IMAGE · ALL 12 PATCHES



What is behind the dog?

ANSWER SO FAR

Behind it are

Read the state at "are".

Toy numbers · one head
Not measured from this photo

QUESTION + ANSWER: READABLE TOKEN LABELS

What is behind the dog? Behind it are

Tokenizer → IDs → embeddings + position handling



Causal self-attention → current text state h_t

$$\begin{array}{c} h_t (1 \times 2) \\ \begin{array}{|c|c|} \hline 3 & 0 \\ \hline \end{array} \end{array} \times \begin{array}{c} W_Q (2 \times 2) \\ \begin{array}{|c|c|} \hline 0 & 1 \\ \hline 0.5 & -0.5 \\ \hline \end{array} \end{array} = \begin{array}{c} q (1 \times 2) \\ \begin{array}{|c|c|} \hline 0 & 3 \\ \hline \end{array} \end{array}$$

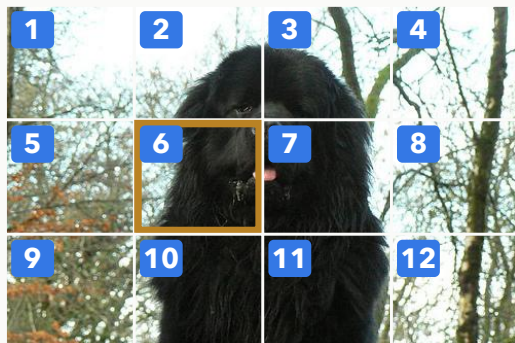
$$q = [3, 0] W_Q = [0, 3]$$

Toy h_t is assumed after the text sublayer; W_Q stays fixed.

The image, keys, values and W_Q stay fixed; the text state changes.

Same patch 6, two different query scores

SAME IMAGE · ALL 12 PATCHES



What animal is shown?

ANSWER SO FAR



Read the state at "a".

Toy numbers · one head

Not measured from this photo

ZOOM INTO PATCH 6 · SAME KEY IN BOTH LOOKUPS

$$h_6 = [0, 3] \rightarrow k_6 = [3, 0]$$

ANIMAL QUESTION

$$[1, 0] \cdot [3, 0] = 3$$

$$\text{Scaled score: } 3 / \sqrt{2} = 2.121$$

BACKGROUND QUESTION

$$[0, 3] \cdot [3, 0] = 0$$

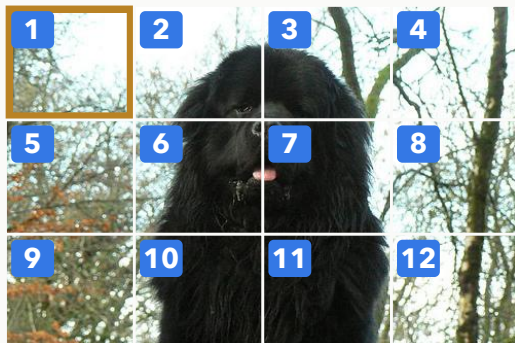
$$\text{Scaled score: } 0 / \sqrt{2} = 0.000$$

This is one row zoom. Both lookups still use all 12.

A fixed image key can match different language queries differently.

Now inspect patch 1 with those same queries

SAME IMAGE · ALL 12 PATCHES



What animal is shown?

ANSWER SO FAR



Read the state at "a".

Toy numbers · one head

Not measured from this photo

ZOOM INTO PATCH 1 · SAME KEY IN BOTH LOOKUPS

$$h1 = [3, 0] \rightarrow k1 = [0, 3]$$

ANIMAL QUESTION

$$[1, 0] \cdot [0, 3] = 0$$

Scaled score: $0 / \sqrt{2} = 0.000$

BACKGROUND QUESTION

$$[0, 3] \cdot [0, 3] = 9$$

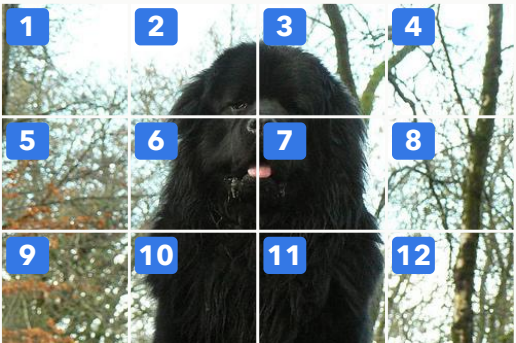
Scaled score: $9 / \sqrt{2} = 6.364$

This is one row zoom. Both lookups still use all 12.

| The query changes which keys receive large scores.

Recompute the whole score row, then softmax

SAME IMAGE · ALL 12 PATCHES



What is behind the dog?

ANSWER SO FAR

Behind it are

Read the state at "are".

Toy numbers · one head

Not measured from this photo

SAME 12 KEYS · DIFFERENT LANGUAGE QUERIES

Patch	Key k_i	$s_i: q = [1, 0]$	Weight	$s_i: q = [0, 3]$	Weight
1	[0, 3]	0.000	0.027	6.364	0.396
2	[1, 2]	0.707	0.055	4.243	0.047
3	[1, 2]	0.707	0.055	4.243	0.047
4	[0, 3]	0.000	0.027	6.364	0.396
5	[1, 1]	0.707	0.055	2.121	0.006
6	[3, 0]	2.121	0.225	0.000	0.001
7	[3, 1]	2.121	0.225	2.121	0.006
8	[1, 2]	0.707	0.055	4.243	0.047
9	[0, 1]	0.000	0.027	2.121	0.006
10	[2, 0]	1.414	0.111	0.000	0.001
11	[2, 0]	1.414	0.111	0.000	0.001
12	[0, 2]	0.000	0.027	4.243	0.047

Softmax runs separately for each query.

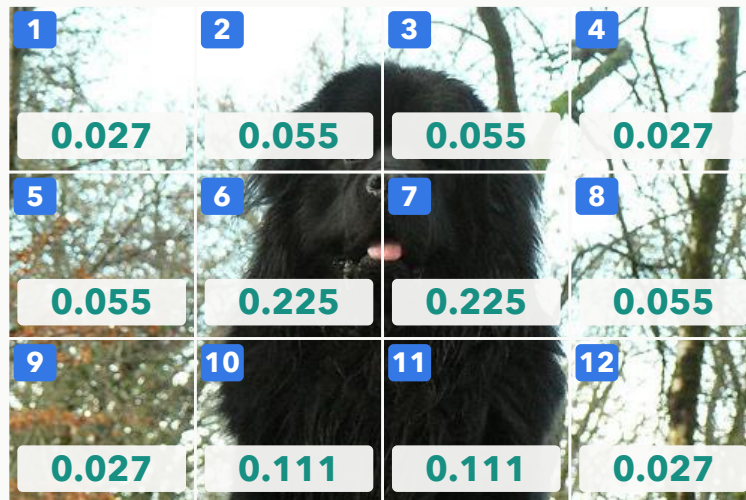
The keys did not change. The comparison with those keys did.

Each query gets its own twelve weights; the image memory is reused.

Put the calculated weights back on the patch grid

What animal is shown?

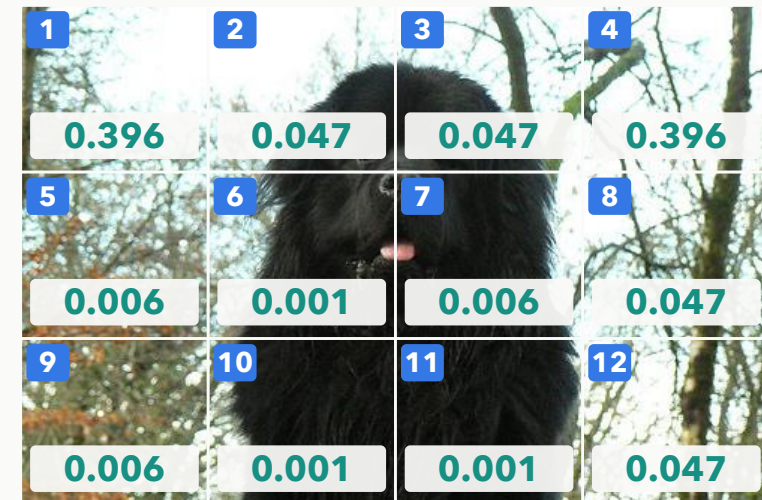
It is a $\rightarrow q = [1, 0]$



$\Sigma \text{ weights} = 1$

What is behind the dog?

Behind it are $\rightarrow q = [0, 3]$



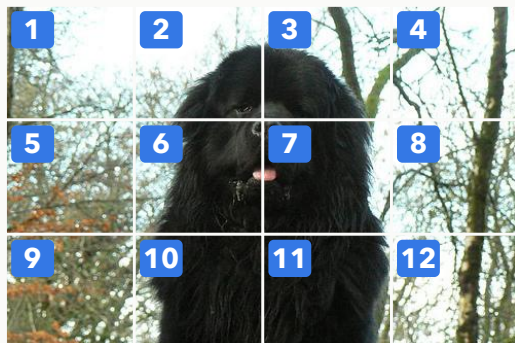
$\Sigma \text{ weights} = 1$

Calculated toy weights on patch positions · not a measured model attention map

Same image and keys; different queries give different mixtures of image values.

Two questions, two messages from the same values

SAME IMAGE · ALL 12 PATCHES



What animal is shown?

ANSWER SO FAR



Read the state at "a".

Toy numbers · one head
Not measured from this photo

V STAYS FIXED; EACH QUERY CHOOSES ITS MIXTURE

Animal question · $q = [1, 0]$

$$c = \sum_{i=1}^{12} a_i v_i = [2.867, 4.031]$$

All 12 rows; weights from the first softmax.

Background question · $q = [0, 3]$

$$c = \sum_{i=1}^{12} a_i v_i = [2.94, 0.34]$$

All 12 rows; weights from the second softmax.

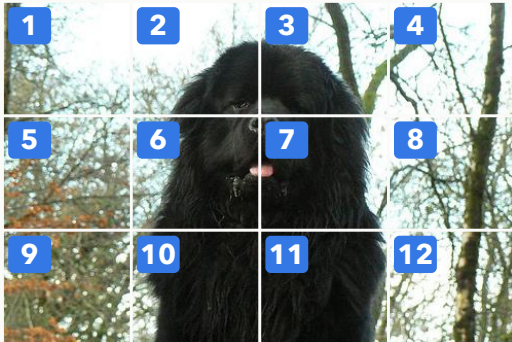
Different messages can update the language state differently.

Neither 2-number vector is the word "dog" or the word "trees".

Changing q can change the visual message without recomputing K or V .

Project the visual message and update the text state

SAME IMAGE · ALL 12 PATCHES



What animal is shown?

ANSWER SO FAR

It is a

Read the state at "a".

Toy numbers · one head

Not measured from this photo

RETURN THE VISUAL MESSAGE TO THE TEXT PATH

$$c = [2.867, 4.031] (1 \times 2)$$

$$c \times \begin{matrix} \begin{matrix} 1 & 0 \\ 0.5 & 1 \end{matrix} \\ W_O (2 \times 2) \end{matrix} = [4.883, 4.031]$$

$$h'_t = h_t + cW_O$$

$$= [1, 2] + [4.883, 4.031] = [5.883, 6.031]$$

Remaining decoder → vocabulary head → next token

The visual message changes the state at "a"; a later vocabulary head scores words.

Check: has cross-attention already answered “dog”?

Lookup result

$$c \approx [2.867, 4.031]$$

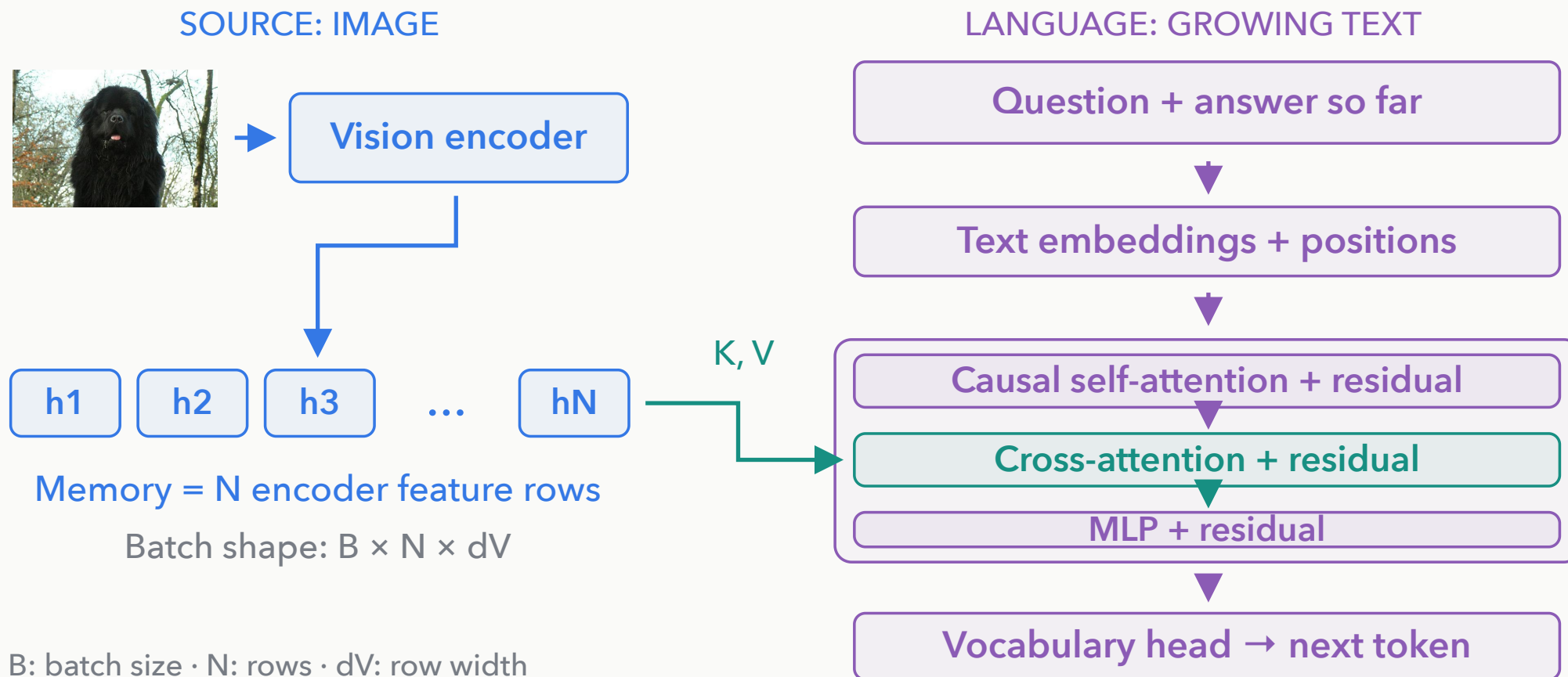
What happens next?

Does c name a word?

Or does it update the decoder’s current state?

| Predict before advancing: where does a vocabulary token actually appear?

The vocabulary head still produces the next-token scores



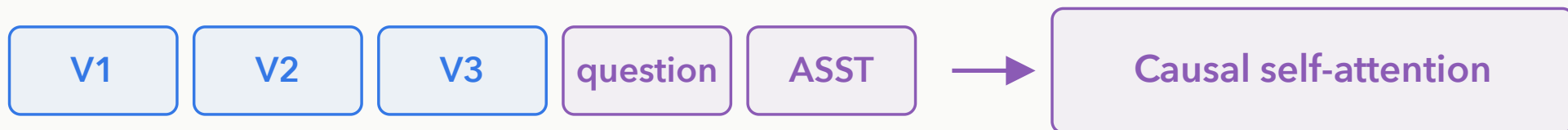
Visual lookup changes the language state; the language head chooses words.

Two ways for language to access vision

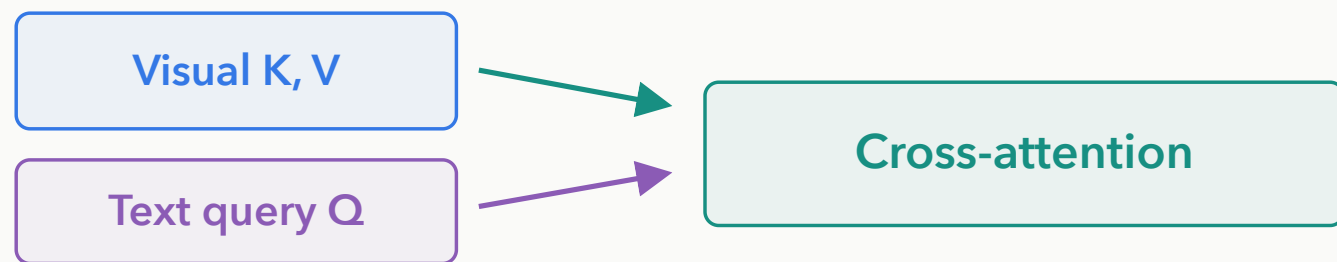


Same image. Same question: What animal is shown?

A · VISUAL PREFIX



B · SEPARATE MEMORY



In-sequence visual context or a separate visual memory.

Idea 2, end to end in one function

**image**

One image

question = "What animal is shown?"**answer_so_far** = "It is a" (empty at the start)

One next-token step · high-level pseudocode

Shapes · batch omitted

<code>def next_token(image, question, answer_so_far):</code>	
<code>memory = vision_encoder(image).features</code>	$N \times d_V$
<code>ids = tokenize(question, answer_so_far)</code>	T token IDs
<code>text = embed(ids)</code>	$T \times d_{LM}$
for block in decoder_blocks:	
<code>text = block.causal_self_attn(text)</code>	$T \times d_{LM}$
<code>text = block.cross_attn(text, memory)</code>	$T \times d_{LM}$
<code>text = block.mlp(text)</code>	$T \times d_{LM}$
<code>return argmax(lm_head(text[-1]))</code>	Next text token ID

cross_attn: Q from text; K and V from image memory. The two sequences stay separate.

Sublayer helpers include residuals and norms. Append the returned token ID to the answer; repeat.

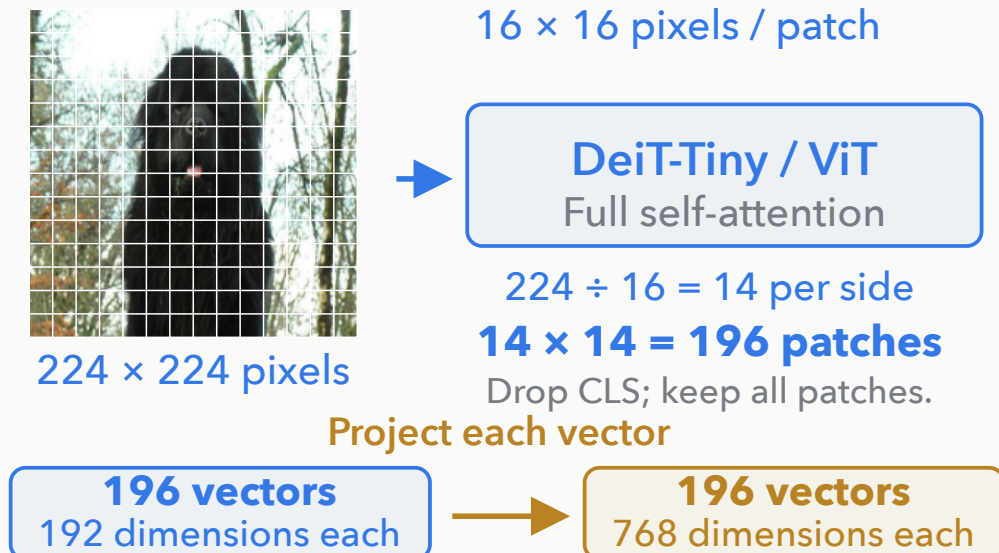
SECTION 05

Do we need every visual token?

Every image adds decoder positions. What do they cost, and can a shorter visual summary help?

How long is this image + question + answer prefix?

ONE IMAGE · AFTER RESIZE / CROP



COMPLETE TEXT SUPPLIED TO THE MODEL

Question: What animal is shown?

Answer: It is a

DistilGPT2 tokenizer → these 13 pieces



· = leading space ↵ = one newline token

13 IDs → 13 text embeddings
768 dimensions per vector

196 image vectors

13

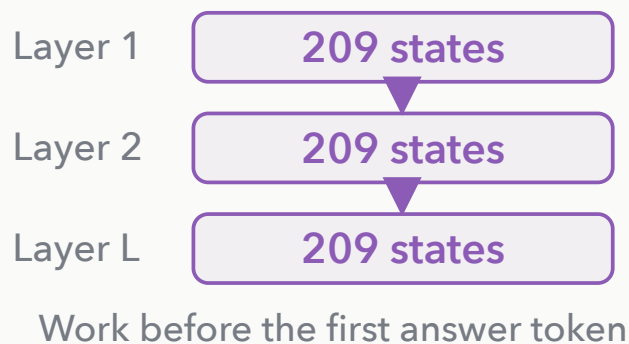
196 visual + 13 text = 209 positions in the causal decoder

Projection changes each vector: 192 → 768 dimensions. We still have 196 vectors.

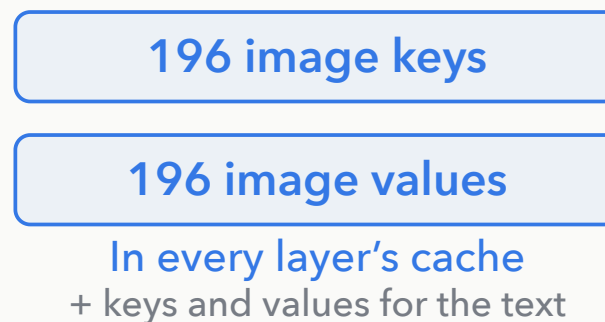
209 positions fit. What do they cost?



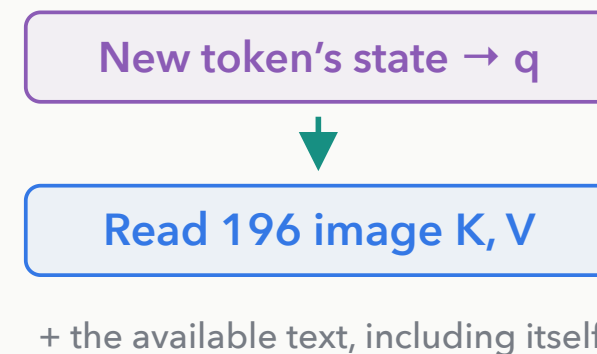
1 · PROCESS THE INPUT



2 · STORE THE IMAGE K, V



3 · READ THEM AGAIN

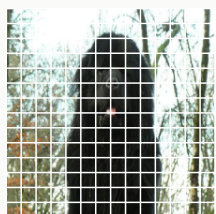


Caching reuses K and V; it does not remove the image positions.

Every image position adds decoder work, cache storage and later attention reads.

Keep the text; shorten the visual prefix

SAME IMAGE + ViT



196

patch rows

Learned summarizer
Choose $M = 8$ readers

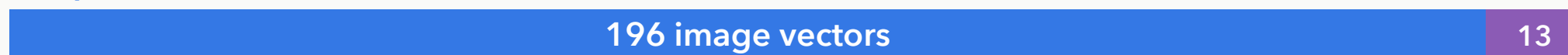
All 196 rows remain available.

8 summary vectors



Project 8 rows to width 768 → prepend them before the same 13 text embeddings

Full prefix: $196 + 13 = 209$



Short prefix: $8 + 13 = 21$



188 fewer decoder positions to process
A smaller cache to store and read · same scale

SAME TEXT

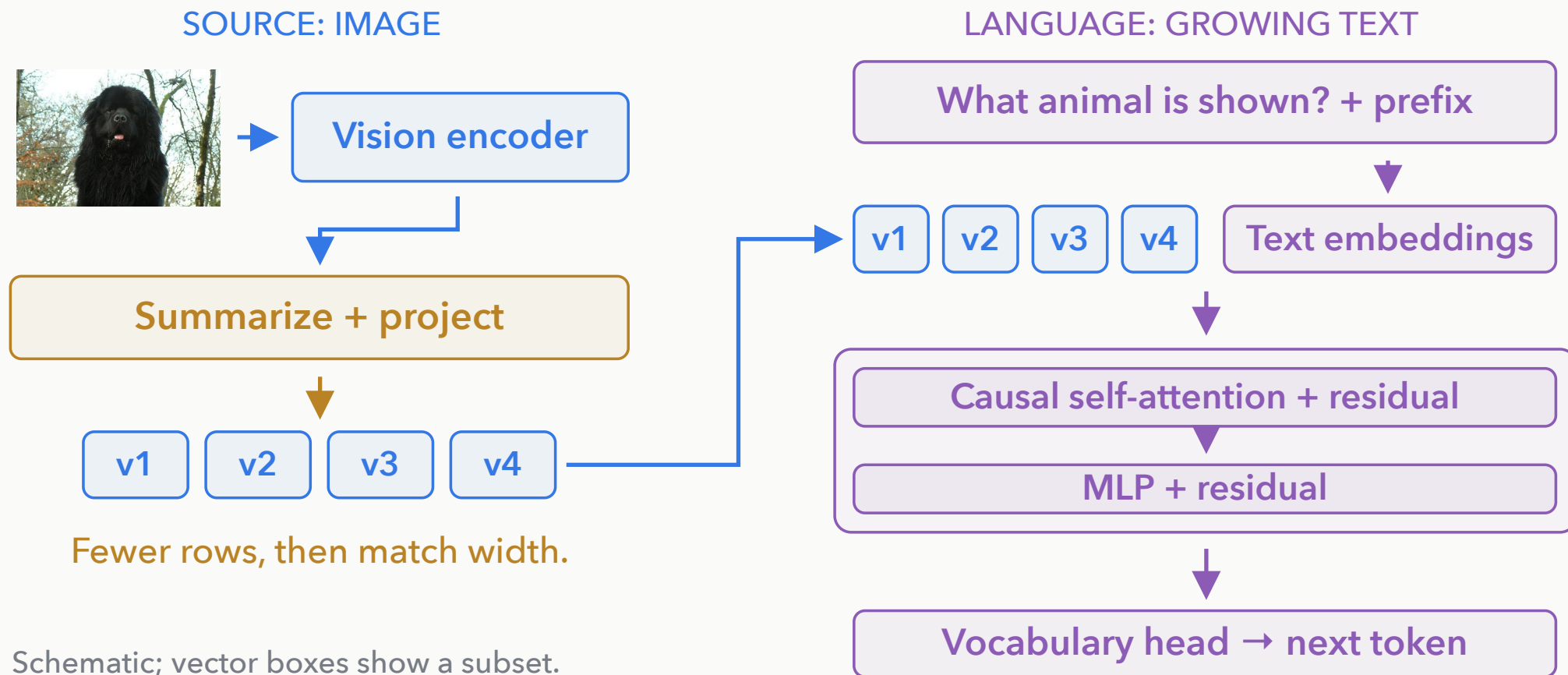
Question: What animal is shown?

Answer: It is a

13 text embeddings

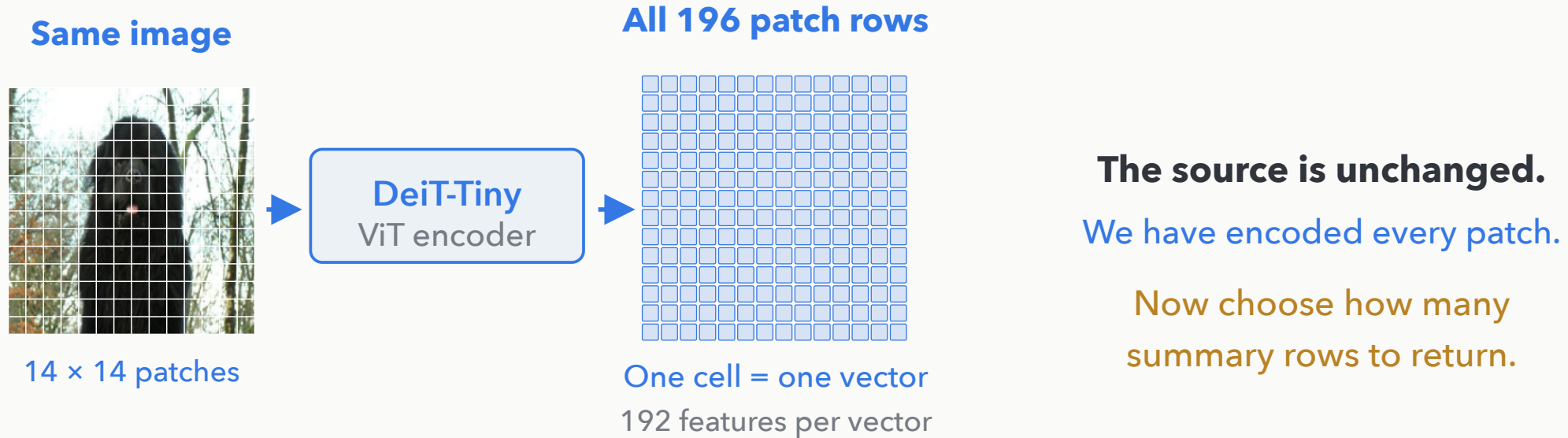
Summarize 196 image vectors into 8: fewer decoder positions, with evidence we must preserve.

Idea 3 in detail · make the visual context shorter



Replace the row-wise projector with a learned summary followed by projection.

Build the summary from all 196 image rows



The grid tracks patch positions; the feature matrix is $H_V \in \mathbb{R}^{196 \times 192}$.

ViT full attention lets each patch row use information from the rest of the image.

| The source memory is the encoded image: one feature vector per patch position.

What is a summary token?

STARTING QUERY SLOTS · S



8 stored vectors × 192 numbers

Shared across images

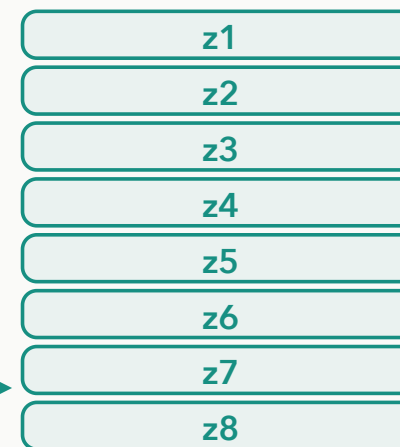


Image features H



Attention reader

COMPUTED SUMMARIES · Z



8 output vectors × 192 numbers

New values for each image

"Summary token" = one vector in Z. It has no vocabulary ID or assigned object name.

| A stored query slot asks for information; a summary vector contains the image-dependent result.

Initialize eight query slots once

CHOOSE 8 SLOTS

A design choice, not an object count

USE THE READER WIDTH: 192

192 scalar coordinates in every slot

```
S = nn.Parameter(0.02 * torch.randn(1, 8, 192))
```

1 shared bank · 8 rows · 192 coordinates per row



$8 \times 192 = 1,536$ trainable numbers in S

Initialize once → train → save → reuse. Do not redraw S for each image.

An existing trained checkpoint loads S. Our notebook creates a new, untrained reader.

| Choose the count and width; start with small random numbers, then learn them.

Where do the reader's Q, K and V come from?

3 HEADS · 192 ÷ 3 = 64 COORDINATES PER HEAD

Stored S: 8×192



W_Q

Q: 8×64

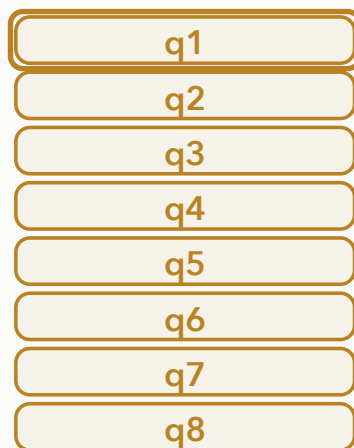


Image H

196×192

W_K

$K \cdot 196 \times 64$
One key per patch

W_V

$V \cdot 196 \times 64$
One value per patch

k_i and v_i come from the same image row h_i .

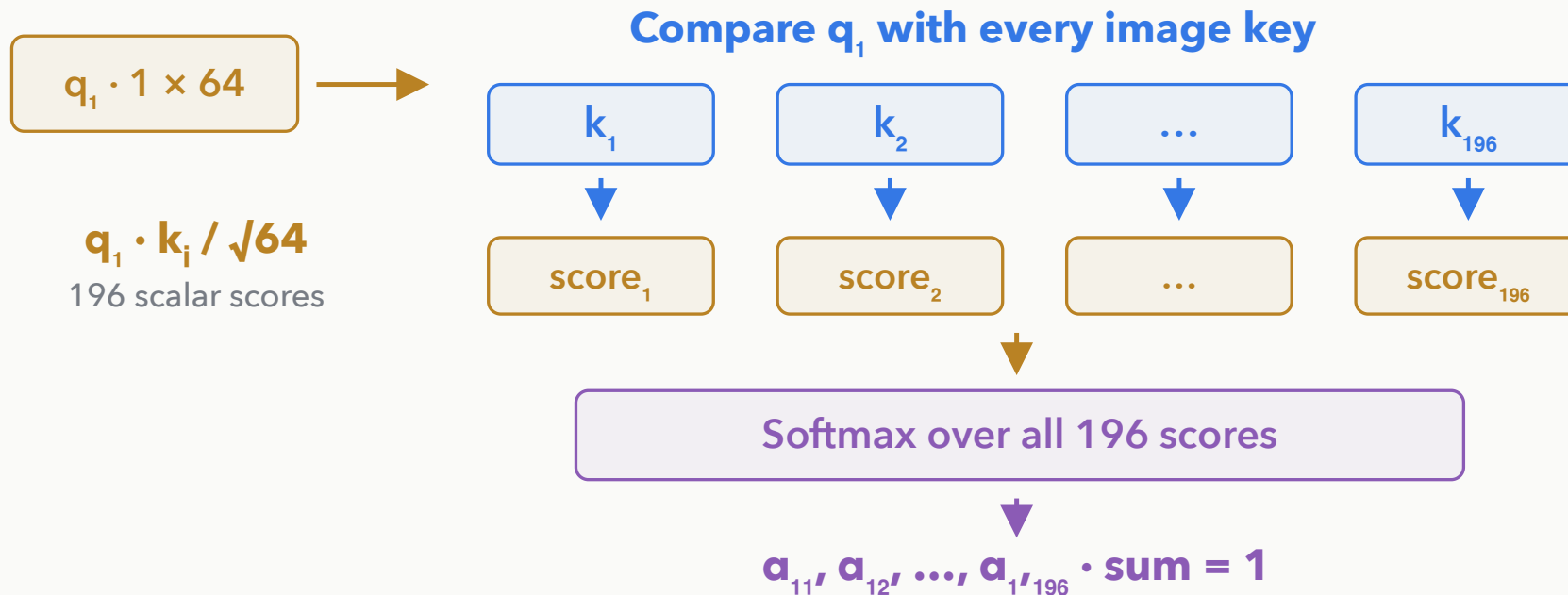
Next: follow q_1 , a 64-dimensional query, through this one head.

The question is added later. It is not an input to this minimal summarizer.

| Slots supply queries; the current image supplies all keys and values.

A soft lookup compares one query with every image key

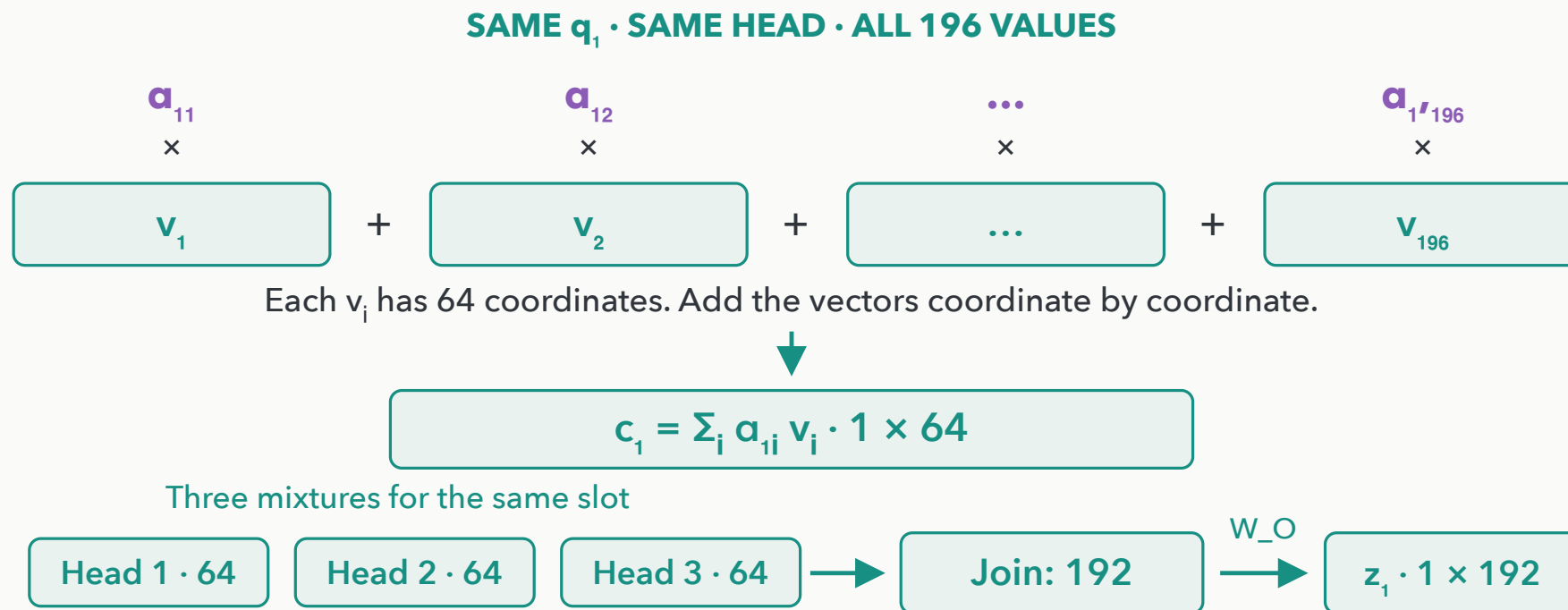
FOLLOW ONE SLOT · ONE HEAD



No vocabulary lookup and no top-1 choice. These weights will mix image values.

Compute 196 scores → softmax → 196 weights that sum to one.

Use the weights to mix image values

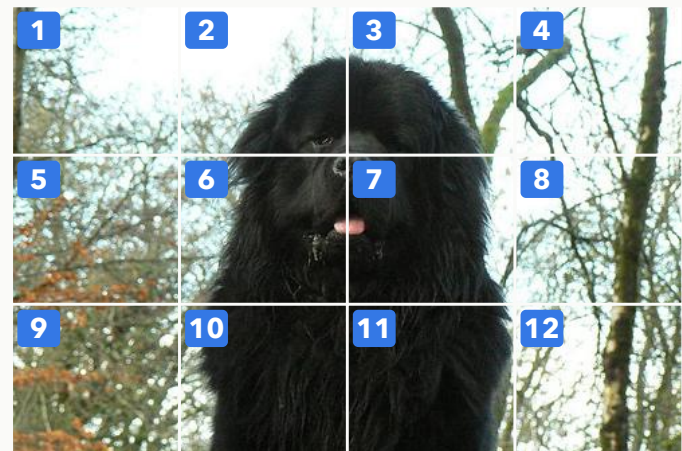


Repeat for all 8 slots → Z has 8 rows, each a 192-dimensional summary.

One slot produces one summary vector, after combining the three attention heads.

Try two numeric readers on the familiar image

TOY: 12 PATCHES → 2 SUMMARIES



Assume two projected reader queries:

$q_1 = [1, 0]$

$q_2 = [0, 3]$

Hand-set values, not photo measurements

Q: 2×2 K, V: $12 \times 2 \rightarrow$ two output vectors, each with 2 features

SAME IMAGE MEMORY AS BEFORE

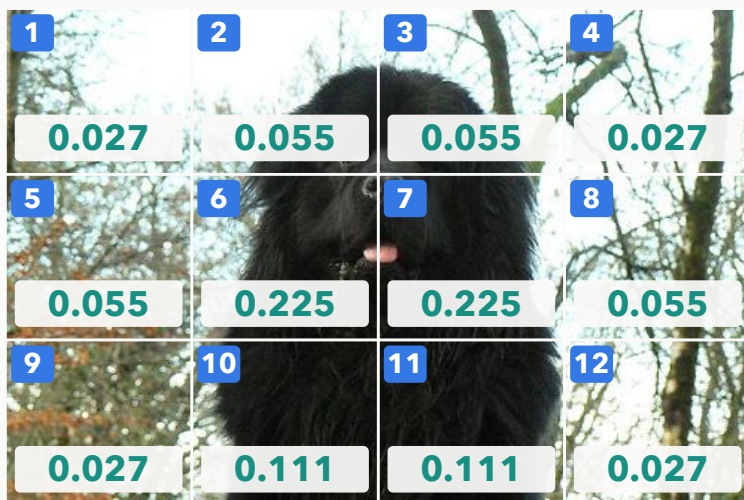
Patch i	Key k_i	Value v_i
1	[0, 3]	[3, 0]
2	[1, 2]	[3, 2]
3	[1, 2]	[3, 2]
4	[0, 3]	[3, 0]
5	[1, 1]	[2, 2]
6	[3, 0]	[3, 6]
7	[3, 1]	[4, 6]
8	[1, 2]	[3, 2]
9	[0, 1]	[1, 0]
10	[2, 0]	[2, 4]
11	[2, 0]	[2, 4]
12	[0, 2]	[2, 0]

For this small calculation, choose two readers; give both all twelve memory rows.

Two readers make two mixtures of the same image

READER 1 · $q_1 = [1, 0]$

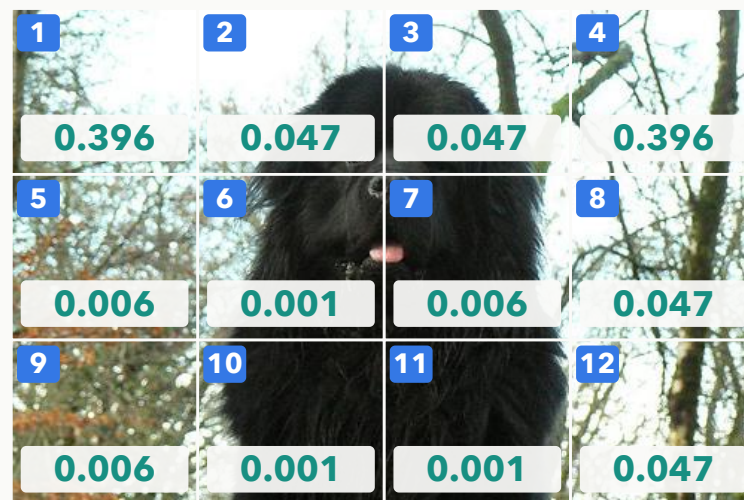
Match = 1 × the key's first coordinate



$$z_1 = \sum_i \alpha_{1i} v_i \approx [2.867, 4.031]$$

READER 2 · $q_2 = [0, 3]$

Match = 3 × the key's second coordinate



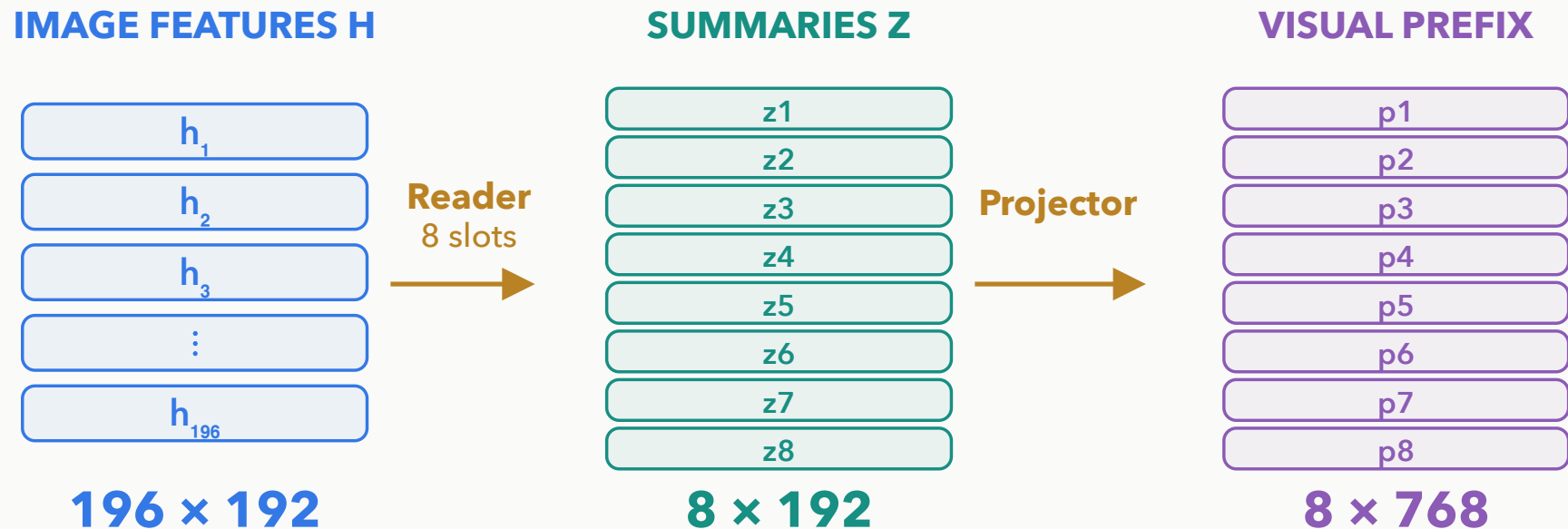
$$z_2 = \sum_i \alpha_{2i} v_i \approx [2.94, 0.34]$$

12 value rows → 2 weighted mixtures · each reader can use every patch

Toy head · $\alpha = \text{softmax}(qK^T / \sqrt{2})$ · output projection = I · not measured attention

A fixed number of readers gives a fixed number of summary vectors.

First reduce the count; then match the language width



Summarize: fewer vectors. Project: wider vectors, same count.

Add text embeddings only after this step. No English caption is created in the reader.

| $196 \times 192 \rightarrow 8 \times 192 \rightarrow 8 \times 768$. Count and width change in different steps.

The summarizer, step by step in Python

PyTorch · inside the summarizer

Shapes / comments

```
# Construct once; register these in your model
S = nn.Parameter(0.02 * torch.randn(1, 8, 192))
reader = nn.MultiheadAttention(192, 3, batch_first=True)
projector = nn.Linear(192, 768)

# Each forward call: H contains encoded patches
slots = S.expand(H.shape[0], -1, -1)
Z, weights = reader(slots, H, H,
                    average_attn_weights=False)

visual = projector(Z)
```

Setup

 $1 \times 8 \times 192$ 3 heads $\times$ 64 $192 \rightarrow 768$ H: $B \times 196 \times 192$ $B \times 8 \times 192$

Query, key, value inputs

Keep weights per head

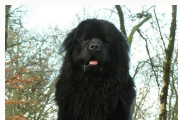
 $B \times 8 \times 768$

reader projects Q/K/V internally, matches keys, applies softmax, mixes values and joins heads.

Z: $B \times 8 \times 192$. **weights**: $B \times 3 \times 8 \times 196$. B is the batch size.

Initialize once; reuse the same reader on each new image.

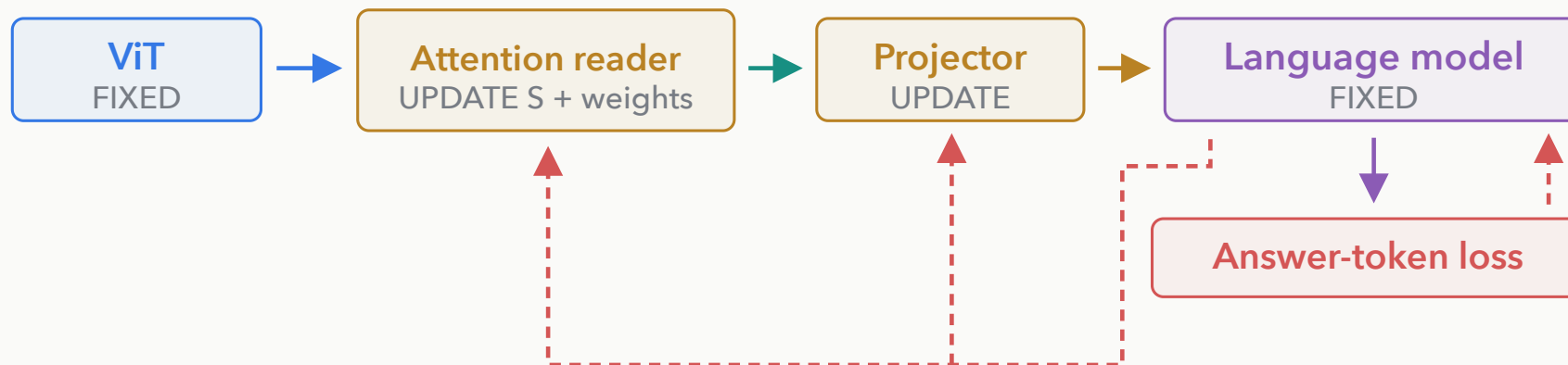
What teaches the slots to extract useful summaries?



Prompt: Describe this image.

Reference: A black dog outdoors.

ILLUSTRATIVE TRAINING RECIPE

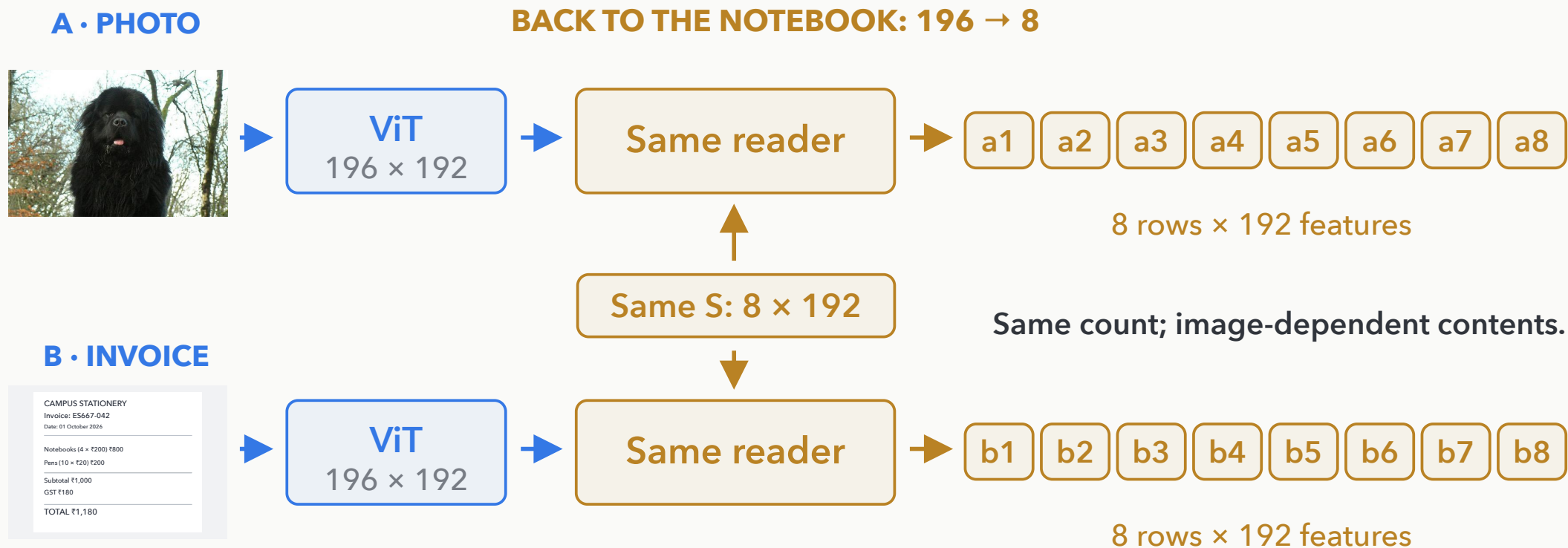


Gradients teach S and the reader which mixtures help predict the reference answer.

No target $z_1 \dots z_8$ is supplied. Repeat over many training examples; save the learned reader.

| The answer loss trains the slots and reader; we never supply a “correct summary vector”.

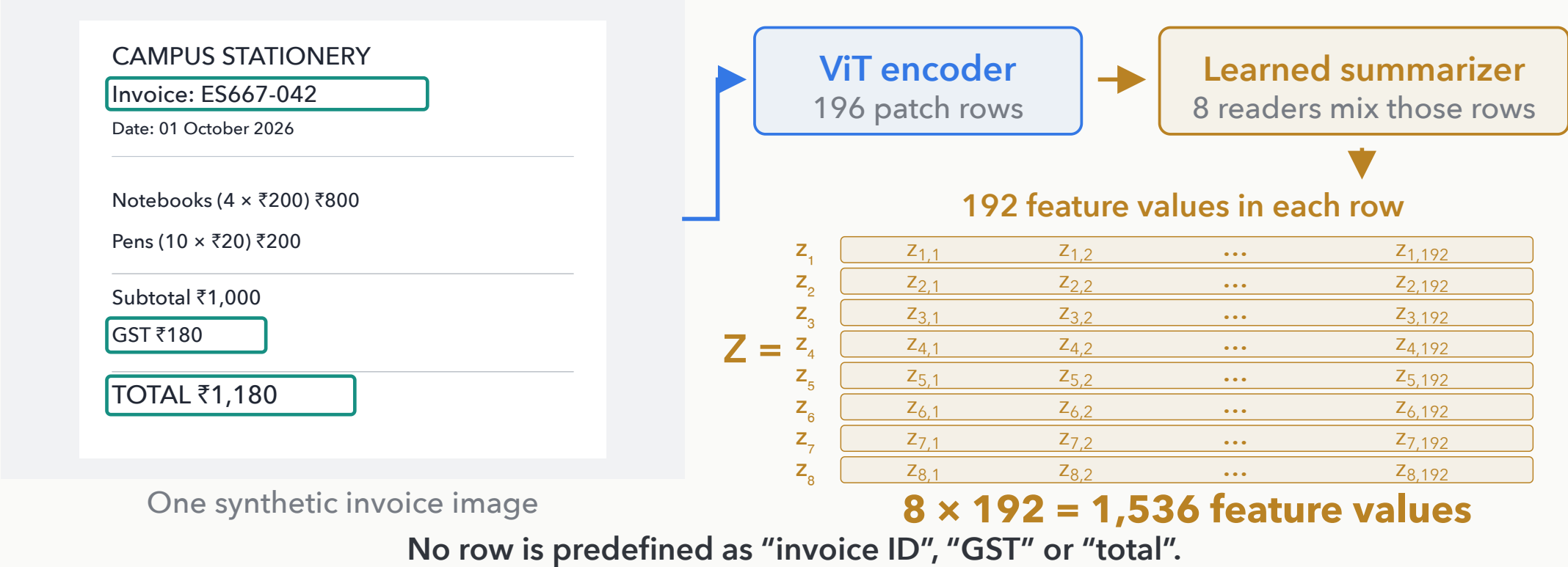
Same readers; a different image to summarize



Project 8 rows → add question + answer prefix → causal decoder → next token

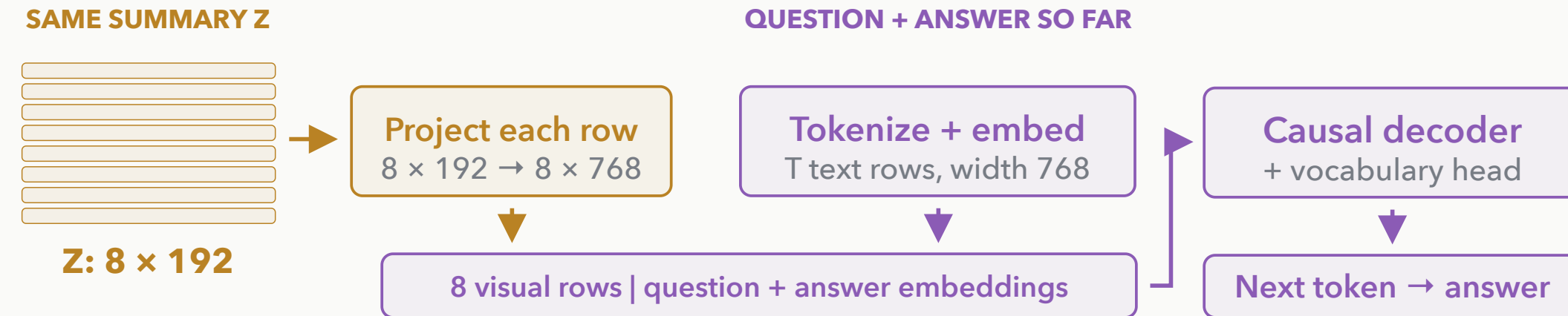
The summary length stays fixed; its feature values depend on the source image.

Eight vectors can represent many image details



A detail can be represented across several rows and feature coordinates.

Reuse the summary; change only the question



REUSE Z · START A FRESH ANSWER FOR EACH QUESTION

Reference answers we want after training

What is the invoice ID?

ES667-042

How much GST is charged?

₹180

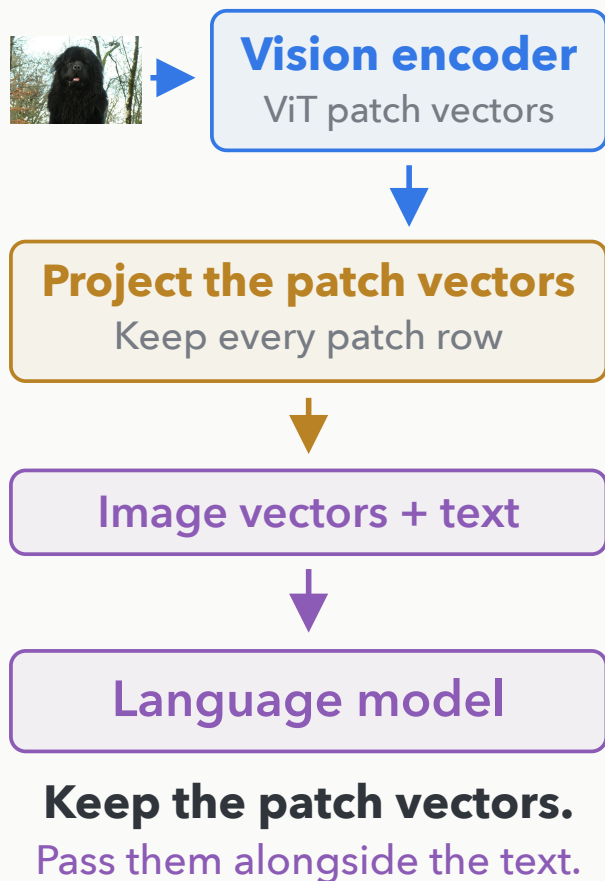
What is the total payable?

₹1,180

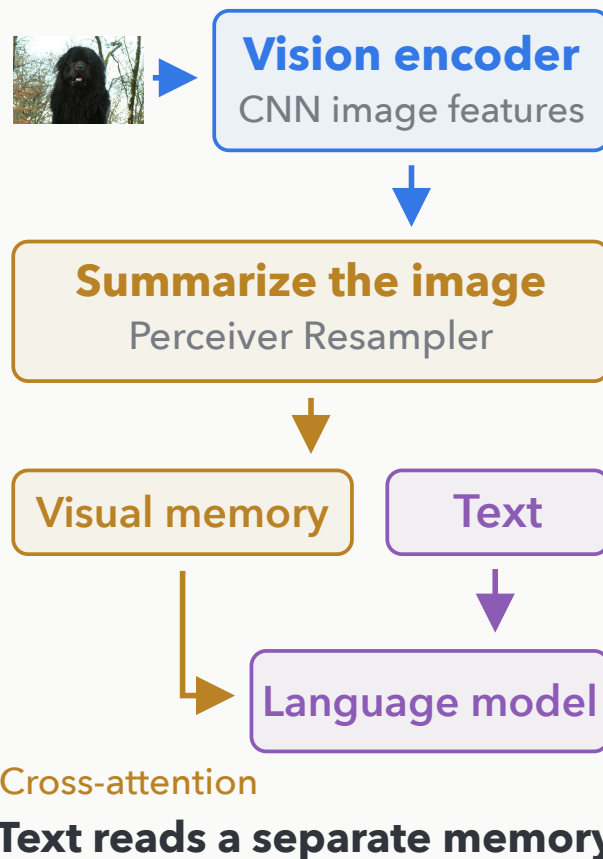
The summary is useful when it preserves evidence for the questions we need to answer.

How each model uses the image

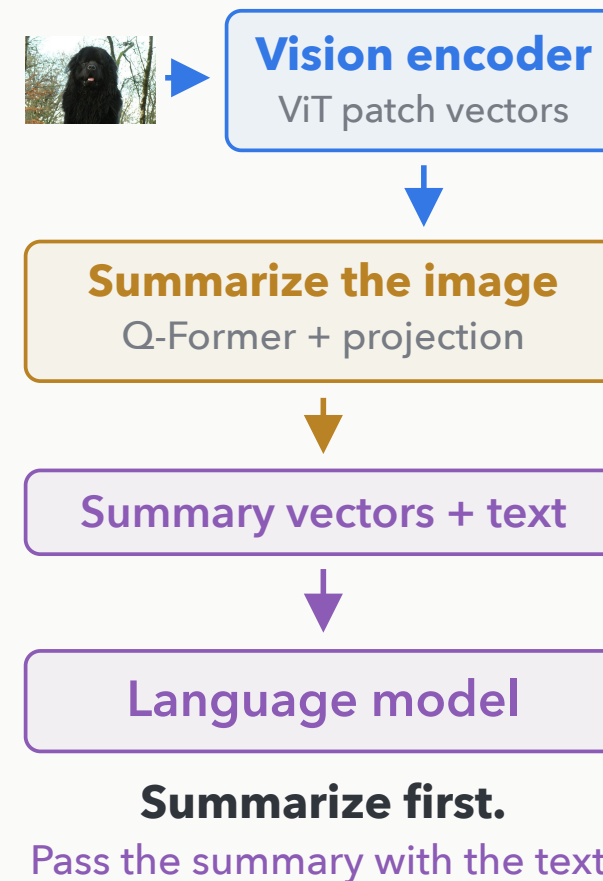
LLaVA



Flamingo

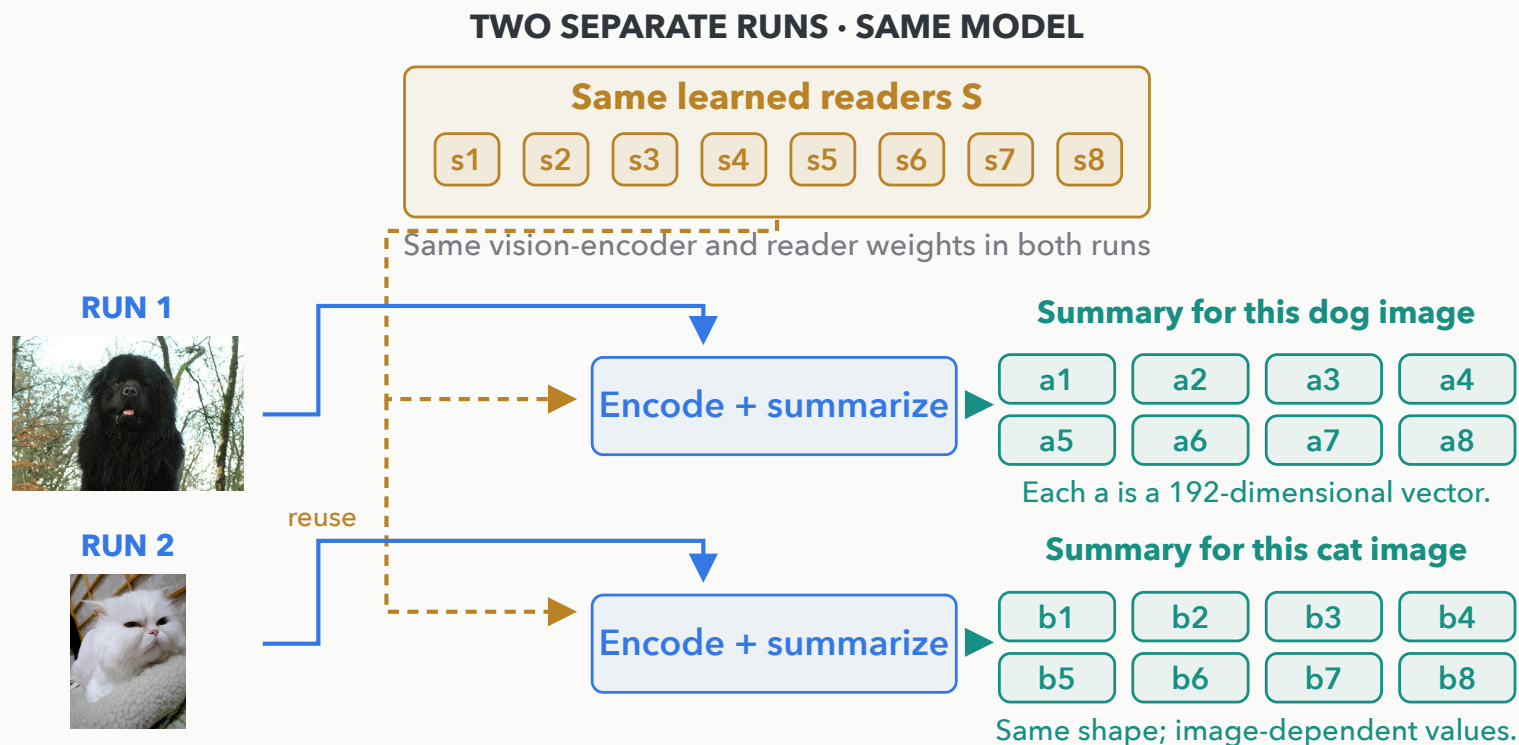


BLIP-2



Text = question + answer so far. BLIP-2 shows its OPT language-model route.

Same readers. New image, new summary.



Training can update S. Inference reuses S and computes new summaries.

The readers are stored parameters; each image produces its own summary vectors.

Idea 3, end to end in one function

**image**

One image

question = "What animal is shown?"**answer_so_far** = "It is a" (empty at the start)

One next-token step · high-level pseudocode

Shapes · batch omitted

<code>def next_token(image, question, answer_so_far):</code>	
<code> patches = vit(image).patch_states</code>	$N \times d_V$
<code> summary = summarizer(queries, patches)</code>	$M \times d_S; M < N$
<code> visual = projector(summary)</code>	$M \times d_{LM}$
<code> ids = tokenize(question, answer_so_far)</code>	T token IDs
<code> text = embed(ids)</code>	$T \times d_{LM}$
<code> inputs = concat([visual, text], axis="tokens")</code>	$(M + T) \times d_{LM}$
<code> hidden = causal_decoder(inputs)</code>	$(M + T) \times d_{LM}$
<code> return argmax(lm_head(hidden[-1]))</code>	Next text token ID

summarizer: M learned queries read all N patch rows through cross-attention; outputs depend on the image.

Shown: BLIP-2 / OPT-style prefix. For a Flamingo-style connection, use the summaries as Idea 2's memory.

SECTION 06

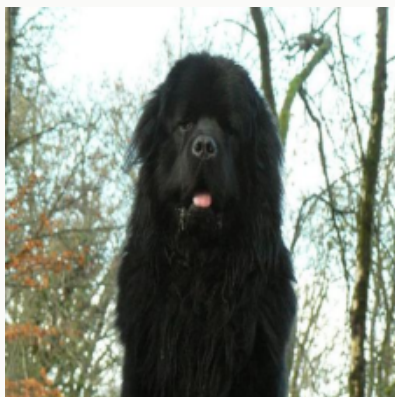
How do we train the connection?

Return to Idea 1's visual prefix and the dog question. Supply a correct answer, score its tokens, then update the projector.

Training starts with a correct answer supplied by us

TRAIN IDEA 1 · THE SAME IMAGE AND QUESTION

One authored reference answer · real token IDs



The input image

What animal is shown?

The input question

A black dog.

Written by the teacher

This is the reference answer.

Training targets

·A

ID 317

·black

ID 2042

·dog

ID 3290

.

ID 13

EOS

ID 50256

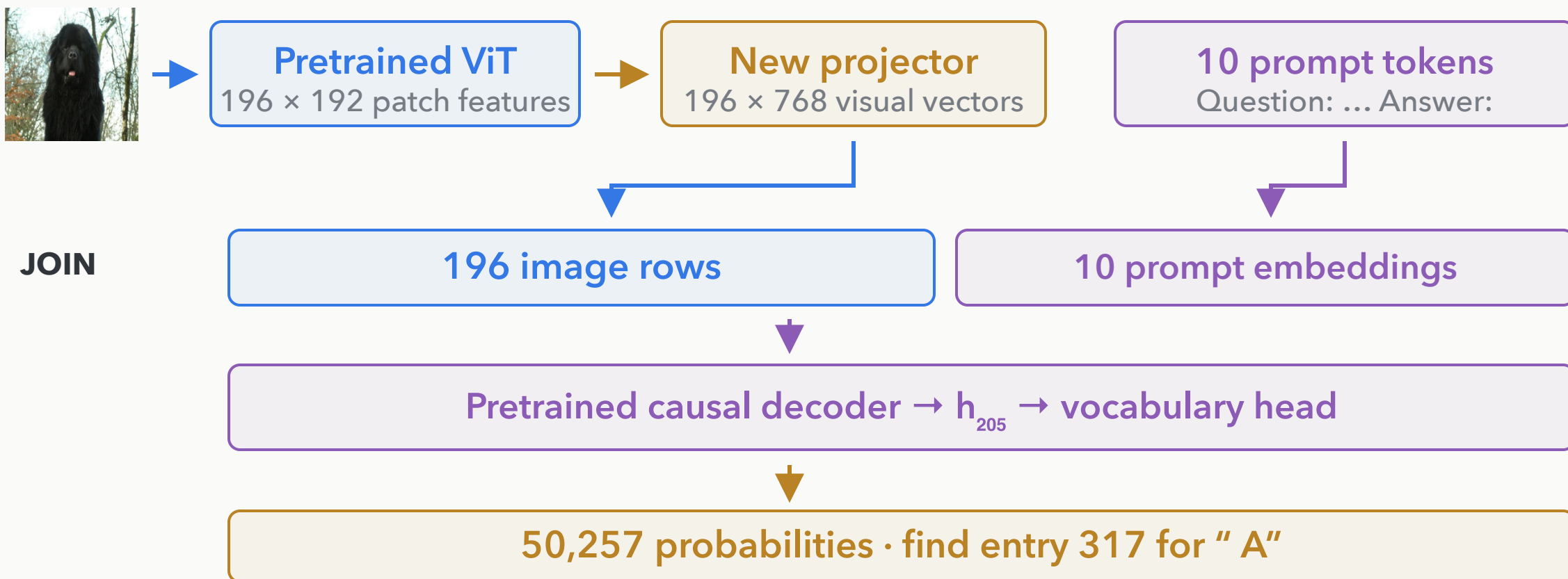
· marks a leading space. EOS says the answer is finished.

The dataset supplies the answer we want the model to learn.

Run the same model; inspect the first answer prediction

TRAIN IDEA 1 · THE SAME IMAGE AND QUESTION

One authored reference answer · real token IDs



The reference ID tells us which vocabulary probability to inspect.

How much probability did the correct token receive?

TRAIN IDEA 1 · THE SAME IMAGE AND QUESTION

Illustrative probabilities · not measured

Reference target: " A"

From the last "Answer:" state

softmax over all 50,257 vocabulary scores

Select $p[317] = 0.20$

All entries sum to 1.

The target stays " A".



Loss for this target = $-\ln(0.20) = 1.609$

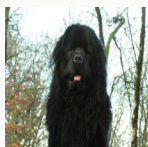
If $p[317]$ were 0.80, its loss would be only $-\ln(0.80) = 0.223$.

Low probability on the correct token gives a large loss.

Which token do we feed back next?

TRAIN IDEA 1 · THE SAME IMAGE AND QUESTION

Illustrative probabilities · not measured



Question: What animal is shown?

Current text ends with: Answer:

$$p(\text{" A"}) = 0.20 \cdot p(\text{" The"}) = 0.30 \cdot \text{suppose " The" is highest}$$

TRAINING · TEACHER FORCING

Reference: "A black dog."

Score A using $-\ln(0.20)$.



Answer:

A

Reference A → predict target black

INFERENCE · GREEDY DECODING

Choose "The": highest probability.

No reference answer is supplied.



Answer:

The

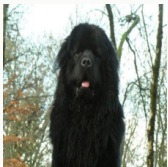
Chosen The → predict the next token

Inference may sample instead of taking the maximum; it still uses its own chosen token.

Training uses the reference token; inference uses the model's selected token.

Now line up all five next-token targets

TRAIN IDEA 1 · THE SAME IMAGE AND QUESTION



Question: What animal is shown?

Reference: A black dog.

One authored reference answer · real token IDs

Same image + question in every row

VISIBLE TEXT ENDS WITH

Answer:
Answer: ·A
Answer: ·A ·black
Answer: ·A ·black ·dog
Answer: ·A ·black ·dog .



NEXT TARGET

·A
·black
·dog
.
EOS

READ POSITION

205

206

207

208

209

All prefixes are known → one causal pass computes all five predictions.

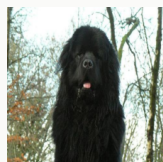
Each predicting state is blocked from its target and later tokens. · = leading space.

Each position predicts the next reference token using only its allowed prefix.

Turn five target probabilities into one answer loss

TRAIN IDEA 1 · THE SAME IMAGE AND QUESTION

Illustrative probabilities · not measured



Question: What animal is shown?

Reference: A black dog.

Reference target	.A	.black	.dog	.	EOS
p(correct target)	0.20	0.40	0.50	0.80	0.90
Loss = $-\ln p$	1.609	0.916	0.693	0.223	0.105

For A: $-\ln(0.20) = 1.609$

Total answer loss = 3.547

Mean answer loss = total ÷ 5 = 0.709

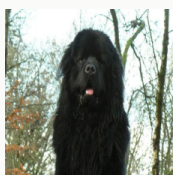
Sum before rounding. Image and question targets are excluded from this average.

Score each correct token, then average the five penalties, including EOS.

One input sequence; five answer targets to score

TRAIN IDEA 1 · THE SAME IMAGE AND QUESTION

One authored reference answer · real token IDs



196 image rows + 10 prompt tokens + 5 reference tokens = 211 positions

Each decoder input row has 768 features. Batch size $B = 1$.

196 image vectors

10 prompt tokens

·A

·black

·dog

.

EOS

Target labels at the same positions:

–100 means “do not score this target”.

–100 at all 206 context positions

317

2042

3290

13

50256

logits at positions 205-209 → labels at positions 206-210

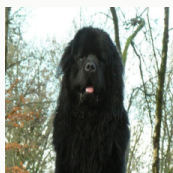
The final prompt “:” predicts the first “A”. The final EOS logit is unused.

Ignore context labels; still use the image and question as context.

Knowing the answer does not let the model peek ahead

TRAIN IDEA 1 · THE SAME IMAGE AND QUESTION

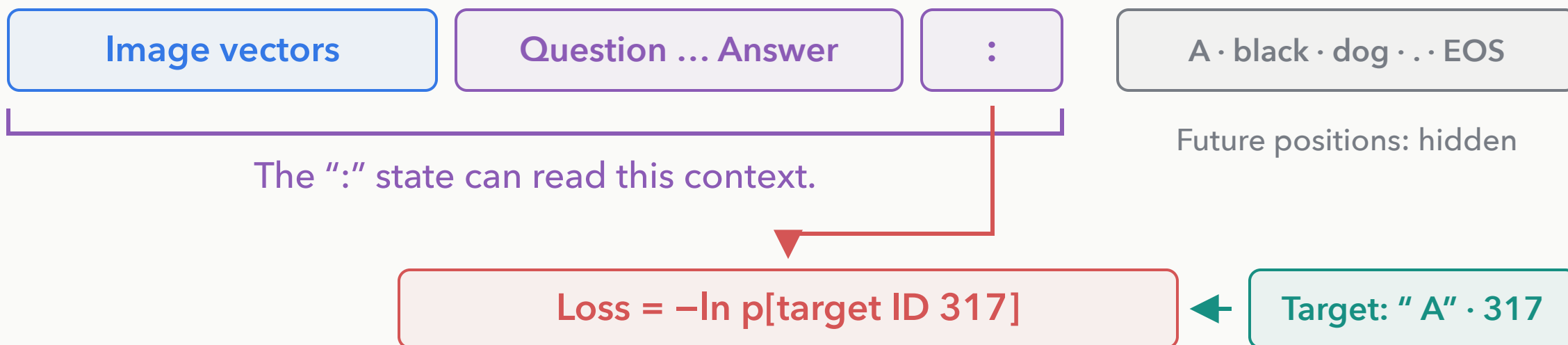
One authored reference answer · real token IDs



Question: What animal is shown?

Answer: empty

Predict "A" at the last prompt ":"



We grade answer targets only. The question remains visible to attention.

Attention controls what a state reads. Loss labels select what we grade.

Use the answer loss to change the projector

TRAIN IDEA 1 · THE SAME IMAGE AND QUESTION

One authored reference answer · real token IDs

FORWARD



Also receives prompt + reference embeddings.

BACKWARD



The LM passes the gradient back; only the projector changes.

$$\text{Projector weights} \leftarrow \text{weights} - \text{learning rate} \times \text{gradient}$$

Frozen LM weights still pass a gradient to the trainable visual connection.

Training adds targets, a loss and a weight update

TRAIN IDEA 1 · ADD SUPERVISION TO THE SAME FORWARD PASS

X = 196 image vectors 10 question / prompt rows 5 reference rows: A black dog . EOS

Prepared X: [1, 211, 768] · answer_ids: [1, 5] · one example, no padding.

```
# 1. Ignore context targets (-100); score the reference answer.
labels = torch.full(X.shape[:2], -100, device=X.device)
labels[:, -answer_ids.shape[1]:] = answer_ids

# 2. Predict next tokens and average the five answer losses.
loss = lm(inputs_embeds=X, labels=labels).loss

# 3. Differentiate through the frozen LM; change the projector.
optimizer.zero_grad() # clear previous gradients
loss.backward()        # compute projector gradients
optimizer.step()       # update projector weights
```

Setup: frozen vit + lm; optimizer owns only projector parameters. Keep LM autograd on.

The reference supplies the targets; only the projector weights change.

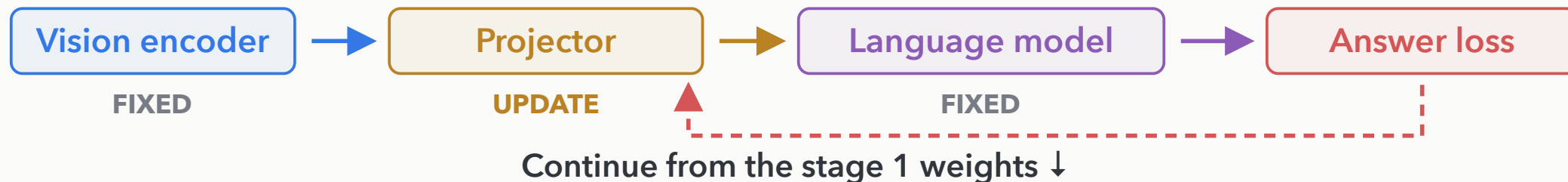
Both stages train on supplied reference answers

1 · TRAIN ON IMAGE-CAPTION PAIRS



Describe this image.

Supplied target: A black dog outdoors.

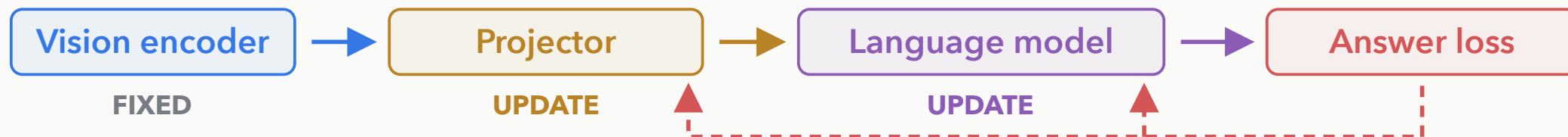


2 · TRAIN ON IMAGE + QUESTION + ANSWER



What colour is the dog?

Supplied target: Black.



The dataset supplies captions first, then question-answer examples.

Predict each next reference token → measure its loss → update the marked weights.

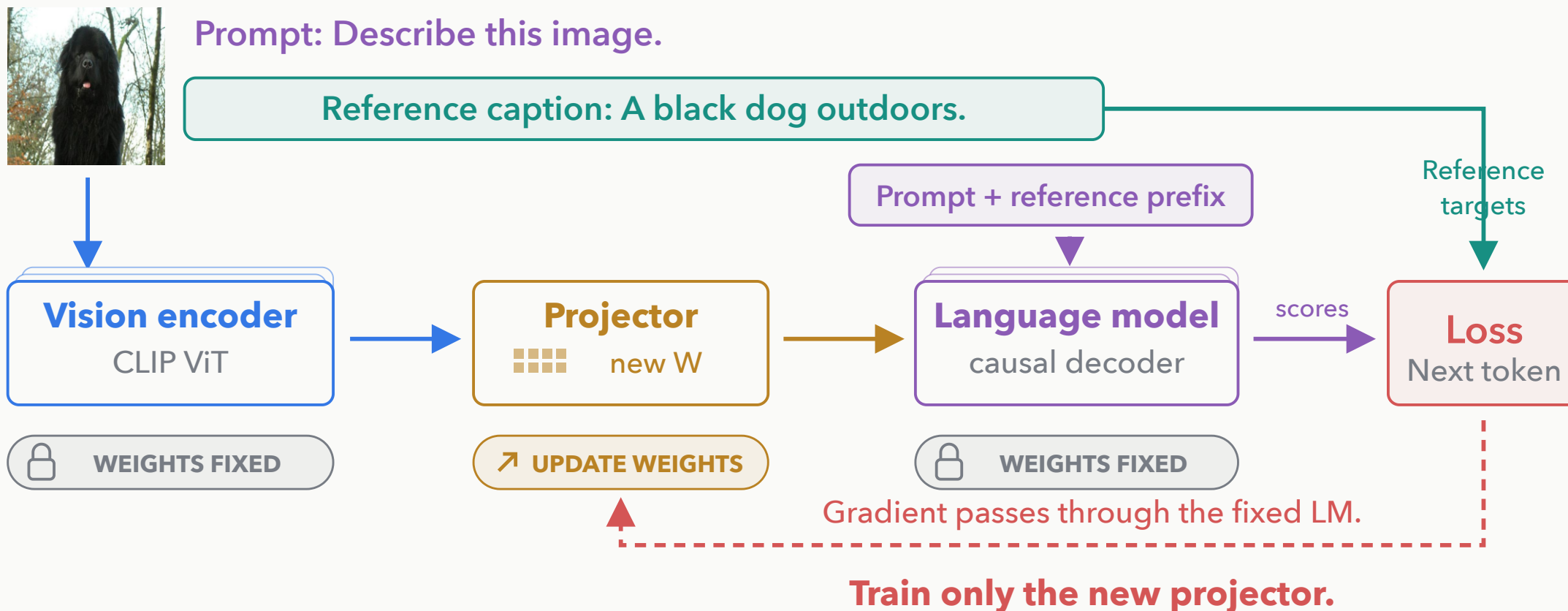
At inference: supply a new image + question; the model generates its own answer.

Captions train the visual connection; varied instructions train the model to answer questions.

Stage 1: learn the visual connection from captions

ORIGINAL LLaVA • STAGE 1 OF 2

Authored example • not a model output



Caption targets train the projector; vision and language weights stay fixed.

Keep the learned connection; change the training examples

ORIGINAL LLaVA · BETWEEN THE TWO STAGES

Authored examples · supplied references



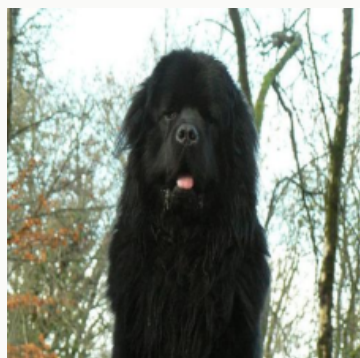
Stage 1 checkpoint

Learned projector + retained vision and LM



Start stage 2 from these weights

Continue with the same model.



Same image

CAPTION EXAMPLE

Describe this image.

SUPPLIED REFERENCE

A black dog outdoors.

INSTRUCTION EXAMPLES

What colour is the dog?

Black.

Where is the dog?

Outdoors.

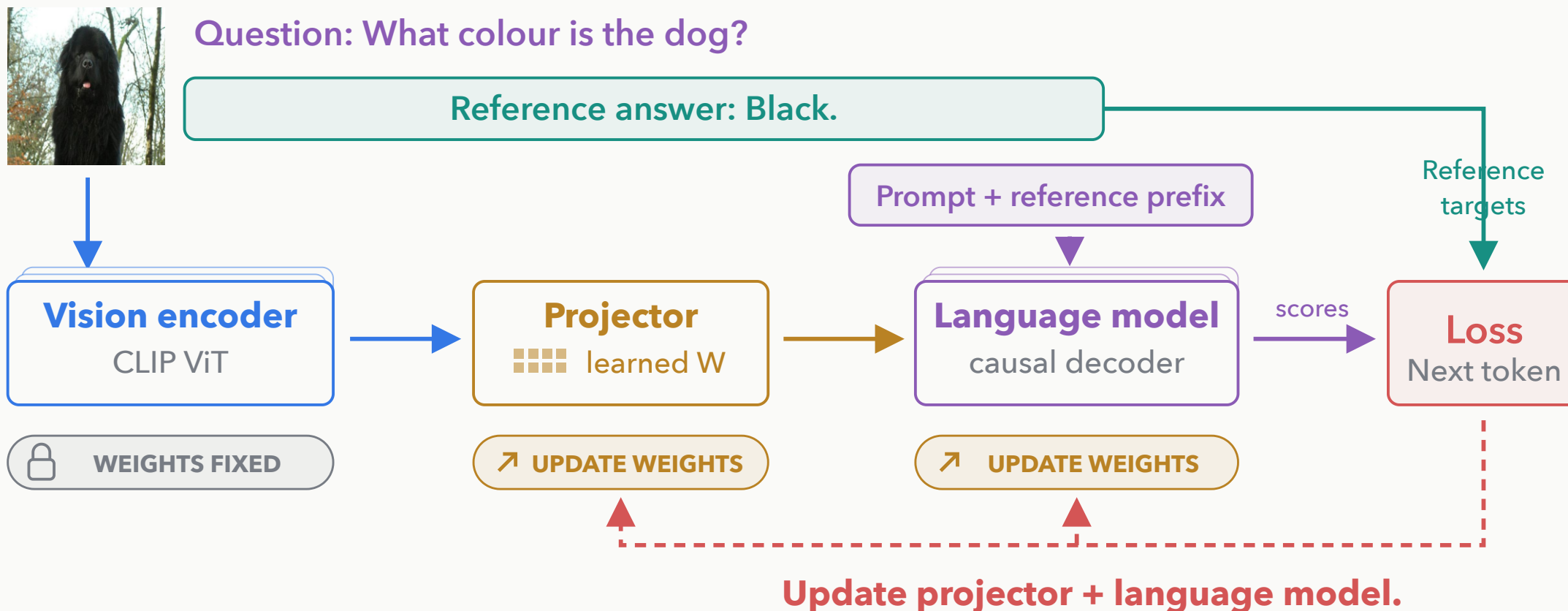
Different questions → different reference answers.

Stage 2 reuses the trained model and broadens the questions it learns to answer.

Stage 2: learn to answer visual instructions

ORIGINAL LLaVA · STAGE 2 OF 2

Authored example · not a model output



The next-token loss stays; the projector and language model can now change.

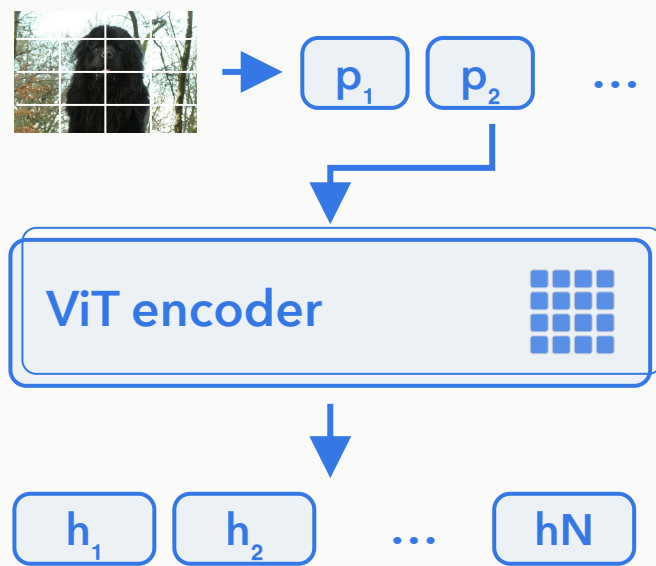
SECTION 07

Summary

Rebuild the whole picture: represent the image, connect it to language, and predict the next word.

ViT represents. CLIP compares. A VLM generates.

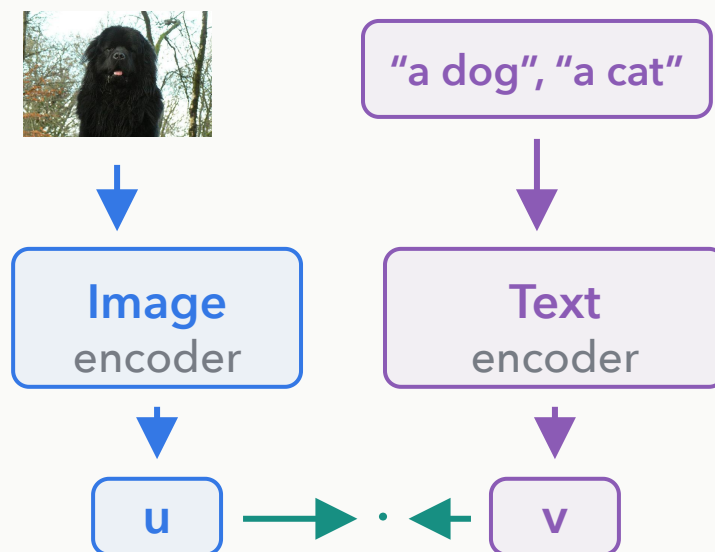
ViT · REPRESENT



One vector per patch

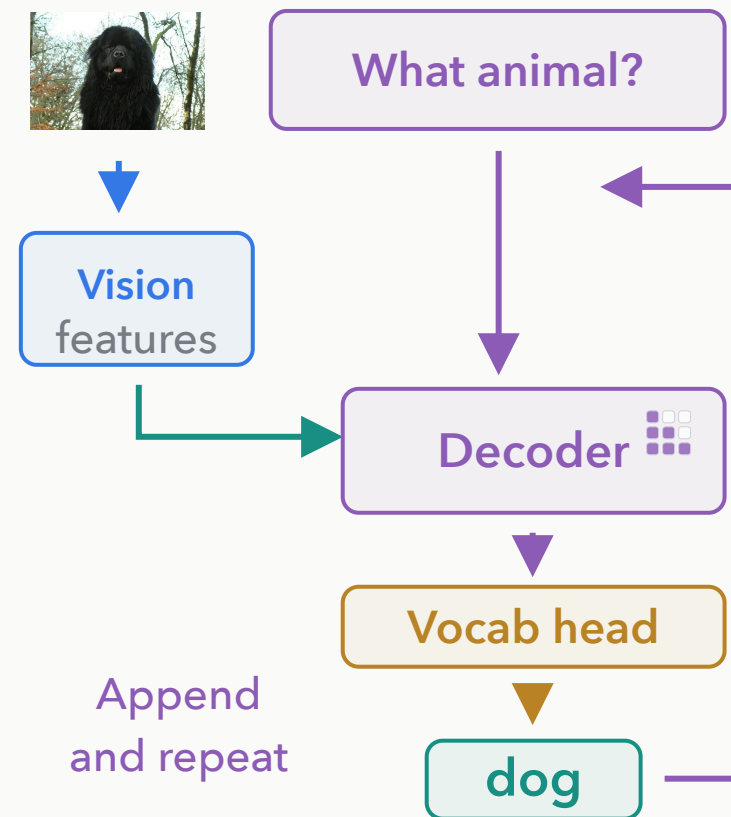
Full attention mixes image context.

CLIP · COMPARE



Best supplied description
"a dog"

GENERATIVE VLM · WRITE

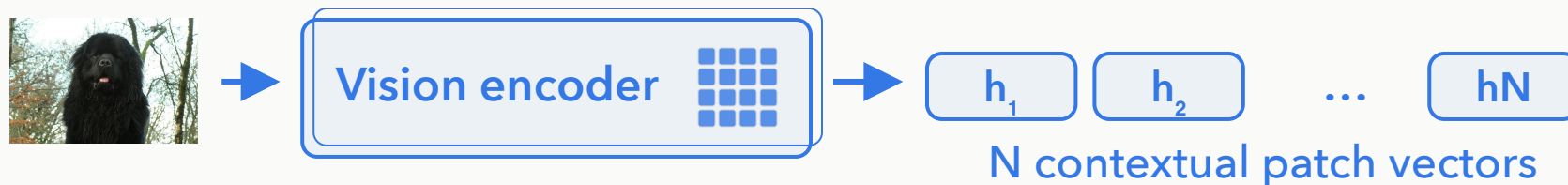


Append
and repeat

Illustrative match and generated word; these are model portraits, not measured outputs.

A generative VLM adds a language decoder and a path from the image to that decoder.

Choose how image features reach the decoder



1 · PREPEND

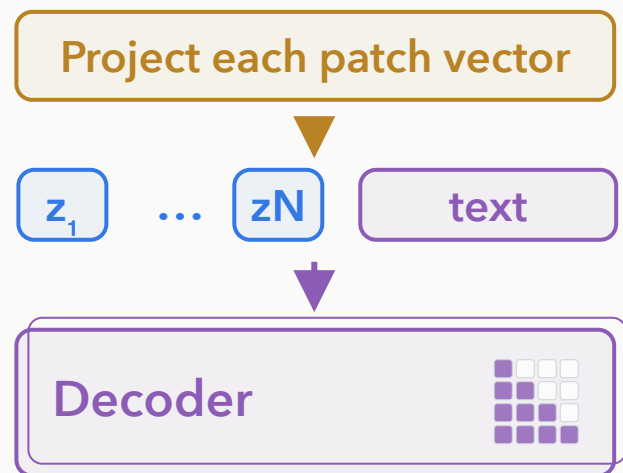
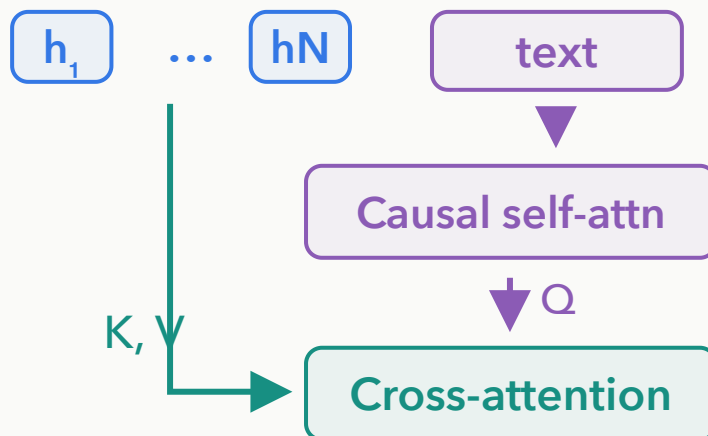


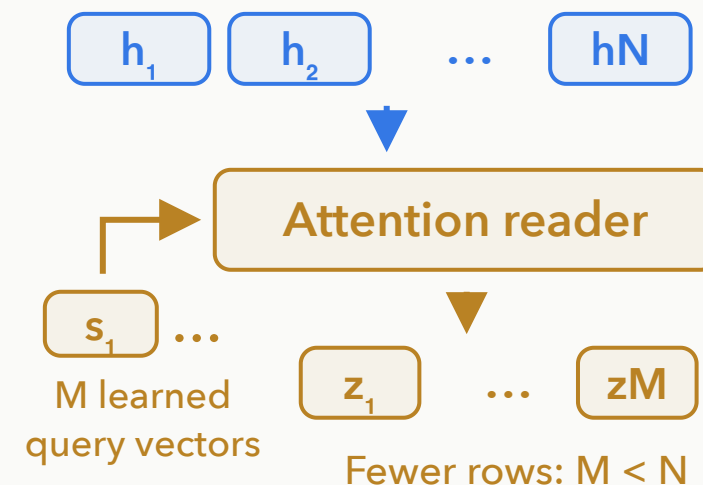
Image + text in one sequence

2 · READ MEMORY



Text reads the image features

3 · SUMMARIZE

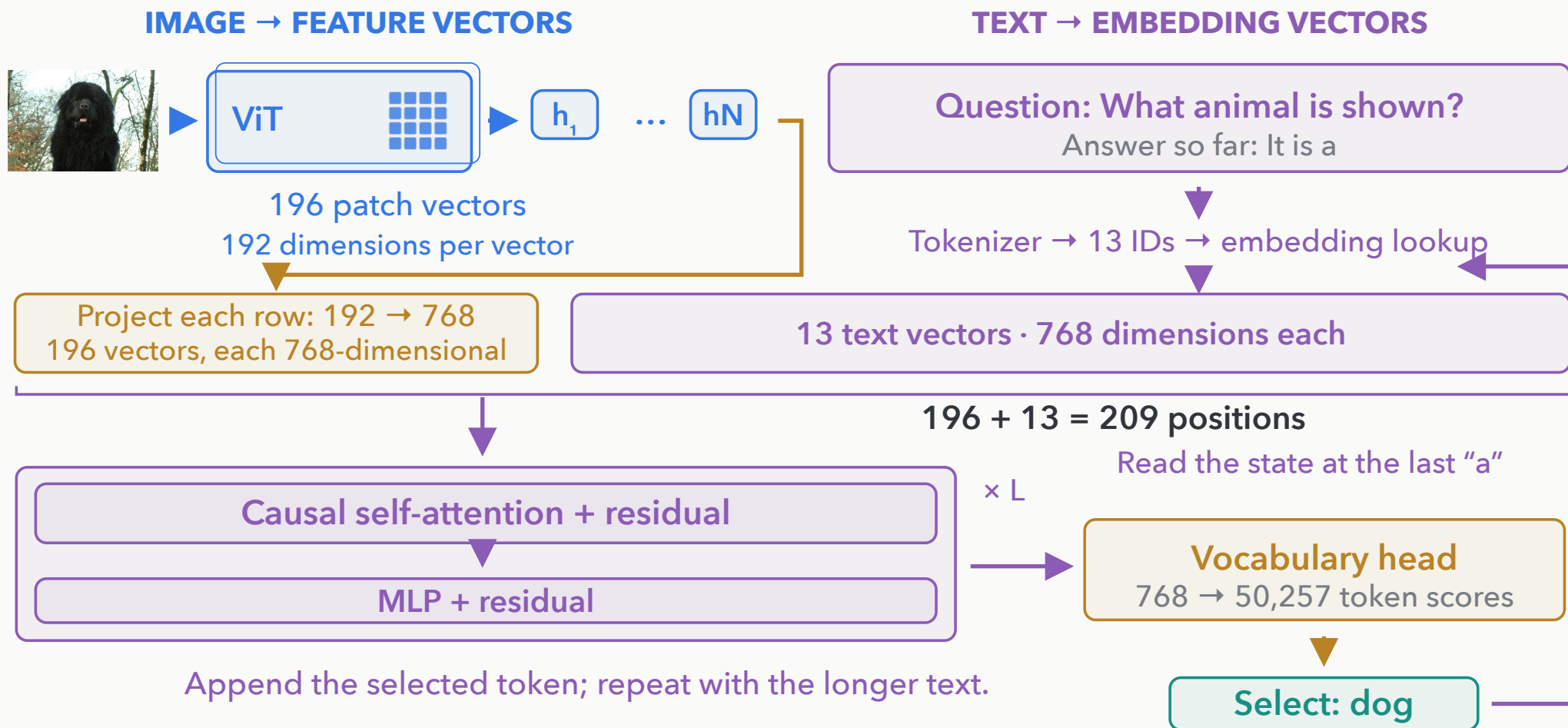


Use M summaries with 1 or 2

Projection changes vector width. Summarization changes row count. Access chooses the path.

Project width. Choose the access path. Optionally shorten the visual sequence.

From this image to the next word



DeiT-Tiny + DistilGPT2 dimensions · dog is an illustrative choice, not a measured prediction.

The image supplies context; the language vocabulary supplies the next token.

Backup and optional reading

The main lecture ends here. Open a topic when needed.

A

Multiple images

Encode, summarize and compare a before/after pair.



B

Inference refresher

Prefill, decoding, KV caching and conversation context.



C

Evaluation

Image evidence, diagnostic examples and metrics.



D

Model details

Flamingo, BLIP-2 and further training choices.



E

Sources

Model papers and evaluation references.



BACKUP A • MULTIPLE IMAGES

Optional • compare multiple images

What changed from before to after?

Follow two images all the way to one answer.

One question, two ordered images

IMAGE A · BEFORE

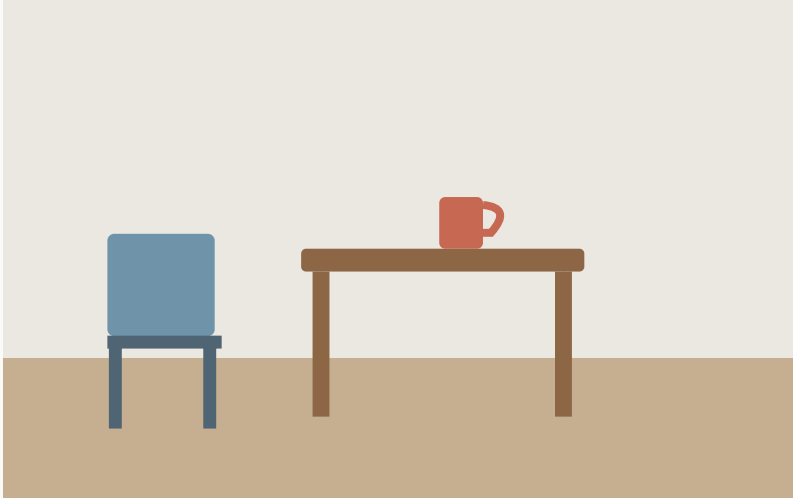
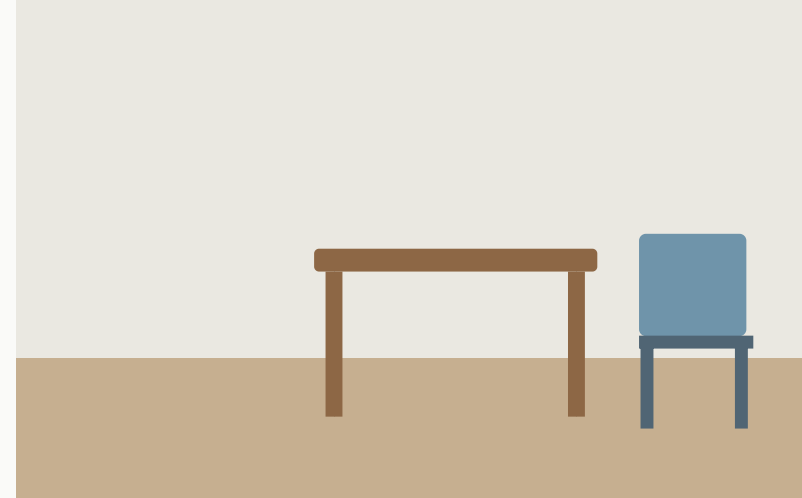


IMAGE B · AFTER



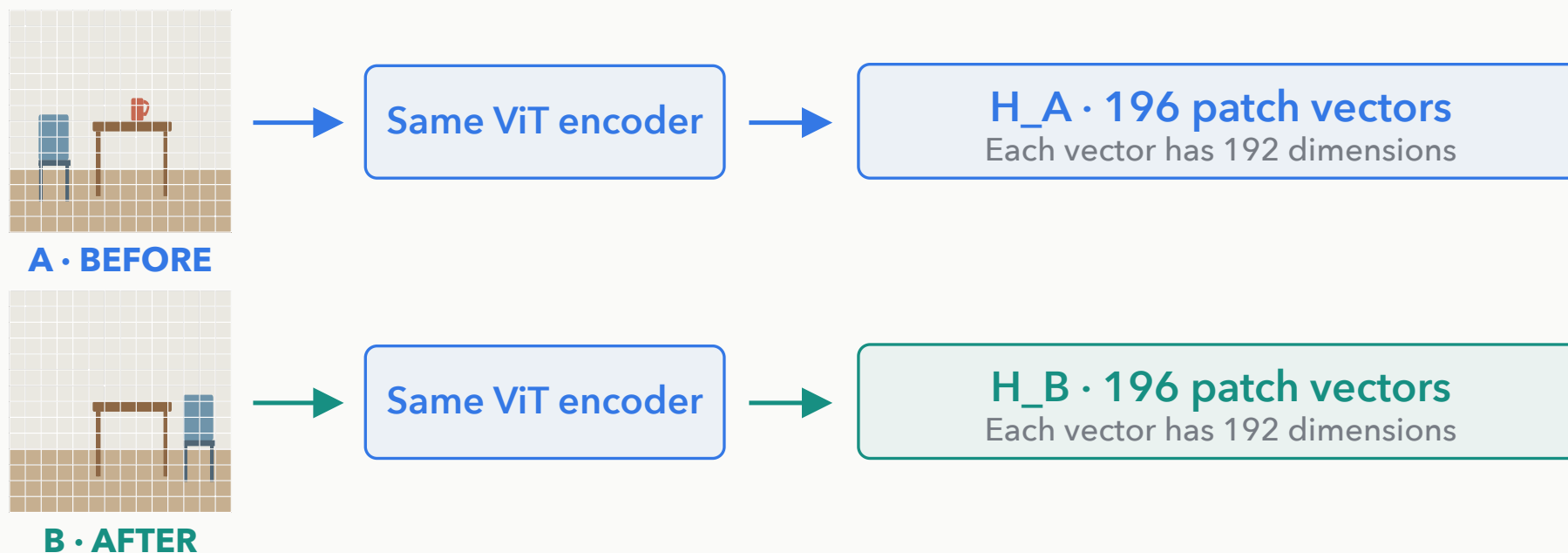
What changed from before to after?

Two image inputs + one question → one answer

| Keep "before" and "after" attached to the correct image.

Encode each image with the same vision model

Teaching setup · 224×224 image · 16×16 pixels per patch

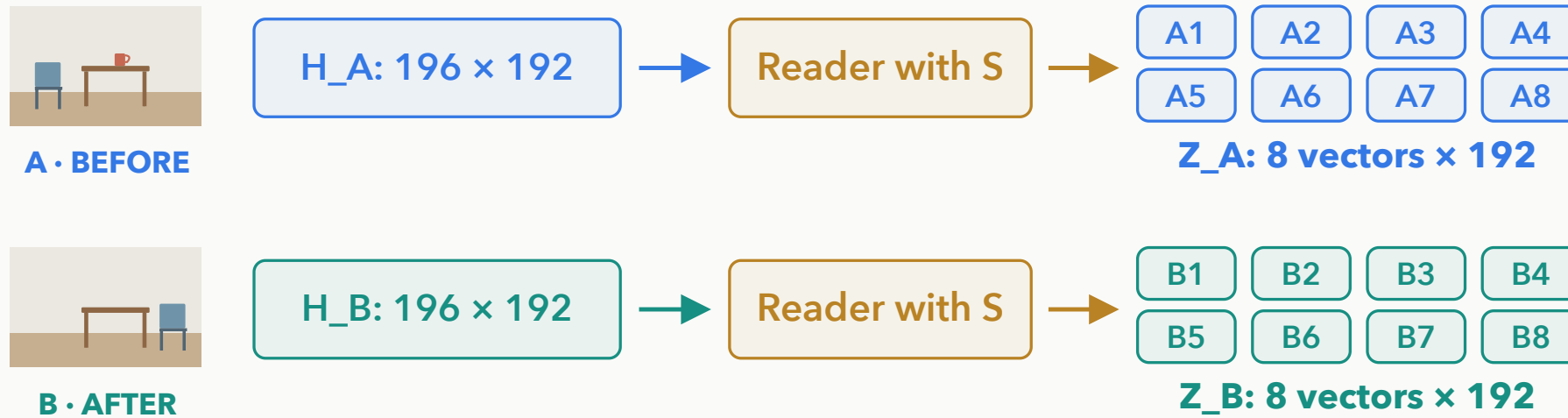


$14 \times 14 = 196$ patches per image · 392 patch vectors across the pair

Keep the two feature groups identifiable from the start.

Summarize A and B separately

Same 8 learned readers S · two separate reads



8 summaries from A + 8 summaries from B = 16 visual vectors

This reader sees one image at a time. The decoder will compare the groups.

| The same readers produce two image-dependent groups.

Text labels and image vectors share one input sequence

TEXT LABELS: "Before:" AND "After:"

Tokenizer → token IDs → embedding table

One 768-dimensional vector per text token

VISUAL SUMMARIES: A1...A8, B1...B8

Project each vector: 192 → 768 dimensions

Still 8 from A + 8 from B · no vocabulary lookup

ONE ORDERED DECODER INPUT



Causal language decoder → next token

Purple boxes show text spans; each may contain several tokenizer pieces.

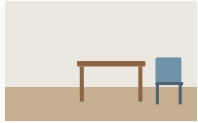
Next: build this same ordered sequence in code.

Before and After are ordinary text; A and B vectors come from the images.

Build the two-image input, line by line



A · before



B · after

question = "What changed from before to after?"

answer_so_far = "The chair moved to the"

Pseudocode · the same ordered input as the preceding diagram

Shapes / comments · batch omitted

```
# summary_A and summary_B are already computed
```

Each: 8×192

```
A = projector(summary_A)
```

8×768

```
B = projector(summary_B)
```

8×768

```
before = embed(tokenize("Before:"))
```

Ordinary text $\rightarrow$ vectors

```
after = embed(tokenize("After:"))
```

Ordinary text $\rightarrow$ vectors

```
text = embed(tokenize(question, answer_so_far))
```

One vector per text token

```
parts = [before, A, after, B, text]
```

Same order as the diagram

```
inputs = concat(parts, axis="tokens")
```

$(16 + T) \times 768$

```
hidden = causal_decoder(inputs)
```

$(16 + T) \times 768$

```
next_id = argmax(lm_head(hidden[-1]))
```

Next text token ID

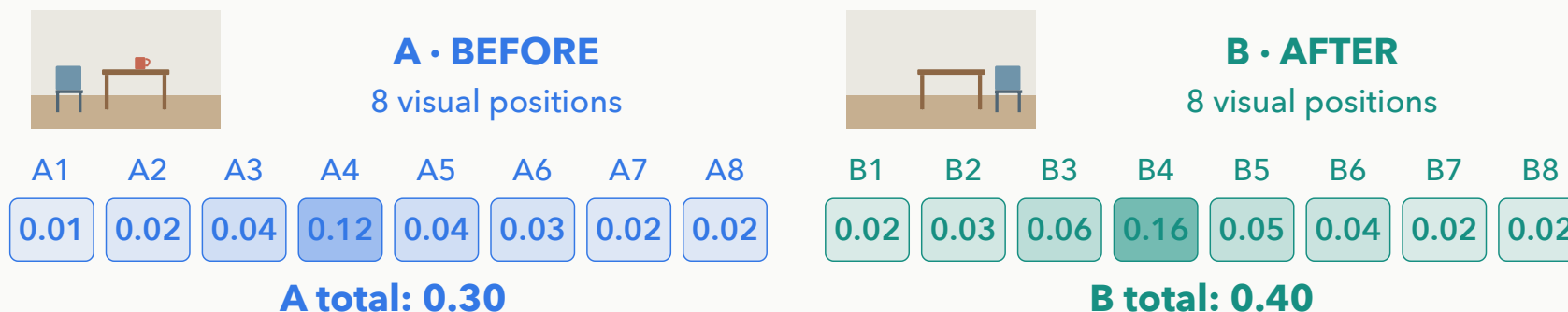
T counts tokens in both labels, the question and the answer so far. Every input vector is 768-dimensional.

The last text state can read both image groups

Answer so far: "The chair moved to the"

State at the last "the" → this head's query q

One softmax over all visible image AND text keys



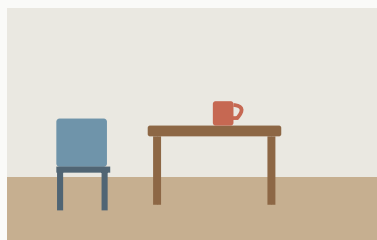
Text positions together: 0.30

$0.30 + 0.40 + 0.30 = 1$ · each weight multiplies its paired value vector

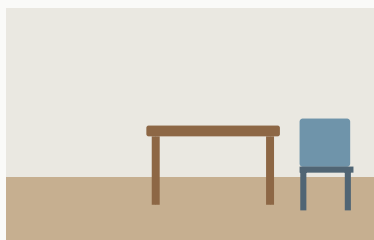
Illustrative weights only · every head and text position can use different weights.

Attention weights are assigned to positions across the permitted context.

Two images, one ordered context, one answer



A · BEFORE



B · AFTER

What changed from
before to after?

1 · Prepare each image
Encode → summarize



2 · Keep their order
A, then B + question



3 · Generate an answer
Use both image groups

"The chair moved to the right; the cup disappeared."

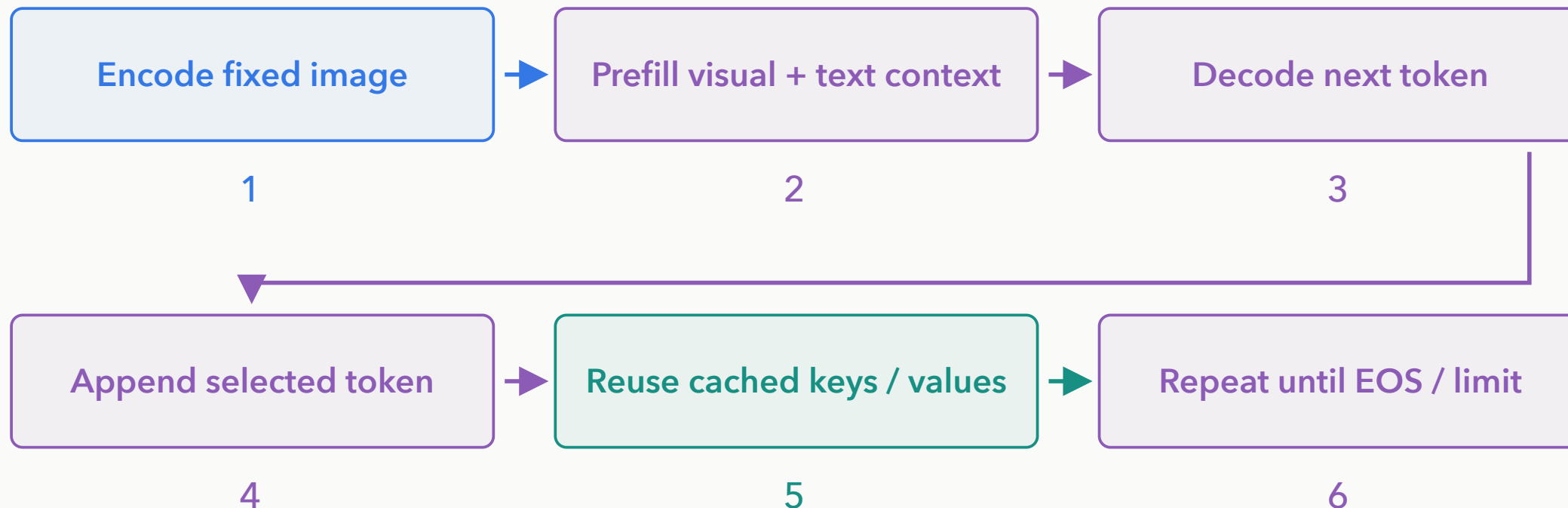
Authored reference answer for these drawings · not a measured model output

Prepare each image, preserve its order, then answer using both.

Optional • inference and caching

A refresher from our LLM lectures: prepare the context, decode tokens and reuse the cache.

The whole generation loop before its individual steps



Prepare the fixed context once; repeatedly predict and append a token.

Encode the fixed image once

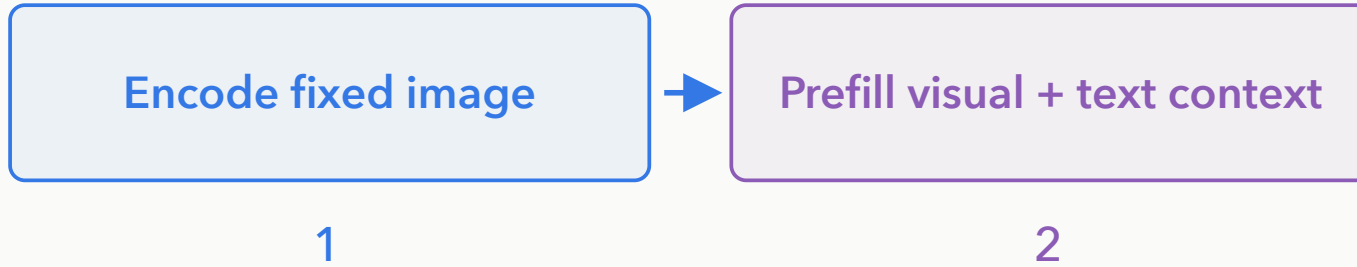


Encode fixed image

1

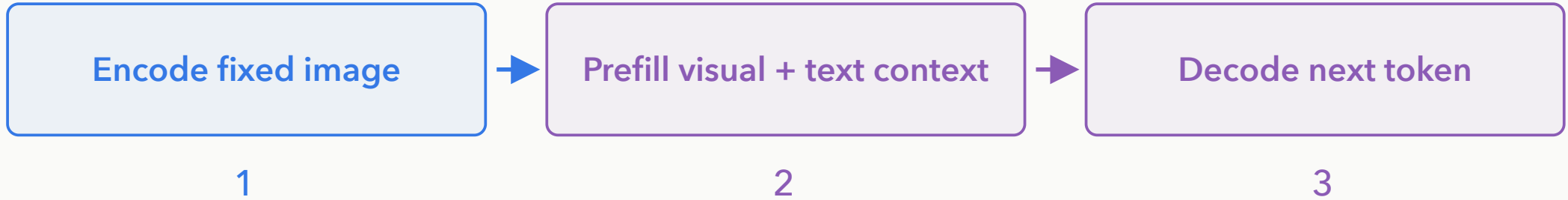
| Only the generated prefix changes at each decoding step.

Prefill the image and question context



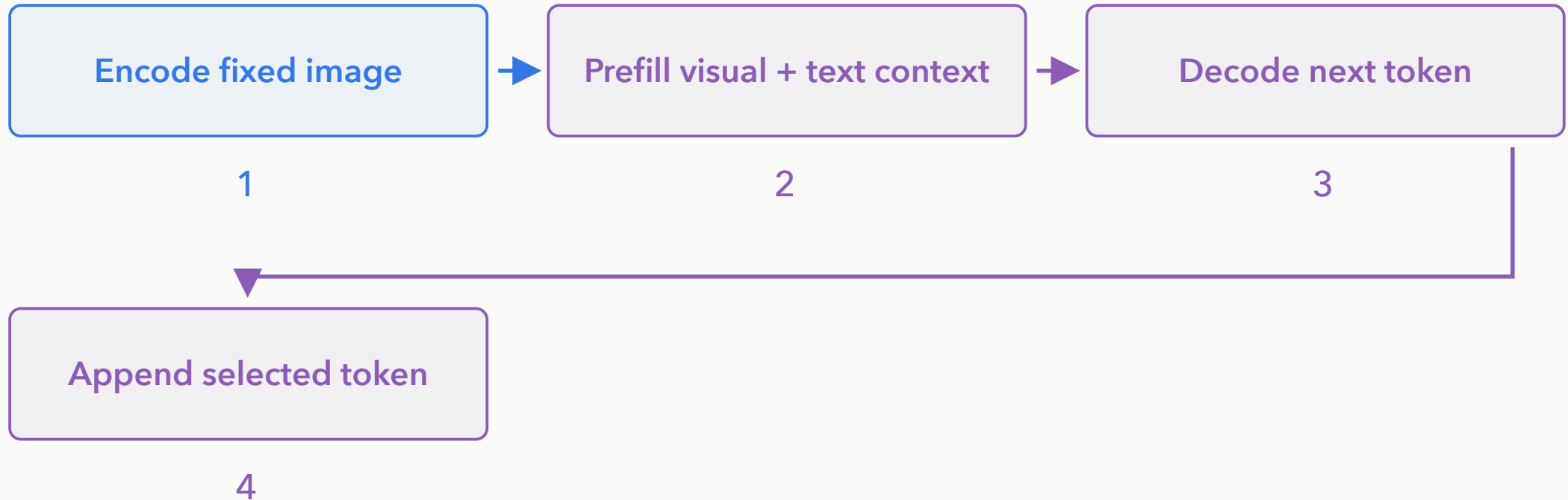
| Only the generated prefix changes at each decoding step.

Decode one new token



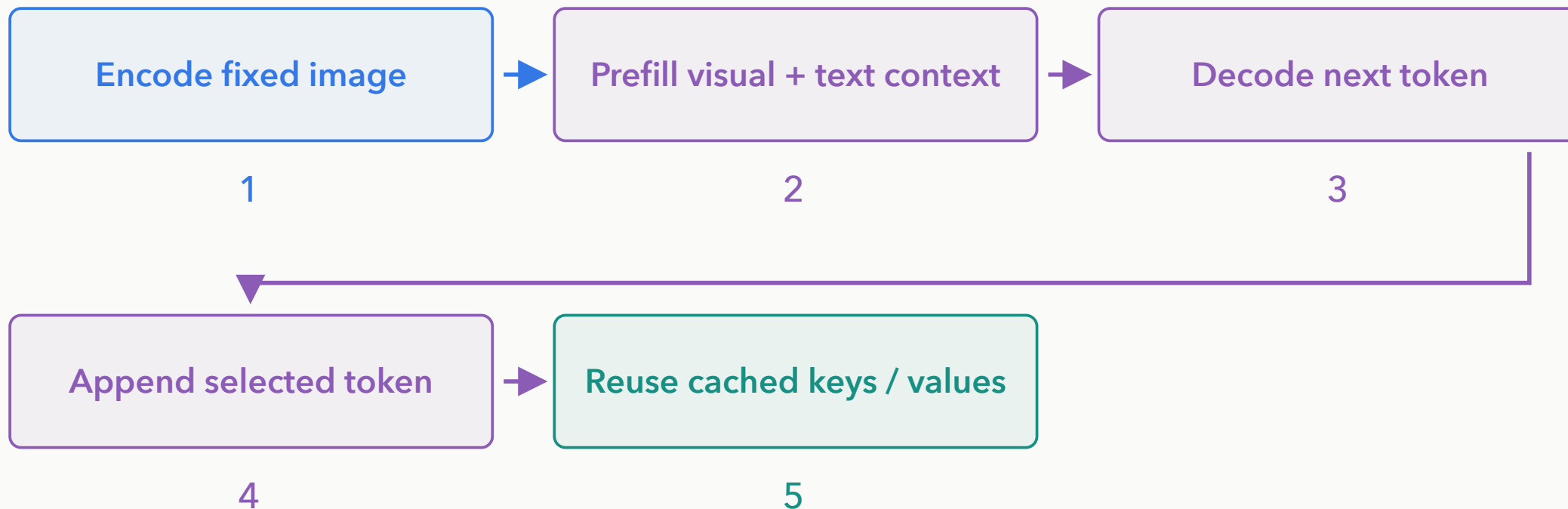
| Only the generated prefix changes at each decoding step.

Append the selected token



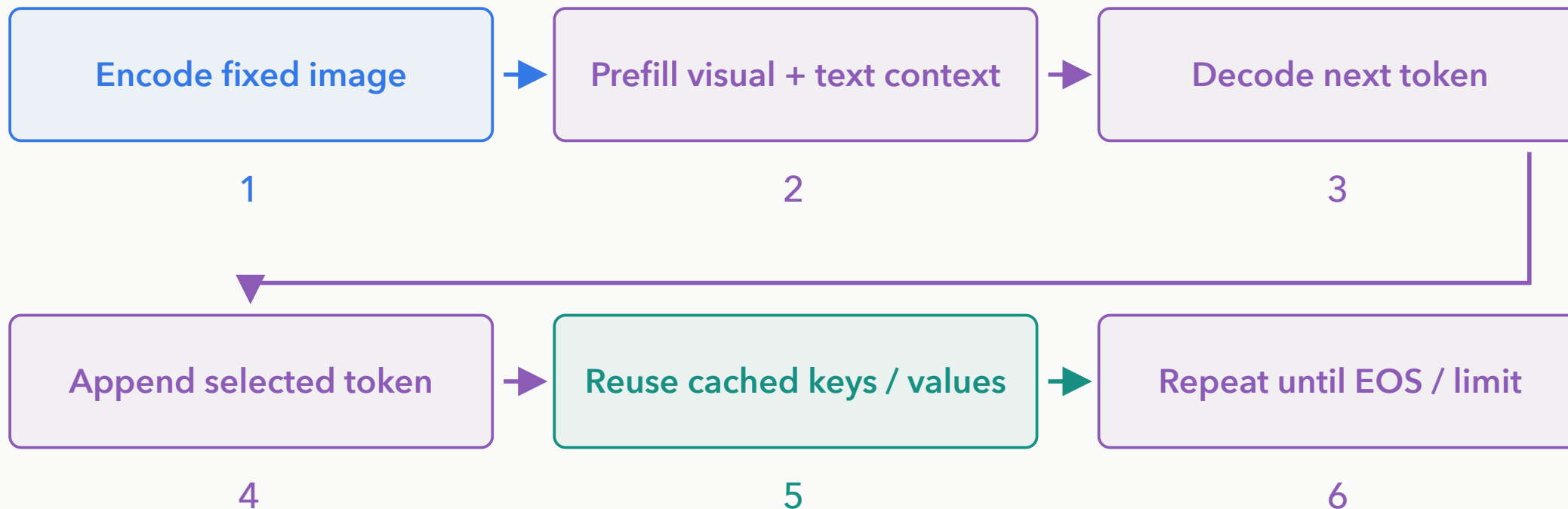
| Only the generated prefix changes at each decoding step.

Reuse the cached keys and values



| Only the generated prefix changes at each decoding step.

Repeat until EOS or the generation limit



| Only the generated prefix changes at each decoding step.

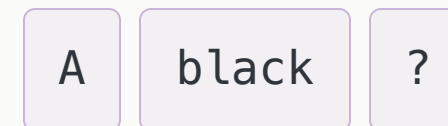
Same conditional model, different available prefix

Training



Reference prefix supplied.
Target loss and gradients.

Inference



Generated prefix grows.
No reference answer or backward pass.

Known targets allow parallel teacher-forced positions; generation is sequential.

Reuse the image; keep the conversation context



User: What animal is this?

Assistant: A dog.

User: What colour is its fur?

Assistant: Black.

Illustrative conversation, not a captured model transcript.

| Visual features can persist. Decoder cache reuse also depends on the exact prefix.

BACKUP C • EVALUATION

Optional • can we trust the answer?

Check evidence use, not just fluent language.

A fluent sentence can have no visual support



What color is the bicycle?

Reference: No bicycle is visible.

| Recorded baseline: "The bicycle is red."

| Fluent ≠ grounded.

Measured check: change only the image.

What animal is shown?



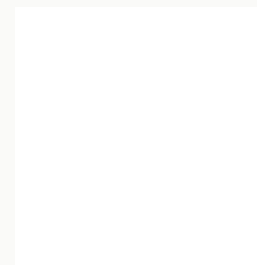
Dog

| There is a black dog in the foreground.



Cat

| A cat is visible.



Blank

| A squirrel.

$$p(y_t \mid y_{<t}, I, P)$$

| Recorded SmolVLM-256M answers: a separate trained model.

Keep a small diagnostic suite

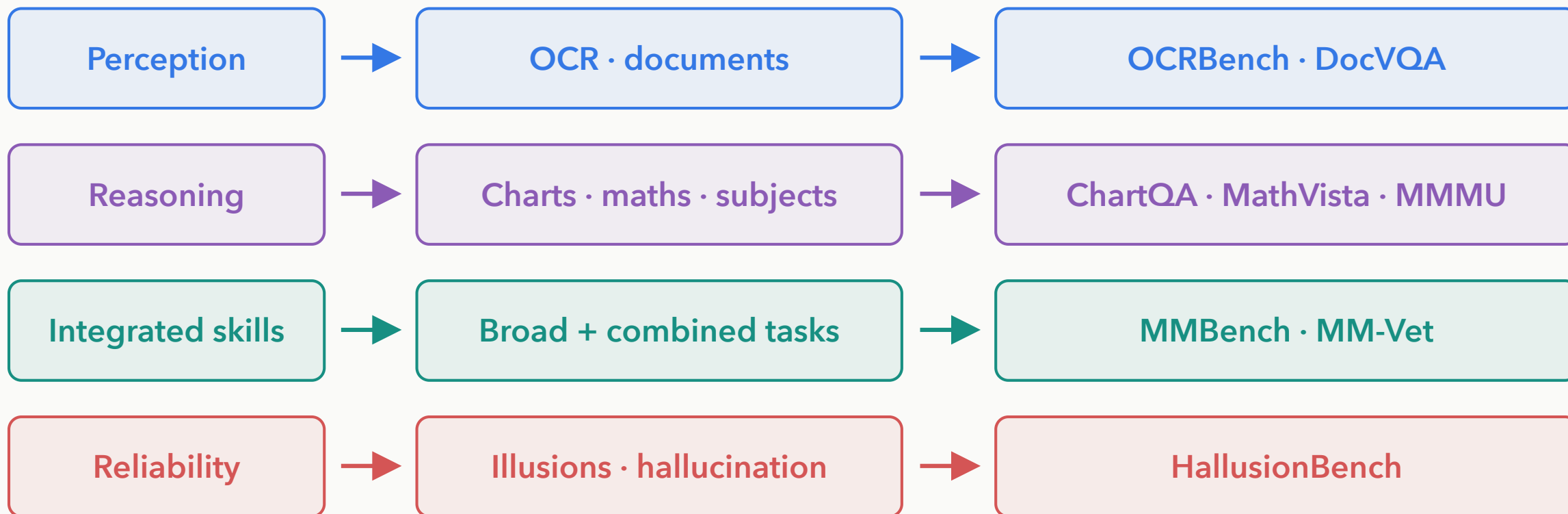
IMAGE	FIXED QUESTION	REFERENCE
Dog	What animal is this?	Dog
Desk	Is the laptop plugged in?	Yes, visible cable connected
Shapes	What is immediately left of the mug?	Red ball
Invoice	What is the invoice number?	ES667-042
Chart	Which month has the highest value?	March
Coffee	What color is the bicycle?	No bicycle visible

Run / record the diagnostic suite ↗

[Offline fallback: recorded model outputs](#)

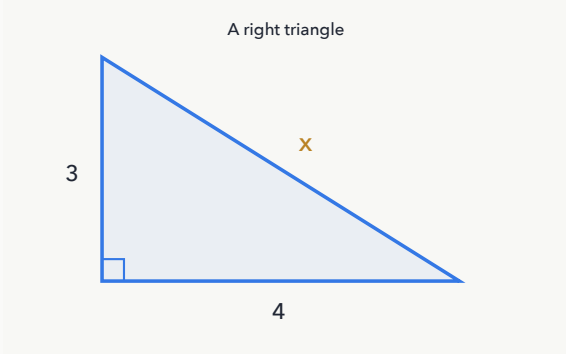
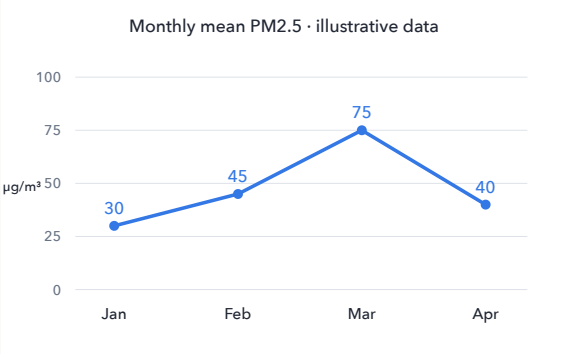
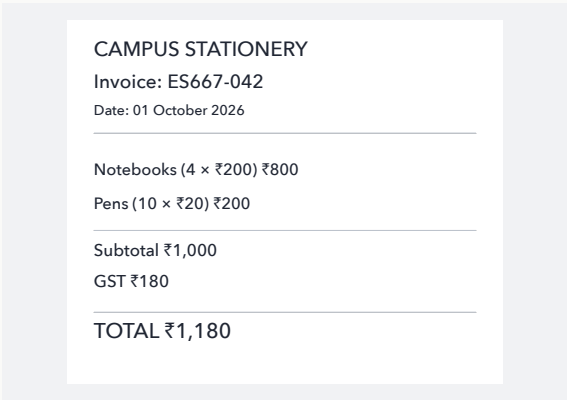
Preserve the raw answer, reveal the reference, then judge support.

Different benchmarks probe different capabilities



| Choose the evaluation from the capability you need.

Inspect the questions before interpreting a score



Read: What is the invoice ID?

Calculate: March minus January?

Reason: What is x?

Reject: What colour is the bicycle?

Authored classroom examples, not copied benchmark items.

A wrong answer may come from perception, reasoning, or accepting a false premise.

Use a metric that matches the answer format

OUTPUT / USE	MEASURE	WHAT STILL NEEDS INSPECTION
Fixed answer	Accuracy	Ambiguity and answer normalization
Document text	Exact / normalized string match; ANLS	OCR errors and layout dependence
Open explanation	Human or specified model rubric	Judge bias and unsupported details
Grounding coordinates	IoU / pointing accuracy	Object identity and spatial accuracy
Deployment	Latency, memory, output tokens/s	Hardware and decoding settings

Report the evaluation protocol along with the number.

Model details

Original architectures and further training choices.
Use these diagrams when a model-specific
question comes up.

Flamingo: a Visual Language Model for Few-Shot Learning

Jean-Baptiste Alayrac · Jeff Donahue · Pauline Luc · et al.

[Read the paper ↗](#)

6 · Read the colors

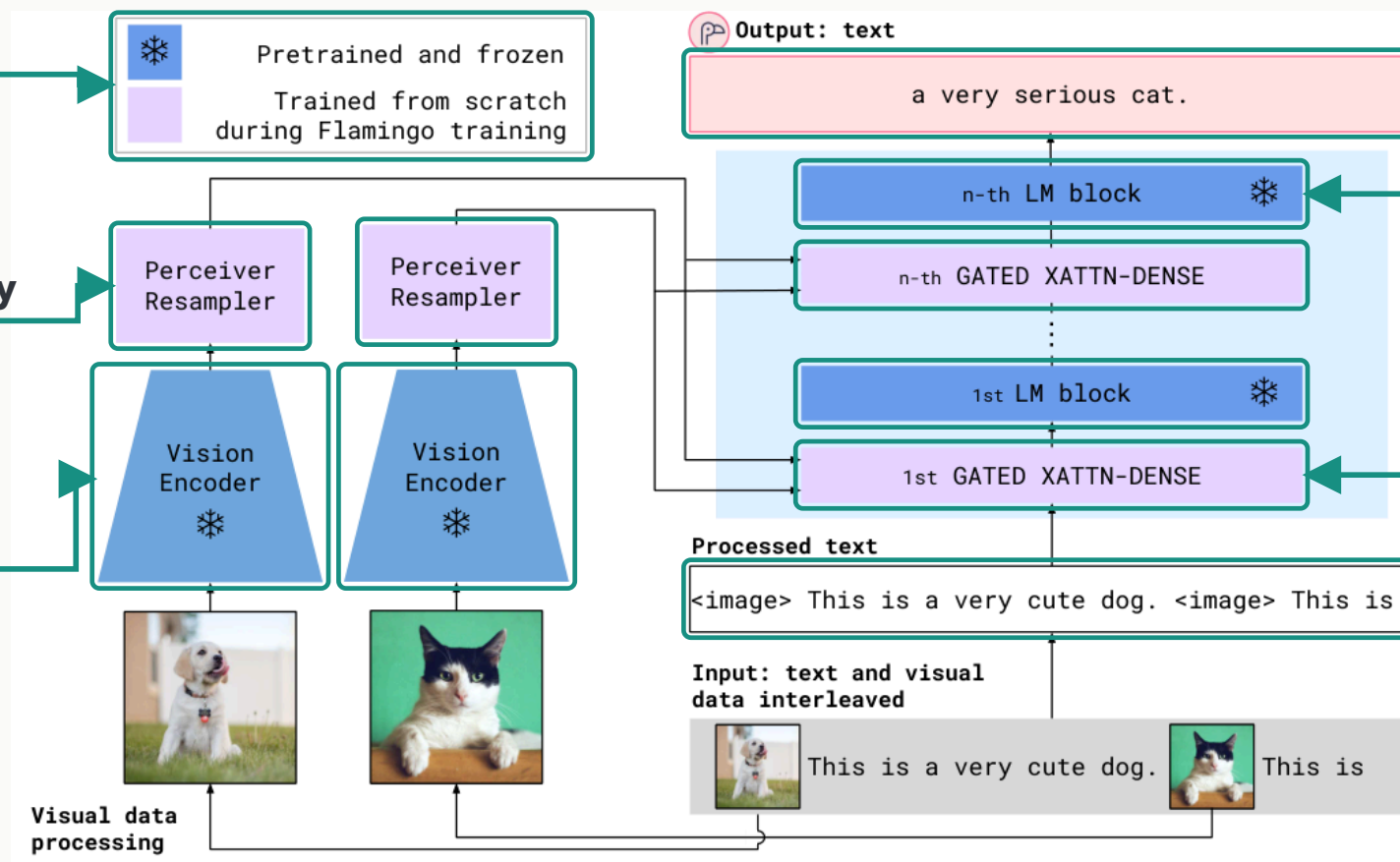
Blue + snowflake: pretrained and frozen.
Purple: trained from scratch during Flamingo training.
Purple: trained here.

2 · Make a summary

The resampler returns 64 vectors per image.

1 · Vision encoder

Each image becomes a grid of features.

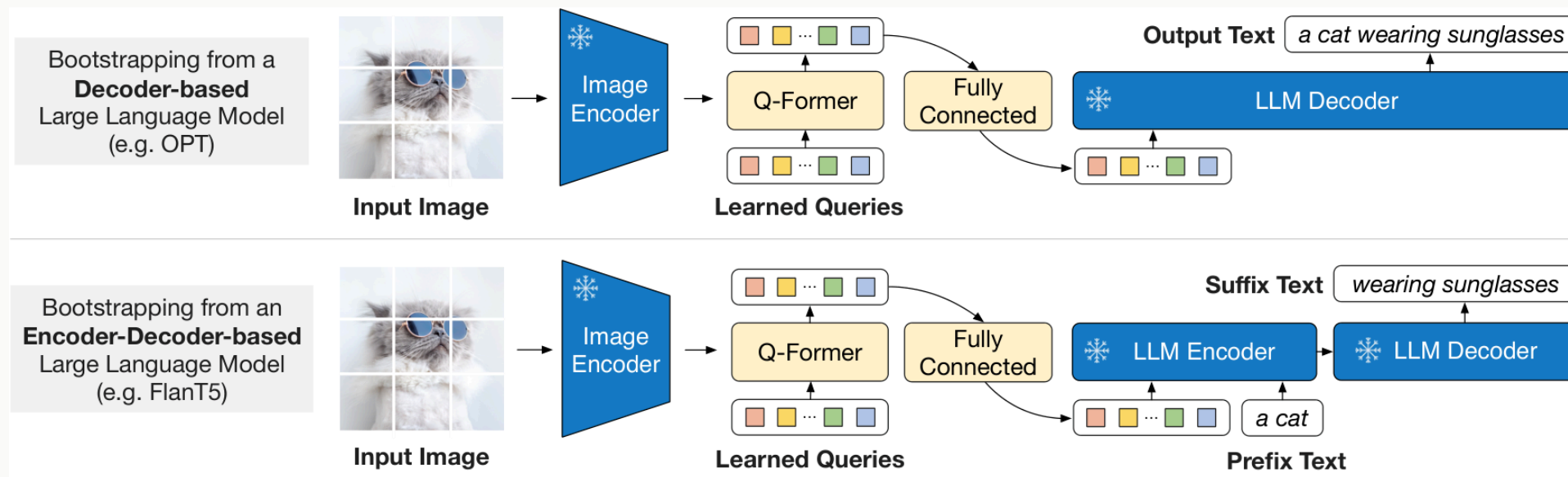


Original Figure 3 + teaching callouts · follow the arrows from the images and text upwards

BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models

Junnan Li · Dongxu Li · Silvio Savarese · Steven Hoi

[Read the paper ↗](#)



Original Figure 3 · Top: decoder-only LLM. Bottom: encoder-decoder LLM.

1 · Idea 3 · learned-query summaries

32 learned queries produce 32 image-dependent output vectors.

2 · Idea 1 · prefix for OPT (top)

Project the summaries; prepend them to text embeddings.

3 · Bottom: encoder-decoder

Visual vectors + text enter the encoder; its decoder generates the answer.

Further training can change how the model answers

Behaviour

Instruction following
Preference optimisation
Safer responses

Evidence use

Better grounding data
Hard negative examples
Missing-evidence responses

| The basic inference target remains image-conditioned next-token prediction.

BACKUP E • SOURCES

Sources

Model papers and evaluation references.

Sources • architectures and browser implementation

[CLIP • Radford et al., 2021](#)

<https://arxiv.org/abs/2103.00020>

[LLaVA • Liu et al., 2023](#)

<https://arxiv.org/abs/2304.08485>

[Flamingo • Alayrac et al., 2022](#)

<https://arxiv.org/abs/2204.14198>

[BLIP-2 • Li et al., 2023](#)

<https://arxiv.org/abs/2301.12597>

[SmolVLM-256M • model card](#)

<https://huggingface.co/HuggingFaceTB/SmolVLM-256M-Instruct>

[Official Transformers.js browser example](#)

<https://github.com/huggingface/transformers.js-examples/tree/main/smolvlm-webgpu>

[Oxford-IIIT Pet • CC BY-SA 4.0](#)

<https://www.robots.ox.ac.uk/~vgg/data/pets/>

Primary sources; no leaderboard scores are used in this lecture.

Sources • evaluation

[OCRBench](#)

<https://arxiv.org/abs/2305.07895>

[DocVQA](#)

<https://arxiv.org/abs/2007.00398>

[ChartQA](#)

<https://arxiv.org/abs/2203.10244>

[MathVista](#)

<https://arxiv.org/abs/2310.02255>

[MMMU](#)

<https://arxiv.org/abs/2311.16502>

[MMBench](#)

<https://arxiv.org/abs/2307.06281>

[MM-Vet](#)

<https://arxiv.org/abs/2308.02490>

[HallusionBench](#)

<https://arxiv.org/abs/2310.14566>

| The local six-image suite is an instructional diagnostic, not one of these benchmarks.